Running Toward the Clinic: Speed, Biological Risk and Health in the AI Transition

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Abstract

The previous paper asked whether faster general AI capability should be welcomed when the capability that arms cyber attackers also brings health benefits, and it counted one kind of harm against the benefit. This paper counts biological misuse, events at research facilities, and attacks on the health systems that deliver the benefit, in the same health units as the benefit itself, with every year of healthy life gained or lost counted in exactly one term. No source we read settles three choices, so the paper carries both sides of each: whether the misuse hazard is set by the capability level reached or by the speed of advance (two co-equal specifications); whether it runs from the start of the capability path or only once the capability is reached (the ramp-on and arrival-on exposure readings); and, for a facility, what bringing misuse capability closer means (two co-equal conventions). Some results hold whatever is chosen. A plateau holds under each specification within its own hazard conditions (§4a): once the misuse capability has been reached, faster progress raises misuse loss. Earlier arrival of a fixed-duration benefit, counted in full with no horizon cutting it off, is worth nothing undiscounted, and discounting favours timing over the probability that a candidate reaches patients, not the reverse. The rest is conditional. Truncation, in which acceleration pulls level-indexed risk inside the horizon, holds under the level-set specification only, and only while the capability lies beyond the horizon. In the defence clock, whether acceleration helps turns, under one specification, on a race between the capability a countermeasure needs and the capability misuse needs, and under the other on the hazard's elasticity against the share of the deployment wait that capability sets. On the ramp-on exposure reading, the two specifications give conditional opposite signs for the misuse marginal in four cases. On the arrival-on reading, compared on one reading throughout, one of those four survives, and the choice of specification also moves a threshold on the health benefit itself whenever the loss per misuse event is positive, under a stated normalisation fixing one specification's hazard against the other's. Opposite signs for the misuse term need not mean opposite rankings of speed. Under level-set on the ramp-on reading, and under conditions §4a.5 states, a facility's effect through the misuse distance flips sign between the two conventions. One result is invariant across all three choices in form rather than in magnitude: at a fixed speed, with the deployment date inside the counted window, the residual hazard fixed and coverage that translates rigidly with the deployment date and completes within it, shortening the distance a countermeasure waits on lowers misuse loss while that wait is capability-set and deployment lowers the hazard, and does nothing once the wait is adoption-set. That invariance is structural rather than three independent confirmations, and read as a shift in time the algebra is elementary; what is this paper's is the division of the deployment wait into a capability-set and an adoption-set part, and the floor that follows from it (§4a.5). The mechanism results are proved in an appendix. No calibrated sign of the whole balance follows: the health gain

from AI, the loss per misuse event and the conversion of hospital-attack deaths into health units are all unmeasured in the sources we located. The misuse evidence stops at measured task performance: a public model outperformed 94 per cent of expert virologists within their own sub-areas on a purpose-built benchmark, the one randomised test could, on its authors' own account, detect only an effect about five times the one it observed, and its results, in their words, "do not and could not disprove the existence of risk"; and the two concern model generations that do not overlap in time. The one channel with a measured causal magnitude, hospital attacks, comes to the order of one excess in-hospital death per attack on our own arithmetic, while no estimate of any kind was located for the misuse channel; a reassuring reading of what is measured reflects which channel can be measured, not what the evidence shows. The specification and the horizon are choices, not gaps a longer search would close.

1. Introduction

The previous paper, "Running Through the Rain" (RTTR), asked whether faster general AI capability should be welcomed when the capability that arms cyber attackers also brings health benefits. Its answer on the health side was narrow. Benefits that arrive with the capability give a second reason to run, but only while the benefit still waits on capability rather than on its own trials and approval, and only if the health gained from earlier arrival exceeds the added cyber harm (RTTR §4.7; Appendix A, Proposition 16). That ranking counted one kind of harm against the benefit. This paper counts biological misuse, events at research facilities and attacks on the health systems that deliver the benefit, in the same health units as the benefit itself. Does the ranking survive? Where it does not, which mechanism reverses it? Does the answer depend on how the misuse hazard is written down? And what would have to be observed to settle the rest?

The units are DALYs. The literature RTTR carries already puts mortality gains and catastrophic risk in one unit [g9], and we keep the additive ledger as a marginal approximation for non-overlapping expected losses [g10]. The misuse hazard can be written two ways. Under a level-set specification it is set by the capability level reached; under a speed-set specification, the form of RTTR's main results, it is set by the speed of advance. RTTR also carried a level-set hazard as a separately scoped alternative (RTTR §4.3; Appendix A, Section A.7). No source we read settles which holds [b12], so the paper carries both, neither as the base. Figure 1 shows what each does to the hazard.

A note on the figures. Most figures in this paper draw the model, not the world: the shapes are the model's own functions at illustrative values, chosen to show a mechanism, and no axis in them carries a measured quantity. Seven do something else, and say so in their captions. Figures 2, 19 and 27 set out the paper's own claims and dependencies rather than drawing anything. Figure 18 reports counts from the project's own numerical check. Figure 22 is our arithmetic over one study's counts, graded WEAK. Figure 24 records what the evidence reaches and where it stops. And Figure 25 alone plots published statistics. Each caption repeats which kind it is, and the reader should take this as governing all twenty-seven.

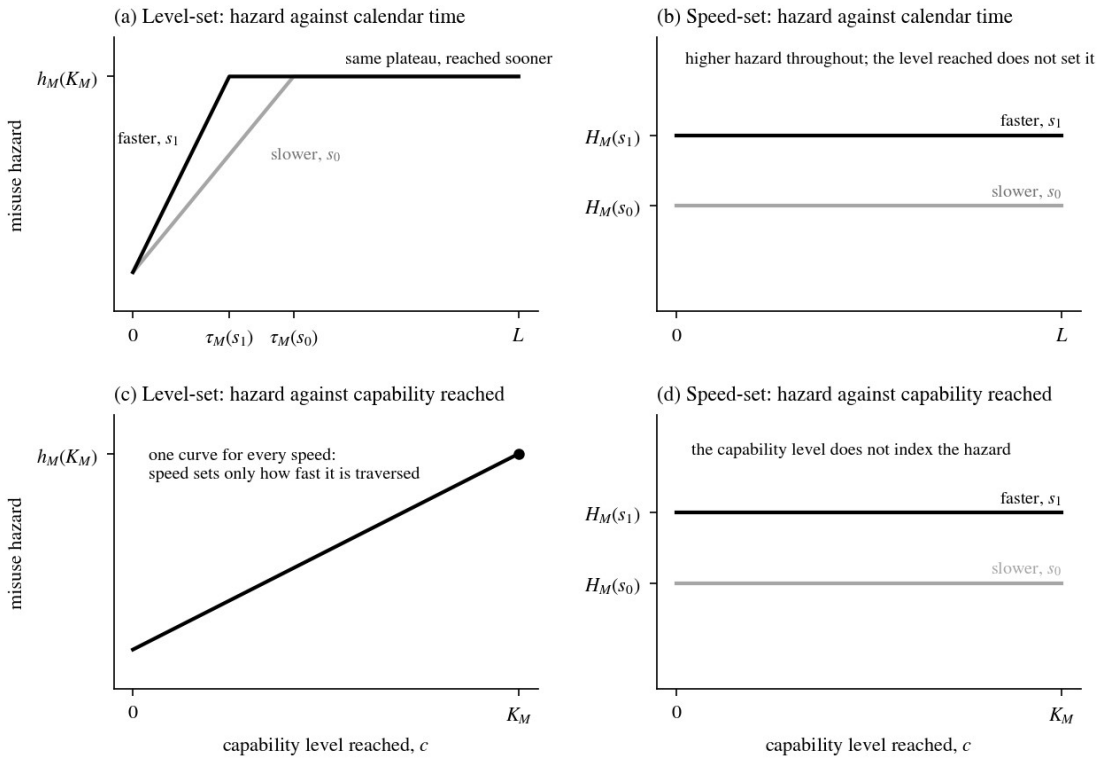


Figure 1. The two hazard specifications. Each panel shows the misuse hazard at two speeds, slower s_0 (grey) and faster s_1 (black). Under the level-set specification (a, c), the hazard is set by the capability level reached, *c(t)* = min(*st*, *K_M*). Faster progress climbs the same curve sooner and reaches the same plateau *h_M(K_M)* earlier, so when plotted against capability (c) a single curve serves every speed. Under the speed-set specification (b, d), the form of the previous paper's main results, the hazard *H_M(s)* is set by speed itself. Faster progress raises the hazard at every date, and the capability level reached does not index it at all. The paper carries both specifications co-equally, and no source we read settles which holds [b12]. Speed-set is drawn here with ramp-on exposure (see Figure 6). These are definitions, not results. Schematic; axes are not to scale. The curves are the model's own hazard function at illustrative values. *Notation:* the panels and this caption write the speed-set hazard *H_M(s)*, the model checks' symbol; the text writes it *Λ_M(s)* (§4a.1, note on notation).

The paper derives three mechanism results. The *plateau* (Figure 7): once the misuse capability has been reached, faster capability raises misuse loss, under hazard conditions §4a states. *Truncation* (Figure 8): when the misuse capability lies beyond the horizon, acceleration pulls level-indexed risk into the window. The *defence clock* (Figure 10): when defence closes the risk at a date of its own, whether acceleration helps turns, under level-set, on a race between two capability distances, the one defence needs and the one misuse needs; under speed-set it turns on the hazard's elasticity against the share of the deployment wait that capability sets. Truncation holds under the level-set specification only; under speed-set it has no counterpart. It also needs an objective that stops counting

at a finite horizon: its regime $sL < K_M$ is empty without one. The plateau and the defence clock exist under both specifications. Speed-set results come in two exposure readings: ramp-on, with exposure counted from the start of the capability path, and arrival-on, with exposure counted from the date the capability is reached. Each result is proved in the appendix on every reading on which it is stated, and §4a gives the conditions each needs.

The paper also sets out two kinds of non-identification, and they do not stand on equal footing. The first is specification-level and derived on the ramp-on reading. In four cases, each under the restrictions §4b states, the two specifications give conditional opposite signs for the misuse marginal. On the arrival-on reading the count is smaller, and the reason is a comparison error now corrected. Read like for like, with the level-set side gated as §4a.2 defines it, **only one of the four survives**: rows 1 and 4 cannot take opposite strict signs there, because with a gated profile each of their divergence conditions would require a negative hazard, and the fourth case has no arrival-on version. Opposite misuse signs do not make opposite rankings of speed. The whole balance ranks speed oppositely under the two specifications only when the rest of the balance, assumed common to both, lies strictly between the two specifications' misuse terms, both taken on the same exposure reading (§4b). The second kind is parameter-level: a construction asking whether values of the unmeasured terms, held inside what the evidence allows, can flip the sign of the balance. §4b states it and computes one part, a bound on the hospital channel graded WEAK; the rest is not yet done.

Two things the paper does not deliver. It gives no calibrated sign of the whole balance, anywhere: the health gain from AI, the loss per misuse event and the conversion of hospital-attack deaths to DALYs are unmeasured, and the probability that an AI-originated candidate reaches patients is measured only at early phases and at abstract level [g7, b1, d4, g2, g5]. And it gives no ranking of a company-run laboratory against a public or academic one, because no source we located supplies one [c9, f6].

Public events prompted the paper. Between August 2025 and April 2026 four kinds of Anthropic-related cyber event were reported: attacks run using its model, a capability it disclosed, leaks of its own material, and a claim of unauthorised access to a model through a vendor's environment, which the company said it was investigating; they differ in who acted and what was exposed, and they are not one series [a7]. Reuters reported the company's biology laboratory on 18 September 2026, after an interview in which the company confirmed it, and the company described it publicly five days later [a4]. The public argument over the laboratory that our search located is thin and mostly not expert, and it is about legitimacy: secrecy, limited outside oversight, and a company that warns of catastrophe expanding into biology [a8, DIS 11]. It is not a comparison of safety; no such comparison was located [f6]. This paper takes no side in that argument. It asks the prior question of what the health balance looks like when every harm is counted in one unit.

The paper runs as follows. Section 1.1, next, states in one place what the paper can and cannot show. Section 2 reports what is known about the laboratory and poses the distance question the model answers; §3 says what seven literatures establish and where each stops. Section 4a derives the mechanism results under each specification, and §4b sets out the two kinds of non-identification. Sections 5 to 8 give the evidence for the misuse,

hospital, defence and benefit terms in turn, each at the strength its sources support. Section 9 takes up the facility, §10 two further levers, and §11 the limits of the search and the quantities that would settle what is left open. Section 12 concludes. Proofs are in the appendix, and the record of how the results were first derived is in a supplement after it.

1.1 What this paper can and cannot show

This subsection gives the paper's bottom line for a reader who will not read the model sections.

What holds whatever one assumes. Four mechanism results hold under every choice the paper declines to make, each within its own restrictions. The defence clock, which signs how misuse loss responds to speed, cannot in general sign whether a facility that moves the two capability distances lowers misuse loss, nor the reverse — proved under speed-set, where the hazard's elasticity is absent from the facility rule, and shown by counterexample under level-set (§4a.5). Once the misuse capability has been reached, faster progress raises misuse loss under each specification's own hazard restriction. Where the countermeasure waits on adoption, both specifications agree that speed hurts. And earlier arrival of a benefit of fixed duration, counted in full, is worth nothing undiscounted, because its end moves forward with its start; discounting favours the timing channel over the probability that a candidate reaches patients, not the reverse. A fifth result holds in the same form under every choice, and that sameness is structural rather than three independent confirmations. At a fixed speed, shortening the capability distance a countermeasure waits on (K_D) lowers misuse loss exactly while that wait is set by capability rather than by deployment, and while the misuse hazard before the countermeasure arrives exceeds the residual hazard after it — with the deployment date interior to the counted window and after the capability is reached ($\tau_M < t_D < L$), the residual fixed, and, where coverage is gradual rather than a step, a schedule that moves rigidly with the date and finishes before the horizon; once the wait is set by adoption, further cuts buy nothing (§4a.5, A4a.7(i)). What is invariant is the form and that sign condition, not the magnitude, which carries each case's own plateau hazard; it holds because K_D enters the loss at one point, the deployment date, while each of the three choices governs a different object (A4a.7(i)). Written as a shift in time the derivative is elementary, and what is this paper's is the division of the deployment wait into a capability-set and an adoption-set part, and the floor that follows from it. The accounting identity that counts each year of healthy life gained or lost in exactly one term also holds under every choice [ACC], as do the negatives: no calibrated sign of the whole balance, no ordering of the interventions, no decision rule for acting while the balance is unidentified, declined explicitly rather than left to inference (§4a.7), and a named list of what would be needed (§11).

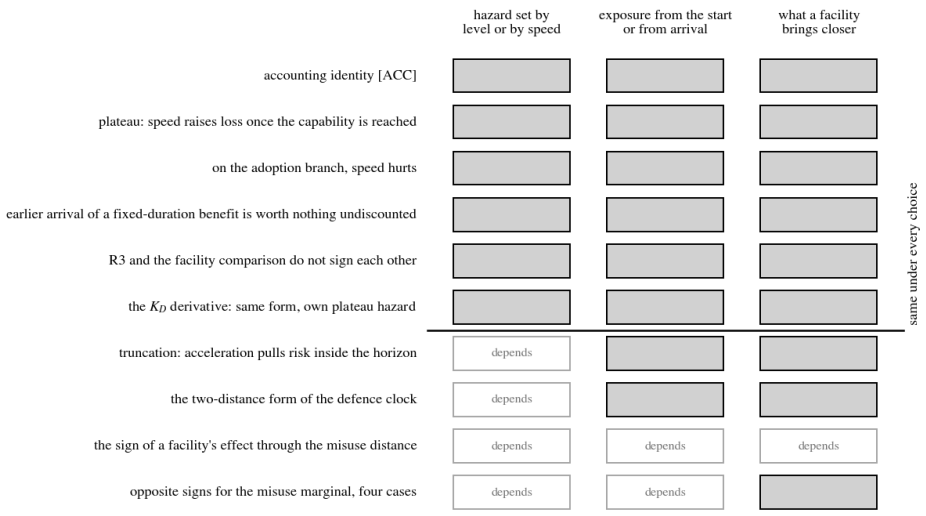


Figure 2. What the three choices do and do not touch. Rows are results, columns the three choices the paper declines to make. A filled cell means the result takes the same form whichever way that choice goes; "depends" means it does not. Six rows are filled across: the accounting identity and five mechanism results — the plateau, the adoption branch, the benefit's undiscounted timing margin, the mutual non-determination of R3 and the facility comparison, and the *K_D* derivative, whose sameness is of form and not of magnitude. The four below the rule are where the paper's non-identification lives. The grid states the paper's own claims; it proves none of them.

What is conditional, and on which choice. Most other results turn on one of three choices. These are modelling choices, not measurement gaps: no further search of the literature produces them, and each is unsettled and changes the answer. The first is whether the misuse hazard is set by the capability level reached (level-set) or by the speed of advance (speed-set, the form of RTTR's main results); the paper carries both, neither as the base. Truncation, in which acceleration pulls level-indexed risk inside the horizon L, exists under level-set only, and only where the misuse capability lies beyond what the horizon reaches at the given speed ($sL < K_M$); the two-distance form of the defence clock is level-set only. The second, a separate axis, is whether the hazard runs from the start of the capability path (ramp-on) or only once the capability is reached (arrival-on). The four cases of conditional opposite signs for the misuse marginal rest on the ramp-on reading alone, and opposite misuse signs need not mean opposite rankings of speed (§4b.1). The misuse distance *K_M* carries zero weight in the facility comparison under speed-set ramp-on only. The third is what it means for a facility to bring misuse capability closer: the same hazard path moved to a lower threshold (convention A), or the hazard at each capability level unchanged so that the path stops lower (convention B), carried co-equally. Under level-set on the ramp-on reading, under the conditions of §4a.5, the same verbal intervention raises misuse loss under A and lowers it under B.

What the state of the art in frontier models cannot settle. The chain from a model to a health loss runs from access, to intent, to measured task performance, to real-world

misuse, to consequences in cases, deaths or years of healthy life. The evidence reaches the third link and stops. Three facts must be held together. On a purpose-built virology benchmark a public model outperformed 94 per cent of expert virologists within their own sub-areas: STRONG for the benchmark, and support for no attack probability [b3]. The one controlled trial with a randomised no-AI arm found no significant difference in the adjudicated viability of written plans, from a design its authors say could detect only a change about five times the one observed; the results, in their words, "do not and could not disprove the existence of risk" (MODERATE; §5.2–§5.4) [b2]. And the two concern different model generations that do not overlap in time, so neither speaks to the other or to the frontier as it now stands [b3, DIS 1]. Developer capability thresholds are, on the developer's own account, governance decisions under acknowledged uncertainty, not measured attack probabilities [b6]. No source read estimates attack probability, casualties or healthy life lost per unit of capability, and the consensus report read says the field has no standard for it, for contingent reasons: secrecy, testing prohibitions and immature evaluation [b1]. No misuse of an AI biological tool that caused real-world harm is recorded in what was read, though one developer reports having disrupted attempts [b10, b14], and no public record is not the same as no record. The paper can write the misuse term; it cannot calibrate it, and the interest on each side — biosecurity organisations behind the benchmark, model developers among the authors of the sceptical review — is named in §3.

What our understanding of the biology cannot settle. Whether misuse is limited by capability or by tacit knowledge and organisational capacity, two different claims, was contested before AI; the strongest critics dispute what binds, not whether capability is rising [b7]. The literature names one mechanism by which an AI-directed laboratory could lower the tacit-knowledge barrier and says the test has not been done, and §2 reports singly the four conditions for carrying it to a facility [b8]. No accident rate by operator type, and no comparison of safety outcomes across privately and publicly owned life-science facilities, was located [c9, f6]. No conversion from an excess hospital death to years of healthy life was located [d4]; the one causal design finds no 30-day mortality effect, consistent, on our reading and not the authors', with deaths brought forward rather than full remaining life lost [d1, d2]. And the loss per misuse event (V_M) is fixed by assumption, not shown: under level-set, capability then indexes how often misuse events occur but not how severe they are, though the catastrophe literature the ledger draws on is about consequence scale growing [g9, g10].

Measurability and magnitude need not align. The only causal finding the paper grades STRONG is in the hospital channel: ransomware attacks raise in-hospital mortality among patients already admitted, STRONG in direction and MODERATE in magnitude [d1], and on our own arithmetic, graded WEAK, of the order of one excess in-hospital death per attack [d3]. Section 4b.2 bounds the mortality part of that harm, graded WEAK, as a function of a conversion from deaths to DALYs for which no measure was located, at anchors from under 0.05 DALY per attack to about 20. That factor of 260 is not the unit's doing and not the data's: both ends use WHO's valuation, and what separates them is whether an attack caused a death or advanced one, which the study was not designed to answer. The low end is a ceiling for the counterfactual in which the deaths were advanced by at most a month; the high end is what the same unit books if they were caused outright. Where misuse and

the hospital channel pull opposite ways and all else is held fixed, misuse loss would then have to change by more than roughly 20 DALYs per unit change in expected hospital attacks, times an unmeasured all-payer scaling, before misuse decides the comparison under every conversion between those anchors. That 20 is on the authors' own death counts; allowing for sampling error in the mortality effect moves it to 25 or 34 depending on which of the study's two control-group specifications is read, and §4b.2 gives each rule's bracket with both ends computed the same way. For the misuse channel, where the paper's concern lies, no estimate of any kind was located, not a probability, a casualty count or a figure in healthy life, and the field says it has no standard for producing one [b1]. A reader who weighs only what is measured therefore finds a small number for the one harm that has a number, nothing for the harm the paper is about, and a balance that looks reassuring. The measured record cannot justify that inference. What it establishes is that the two channels are unequally observable, not that the unmeasured one is larger: the one channel with a measured causal magnitude is not the channel this paper is about, and the channel it is about has no estimate of any kind. A reassuring aggregate reading is therefore manufactured by what happens to be measurable rather than supported by it — and by the same token nothing here licenses an alarming one, since neither the benchmark evidence nor the hospital evidence identifies the balance.

Section 11 lists what would settle each open question and which result each quantity decides. One item there is not a search result at all: the horizon L is a choice, and it decides whether truncation exists. A second, the specification, is listed there among what no source read supplies. It differs from the other gaps in being a modelling choice the paper makes both ways, which a measurement of whether misuse risk scales with the level reached or with speed could inform. None was located; until one is, the choice decides the sign of the misuse marginal in the four cases.

2. The facility as reported

That the laboratory exists is independently reported: investigative reporting from two outlets, confirmed by the company [a1]. Its size, staff and opening date are not stated. What it does is known only from the company's own account, and everything below is labelled as such.

By the company's account, the laboratory works on fundamental biology, not on drug discovery specifically, and runs no clinical work for now [a3]. The company's wider life-sciences unit, by its own description, includes drug discovery; nothing read establishes where that work is done [a3]. The company states that it works only at biosafety levels 1 and 2, handles no pathogen that can infect humans, and has all bench work done by human scientists [a2]. On those statements the laboratory lies outside the population the high-containment biosafety literature studies, which is maximum- and high-containment facilities; that placement rests on the company's account alone [a2, DIS 11]. The company describes model-directed control of laboratory equipment as an aim, not current practice [a5]. No independent inspection, permit, registration or institutional biosafety committee record was located [a2, DIS 11]. Whether the operating entity falls inside the US research-oversight regimes is not determined on this record. The company's statements on containment and agents bear only on the first of two US triggers, possession of a listed

agent, and are its own account; nothing read establishes the second, federal life-sciences funding [a2, c4]. No measured contribution of the laboratory to any therapy's arrival was located either: an absence, which does not show the laboratory is irrelevant to therapies [a3].

The literature on AI-directed laboratories names one mechanism that could lower the tacit-knowledge barrier to misuse, and says the test has not been done [b8]. Carrying that concern to a particular facility needs four conditions. We report each singly, as the company reports it, and do not combine them.

1. Biological rather than chemical work: met [b8].
2. Automation that substitutes for skilled execution rather than supervising it: unmet as company-reported. Bench work is reported as human-performed and model direction as an aim [a2, a5, b8].
3. Acceptance of experiments specified by outsiders: unknown. Nothing read addresses it [b8].
4. Work with agents anywhere near those this literature concerns: unmet as company-reported [a2, b8].

Both "unmet" entries rest on the company's own statements, with no independent inspection, permit or registration located [a2, DIS 11]. If bench work is human-performed, as the company reports, the route this literature names, automation that substitutes for skilled execution, does not apply to that work; that would not establish the absence of AI assistance or of every tacit-knowledge effect [a2, b8].

The paper does not ask whether this laboratory is safe, or safer than a public one; nothing we located can answer that [c9, f6]. It asks a distance question. Intervention I3, a developer-owned laboratory taken as reported, might move two capability distances: K_D, the capability a countermeasure needs, and K_M, the capability misuse needs. At a given speed s, does the move lower misuse loss? Section 4a.5 derives the comparison, with its conditions, Figure 11 draws it, and §9 takes the question up. This section says nothing about which distance the laboratory moves, by how much, or where its event rate sits [a2, a3].

The question is not the defence clock's. R3 (§4a.5) signs how misuse loss responds to speed, and its inequality compares the two distances' levels. The facility question compares changes in the distances at a fixed speed. R3 therefore cannot sign the facility comparison in general, and the facility comparison cannot sign R3; the single mechanical exception is footnoted at §4a.5. What a longer K_D does to each is stated at §4a.5: two true statements about different objects, and neither is advice about K_D.

Two features of the comparison shape what this section can say. Under speed-set on the ramp-on reading, the misuse distance carries zero weight, so the question is about the defensive distance alone. Under level-set the answer turns on what it means for a facility to bring misuse capability closer, and no source we read fixes that meaning [b12]: the paper carries two conventions co-equal, neither as the base, and on the ramp-on reading the sign

of the *K_M* effect flips between them. Figure 26 (§9) shows what each convention does to the hazard path, and §4a.5 states both in symbols.

Each of those conditions signs misuse loss alone. Every other term of the balance, $J_I3 - J_0$, is held fixed by assumption, and the sign of that whole difference is not derived (§9).

3. Literature

Our search covered seven literatures (its scope and gaps are in §11). For each we say what it establishes and where it stops.

Growth, catastrophe and speed. This is the literature RTTR built on. It puts mortality gains and catastrophic risk in one unit, which licenses the DALY ledger, but its catastrophe is a generic one-time event, not a hazard tied to the capability that produces the gain [g9]. The result that costs of catastrophes multiply while benefits add binds only under its own assumptions about consumption and risk aversion [g10]. The strongest objection to speed in it, that optimal adoption is gradual when damages scale with capability and society learns, rests on learning, which RTTR's model does not contain [g11].

AI and biological misuse. No source read estimates the change in attack probability, casualties or DALYs per unit of capability, and the consensus report read says the field has no standard for it [b1]. The evidence reaches access and task performance in evaluations and stops. A purpose-built expert benchmark places a public model above most expert humans on its own items [b3], and the one controlled comparison returned a null from a design its authors say could detect only a change about five times the difference observed [b2]. The two concern different model generations and do not overlap in time, so neither speaks to the other [b3, DIS 1]. No misuse of an AI biological tool that caused real-world harm is recorded in what was read, though one developer reports having disrupted attempts [b10, b14]. Interest runs both ways in this literature, and the paper discloses it rather than grading around it. The benchmark's authors are at SecureBio and the Center for AI Safety, both biosecurity organisations [b3]; the review that finds no recorded harm is led from Cohere For AI and has a co-author at Mistral AI, two model developers among otherwise academic co-authors, and its case is that the available literature is methodologically immature and not transparent enough to settle the question [b10]; and the report of disrupted attempts is a vendor's account of its own product [b14]. None of this bears on the design of any of the three, and none of it is a reason to set one aside; all of it is a reason not to read agreement inside one camp as independent corroboration. An insensitive null is not a tight zero; §5 and Figure 25 give the figures. Developer capability thresholds are governance decisions, not measured probabilities [b6], and the field splits on whether capability or organisation is the binding constraint [b7].

Biosafety and dual-use governance. This literature states the regimes precisely. US law reaches a laboratory through two separate triggers, and a privately funded laboratory holding nothing listed falls inside neither research-oversight regime (§9) [c4]; Canada licenses the activity whatever the funding source [c5]; the treaty read sets duties and has no facility-level verification [c6]. What we read of it does not give an accident rate. The two

published calculations on the one dataset are not estimates of the same quantity [c1], and no rate by operator type was located [c9].

Cyberattacks on health systems. One causal design finds that ransomware attacks cut hospital volume and raise in-hospital mortality among patients already admitted [d1], and null findings from single incidents are part of the record [d5]. No conversion of any of it to DALYs was located [d4]. What is measured is loss to existing care. Only one abstract, on disrupted clinical trials, bears on the benefit a new therapy loses when the systems that deliver it are attacked [d7].

Offence, defence and countermeasures. Theory makes the balance a function of investment rather than a property of a domain, and that analysis was not located for biology [e3]. The one study located that asks the question directly finds attackers favoured on four of five asymmetries, does no quantitative analysis, and restricts defence to pharmaceuticals by choice [e1]. Countermeasure timing is dated for one vaccine programme [e5]. The one control criterion read applies to a single intervention within one model and is not a general limit on defence [e7].

Trust and ownership. Economics gives the ownership question a decision rule about non-contractible quality and the operator's replaceability, not about trustworthiness [f1]. Public ownership does not remove insider risk: the two attributed events these sources record trace to public, military facilities [f2]. One, the 2001 mailings, was attributed in a prosecutorial summary never tested at trial, and a National Academies review found that the scientific evidence alone could not settle the origin [f2]. The other attribution rests on an epidemiological investigation we hold at abstract level only [f2]. Trust surveys are evidence about legitimacy, not safety, and do not rank ownership forms [f5]. No comparison of safety outcomes across privately and publicly owned life-science facilities was located [f6].

The benefit premise. Discovery-stage speed-ups are reported, but only by developers or at second hand, and no cross-drug estimate of AI's effect on total development time was located [g1]. Most of the wait is clinical and after approval [g3]. Realised output comes mainly from narrow tools [g6], three sources from different directions suggest that the probability of picking the right compound may matter more than the speed of picking it [g5], and no DALY or population QALY figure for health gained from AI was located [g7].

What is missing. Among the sources we read, none models benefit and misuse together in health units [g9, b1, g7]. The growth literature has the units but a generic catastrophe; the misuse literature has no health unit; the benefit literature has no misuse term. That is a statement about what we read.

The specification. The literature also leaves open how to write the misuse hazard. A state-risk reading, with hazard set by the capability level reached, appears in theory RTTR cites [b12], and RTTR itself carries it as a separately scoped alternative (RTTR §4.3; Appendix A.7); RTTR's main results set hazard by speed. The two give different mechanism results (§4a) and can give opposite signs for the misuse marginal (§4b, Figure 17). No source read settles the choice, and nothing located measures whether misuse risk scales with the level

reached or with speed [b12, b1]. We treat it as a live modelling choice, carry both specifications, and list the question as an estimand in §11.

4a. The model: mechanism

4a.1 Set-up

We start from the previous paper's Proposition 16 (RTTR, Appendix A.3.5): $W_L(s) = y(L - t_B(s))^{+} - V_cC_c(s)$, under A15–A18. A15 makes the health flow y constant and the terms additive, A16 lets one speed s advance both of RTTR's capability distances, K and K_B (this paper extends it to K_M and K_D), A17 fixes the adoption times and shares in speed, and A18 forbids counting a cyber harm in V_cC_c and again as a cut in y. This paper's balance is $W_L\text{^}X$ ($W_r\text{^}X$ discounted). It recovers W_L exactly when $p = 1$, $\omega = \delta_T = 0$, every added channel is zero and $J_0 = 0$; with the comparator retained it recovers $W_L - J_0$.

The scenario balance. We keep y as the flow to patients and carry foregone benefit as its own term [ACC]. These are definitions only:

$$J_i(s) = p \cdot y \cdot (1 - \omega) \cdot (L - a(s))^{+} - \Phi(s) - V_cC_c(s) - V_M[C_M(s) + C_F(s)] - V_OC_O(s) + G_nat(s) - V_cC_rem(s),\ W_L\text{^}X = J_i - J_0,$$

with J_0 the comparator and a positive $W_L\text{^}X$ an improvement in health. The terms:

- p is the probability that an AI-originated candidate reaches patients, and y is the flow conditional on success. We treat p as a level lever on the strength of [g5] (A19: p fixed, or a function of the level reached). This is a level-set reading of the benefit side, taken while the misuse side carries both specifications co-equally; nothing read indexes p either way [g5], and a speed-set reading of p is left open (§4a.8).
- ω is the proportional loss of expected benefit under a developer-run laboratory, for intervention I3 only; nothing read maps withholding to health [a9].
- $a(s) = t_B(s) + \delta_T(s)$ is actual arrival, with δ_T the delay attacks impose on the benefit [d7].
- $\Phi(s) = p \cdot y \cdot (1 - \omega) \int_0\text{^}\{L\}\ 1\{t \geq a(s)\} A_\Phi(t, s) dt$ is foregone incremental benefit, the new therapy's gain not delivered because of attacks. $A_\Phi(t, s) \in [0, 1]$ is the share of available patient-time on the therapy lost to outages at date t, with overlapping outages combined before integration, so Φ never exceeds the benefit available, $p \cdot y \cdot (1 - \omega)(L - a)^{+}$. Arrival delay is already in a. When A_Φ is a constant share over an attack window ending at $T(s) \leq L$, $\Phi = p \cdot y \cdot (1 - \omega) A_\Phi [T - a]^{+}$.
- V_cC_c is baseline care loss only, with $V_c = E[B_e]$, the quantity the hospital studies estimate [d1–d4].
- V_MC_M is misuse; V_MC_F and V_OC_O are facility events, with each event in exactly one pathway (A21, A22) [c2, ACC (5)]; one severity class, with consequence V_M fixed per class (A20); Figure 5 shows what that holds fixed and what it leaves out.
- G_nat is countermeasure gains against natural outbreaks, credited only here [e6]; gains against capability-induced events reduce C_M and C_F, never y [ACC].
- V_cC_rem is remediation harm, counted once [d8].

Double counting is about lives, not causes. For each attack *e*, the health loss among patients in its outage is *D_e = B_e + Φ_e*: baseline care loss plus foregone incremental benefit, with each unit of patient-time allocated to exactly one attack, so a patient who dies during an attack is counted in *B_e* or in *Φ_e*, never both [ACC]. The additive ledger is a marginal approximation for non-overlapping expected losses [g10, DIS 9]. Figure 3 shows where each health change is counted, and the second place each is kept out of.

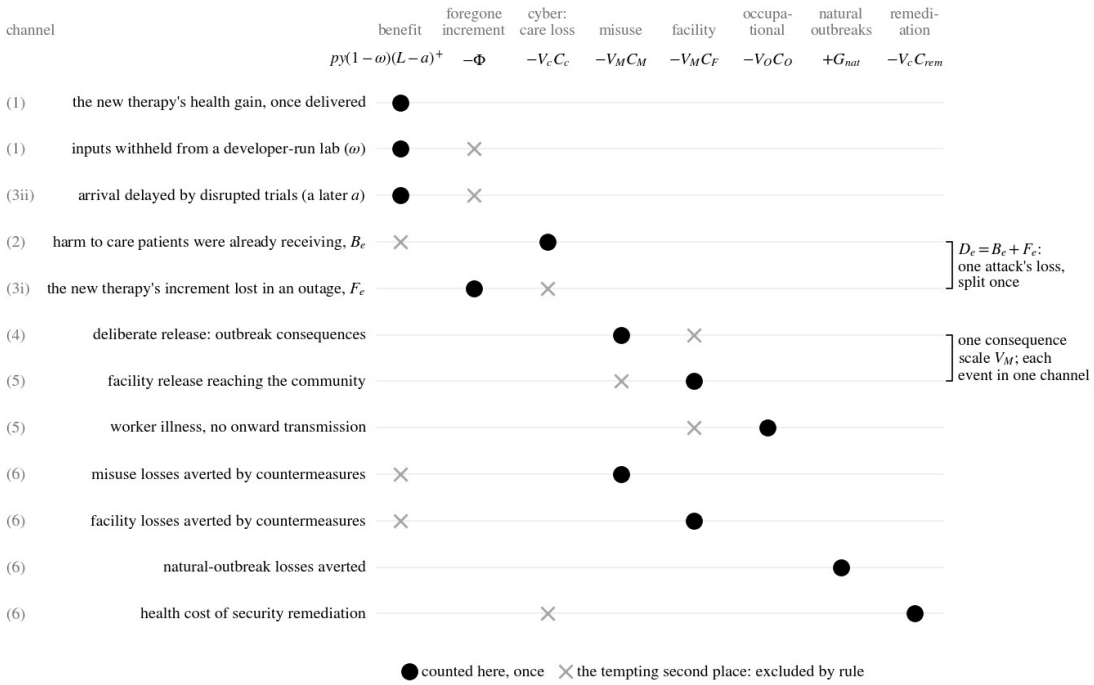


Figure 3. The accounting identity: where each lost or gained year of healthy life is counted. Rows are the ways the capability changes health. The number at left is the accounting channel [ACC]: (1) benefit; (2) harm to care already being delivered; (3) harm to the new therapy's delivery, (i) in an outage and (ii) through delayed trials; (4) deliberate misuse; (5) facility events; (6) countermeasures and remediation. Columns are the terms of the scenario balance *J_i* (§4a.1). Each row has exactly one dot, so every health change is counted in exactly one term. A cross marks the tempting second place that the rule excludes: - Harm to the care patients were already receiving (*B_e*) is counted in *V_cC_c*. It is not also subtracted from the benefit, the previous paper's assumption A18. - The new therapy's increment lost in an outage (*F_e*) is the foregone-increment term Φ, not part of *V_c*. One attack's health loss is therefore *D_e = B_e + F_e*, split once. - Deliberate and accidental outbreaks share one consequence scale, *V_M*, and each event falls in exactly one channel. - Losses averted by countermeasures reduce the misuse and facility terms; they are not added to the benefit *y*. - Natural-outbreak losses averted, which neither of those channels contains, are a separate gain.

A structural diagram; it carries no magnitudes. *Notation:* the panel and this caption write the foregone increment of attack *e* as *F_e*, the accounting ledger's symbol; the text writes it *Φ_e*, so *D_e = B_e + Φ_e*.

The decision problem. Each intervention moves named primitives, and all others are held fixed by assumption, as in RTTR's assumed intervention mappings (A.3.5). I1, speed, moves *s*. I2, release, moves the misuse hazard. I3, a developer-owned laboratory taken as reported, moves the facility terms and ω, possibly p and K_B, and, as a distance intervention, possibly K_D and K_M (§2, §4a.5, §9). I4, a public or academic equivalent, moves the facility terms, with $\omega = 0$ by assumption: nothing read measures withholding under either kind of operator [a9] (§9). I5, a countermeasure programme, moves T_D, m_D and K_D. I6, oversight and interface screening, moves the facility event rate or the misuse hazard (§10). The comparator is J_0. The population is global, with US-only as the alternative that matches the evidence [d1–d9, c4, c5]. The horizon is L. The objective is the undiscounted balance, with the benefit side also stated discounted (§4a.7) [DIS 9, g10].

The defensive schedule. A recognition date $t_R(s)$, a coverage schedule $u(t, s)$ rising from zero, a completion date $t_D(s)$ and a residual hazard after completion, h_M^{post}. We write the completion date on RTTR's mixed timing,

$$t_D(s) = \max\{K_D/s, (1 - m_D)T_D\} + m_D T_D,$$

with K_D the capability distance the countermeasure waits on, T_D its own deployment time, m_D the share of deployment that waits for the capability [e5], $\tau_D = K_D/s$, and $\chi_D = 1$ where the capability term binds (the capability branch), 0 where $(1 - m_D)T_D$ binds (the adoption branch). "Paced by K_M" is the special case $K_D = K_M$. At t_D the hazard steps to the residual h_M^{post}; **complete closure** is the case $h_M^{post} = 0$. The step is an assumption (A23): one framework argues that some biological threats still lack any countermeasure [e4], the one control criterion read applies to one intervention in one model [e7], and nothing read shows either that defence closes the risk or that it cannot.

The misuse path. Under level-set the capability path is $c(t) = \min(st, K_M)$, the hazard is $h_M(c(t))$, $\tau_M = K_M/s$, and $*\bar{h}_M = K_M^{-1}\int_0^{K_M} h_M(c)dc$, *with* $\bar{h}_M(0, x)*$ the average over $[0, x]$. Under speed-set we follow RTTR's loss index: a hazard $\Lambda_M(s)$ constant in calendar time over an exposure length $\ell_M(s)$, so $C_M = \Lambda_M \ell_M$ and

$$sC_M' = \Lambda_M \ell_M(\varepsilon_\Lambda + \varepsilon_{\ell_M}), \ \varepsilon_\Lambda = s\Lambda_M'/\Lambda_M, \ \varepsilon_{\ell_M} = s\ell_M'/\ell_M,$$

with RTTR's A4 applied to Λ_M ($\Lambda_M' > 0$) unless stated otherwise.

The marginal form. Away from kinks, with $\omega = 0$, p, y and the severity class fixed, J_0 independent of s, and P16's conditions for one calendar ledger ($L \geq \sup_s T(s)$, every counted step loss inside L, and A2's fixed step),

$$sW_L^X' = p{\cdot}y{\cdot}\eta_L^a{\cdot}(q\chi_B - s\delta_T') - s\Phi' - V_{cH}(\varepsilon_{HT} - \chi_c\tau) - V_M{\cdot}sC_M' - s(V_{MC_F}' + V_{OC_O}') + sG_{nat}' - V_c{\cdot}sC_{rem}',$$

where $\eta_L^a = 1$ if $a(s) < L$. If T can exceed L, the cyber term must be truncated at the horizon and differentiated afresh (Appendix A4a.0). Everything in this section is about the misuse term sC_M' and the benefit term. We give sC_M' separately under each specification.

Notation. Symbols follow RTTR's Appendix A.2.4. Every collision with RTTR is settled once, for the whole paper, in the note.[1] All results are local derivatives away from kinks, undiscounted unless stated, and with *V_M* fixed (A20) — an assumption, not a scope note: the loss per misuse event is held independent of the capability level reached, so that under level-set the capability indexes the frequency of events and not their severity. Figure 4 draws the model's clocks at two speeds, showing which segments speed compresses and which it does not; the results below turn on which segment binds.

[1] Collision decisions, made once for the whole paper; every section uses these names. (1) The speed-set misuse hazard is *Λ_M(s)*, with elasticity *ε_Λ*; the model checks write *H_M* and *ε_{H_M}*. *H_M* and the level-set *h_M* differed only in case, and *H* is RTTR's cyber hazard. *Λ_M* is a rate per unit of calendar time, not a cumulative hazard; the cumulative is *C_M*. It is distinct from RTTR's window hazard *λ* and from the facility event rate *λ_F*. The exposure-length elasticity is *ε_{ℓ_M}*, so that a bare subscript *ℓ* always means level-set. (2) The residual hazard after the step at *t_D* is *h_M^post*; the checks' *P* is dropped, because it collides with the success probability *p* and with RTTR's labels P1–P16. (3) The fixed benefit duration is *ℓ_B*; the checks write *D*, which is RTTR's positional stake and which, as a subscript here, marks the defence clock (*K_D, T_D, m_D, α_D, τ_D, χ_D*). (4) The two misuse derivatives compared in §4b are *Ξ_ℓ* and *Ξ_s* (level-set, speed-set), with *Ξ_s^ramp* and *Ξ_s^arr* where the reading matters; the checks' *D_ℓ*, *D_s* collide with RTTR's step prize *D_s* and with the defence subscript. *Ξ* does not appear in RTTR; RTTR's lowercase *ξ*, a change in the loss index, is unrelated. (5) The thresholds on *α_D* are *ᾱ_ℓ*, *ᾱ_s* and *ᾱ_s^arr*, and the elasticity threshold of the separable mixed hazard is **ε̄_g; the checks write* θ_ℓ, θ_s* and *T_g*, and *θ* is RTTR's catastrophe term. (6) The evidence set of §4b.2 is *Ω_E*, and the change in the balance between two speeds is written *Δ*, a function on *Ω_E*. (7) The non-misuse remainder of §4b is *Υ*; the checks' *R* collides with RTTR's *R_c* and *R_A*. (8) The plateau hazard in the facility comparison is *h_M^plat*, and convention A's profile is $\hat{h}_M$; the check writes *Γ*, which is RTTR's defensive release term *Γ_def*. Facility changes are *ΔK_D*, *ΔK_M*; the check's *δ_D*, *δ_M* would sit beside the attack delay *δ_T*. (9) The two benefit margins are written as partial derivatives of *B_r*. (10) RTTR's derivative-weighted average keeps its (C3) name, *h̄_{M,r,D}*; its* D* means "derivative-weighted", never the defence clock. (11) *ω* is reserved for withholding; RTTR's subscripted *ω_C*, *ω_A* (first-event times) do not appear here. Statistical p-values and t-statistics are set in roman and named in words, never as italic *p* or *t*. (12) Further distinctions: the countermeasure date is *t_D(s)*, not RTTR's durable share *d* or *d_f*; RTTR writes the benefit adoption time with a subscript distinct from that of its *K_B*, *t_B* and *m_B*; we write it *T_b*; *Φ*'s outage share is *A_Φ(t, s)*, not RTTR's *A* or *A_r*; *q* is always *K_B/s*; *K_D* and *χ_D* are new; *p* is free in RTTR; *α_D* is not *m_D*; *σ_M* is not RTTR's *σ_s*; *ℓ_M* is not RTTR's exploit lag *ℓ_x*; *τ_D* is not RTTR's deployment time *τ_j*; *E* is not used as a symbol, and the subscript in *Ω_E* is a label. The checks' *h_K*, *h_post* and *U* are written *h_M(K_M)*, *h_M^post* and *t_D*.

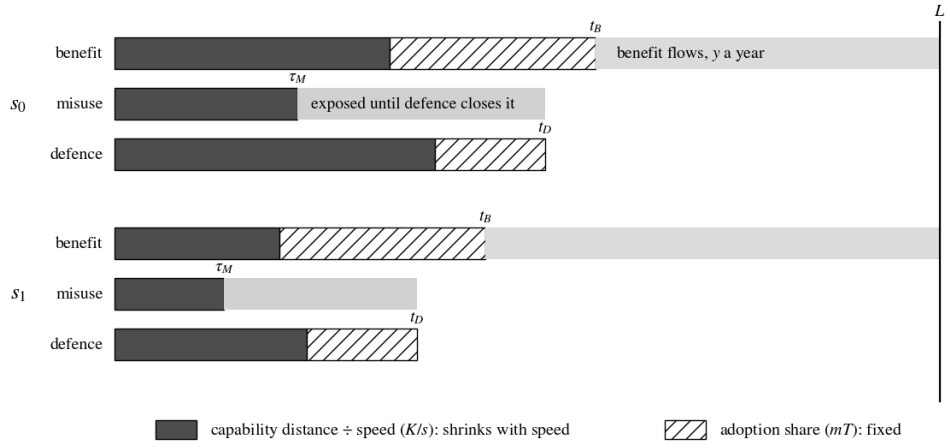


Figure 4. The model's clocks, at two speeds. Each outcome waits first on a capability distance, which speed converts into calendar time (dark: K/s), and then on an adoption share that speed does not compress (hatched: $m \cdot T$). The benefit arrives at t_B and then flows at y a year to the horizon L. The misuse capability is reached at $\tau_M = K_M/s$. It exposes the population until the defence, deployed at t_D, closes it (shaded). Faster progress (s_1) shortens every dark segment in proportion and leaves the hatched ones unchanged. That is why speed's effect on each term of the balance depends on which segment binds. Each clock is drawn on its capability branch. The cyber transition and attack-induced delay, in $a(s) = t_B(s) + \delta_T(s)$, are omitted. Definitions, not results. Schematic; axes are not to scale.

Proofs and checks. Every result in this section is proved in the Appendix from the set-up stated there (A0), with the conditions each proof needs. Where a result has also been checked numerically, against finite differences of a direct numerical integration of the loss [NUMCHECK] or of a separately written exact construction of it, the Appendix gives the worst error beside the proof; a numerical check is evidence of a different kind from a proof and never replaces it. What the Appendix does not prove is listed in its closing section, A-open, which says in each case what is open. **[J]** marks our own judgment, not a derived result. How each result was first derived, by two models working separately on each question, is recorded in the Supplement; no result rests on that record.

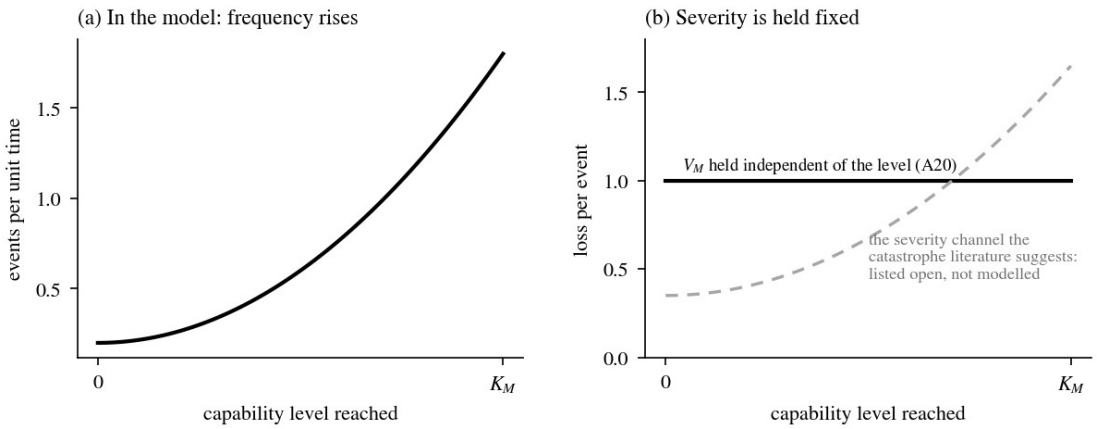


Figure 5. What A20 assumes: capability indexes how often, not how badly. (a) In the model the event rate rises with the capability level reached. (b) The loss per event *V_M* is held independent of that level (A20), so under level-set the capability indexes frequency alone. The dashed curve is the severity channel the catastrophe literature the ledger draws on is about [g9, g10]; it is listed open and is not modelled here. Schematic; the shapes are illustrative and no axis carries a measured quantity.

4a.2 Two specifications, co-equal

Figure 1 shows the two specifications: the misuse hazard set by the capability level reached (**level-set**, *h_M(c)*) or by speed (**speed-set**, *Λ_M(s)*, the form of RTTR's main results). We carry the two co-equal, with neither as base, because no source in the documented search settles the choice [b12] and the search located no estimate of misuse risk per unit of capability under either [b1]; the state-risk reading behind level-set is in two theoretical works RTTR cites, with no estimation, and RTTR carries it as a scoped alternative (§3, §5.5) [b12].

Speed-set needs a second choice. RTTR's set-up does not say when misuse exposure starts. Figure 6 shows the two readings: under **ramp-on**, exposure starts at time zero; under **arrival-on**, it starts when the capability is reached, at τ_M. The paper carries both readings and states each speed-set result under each.

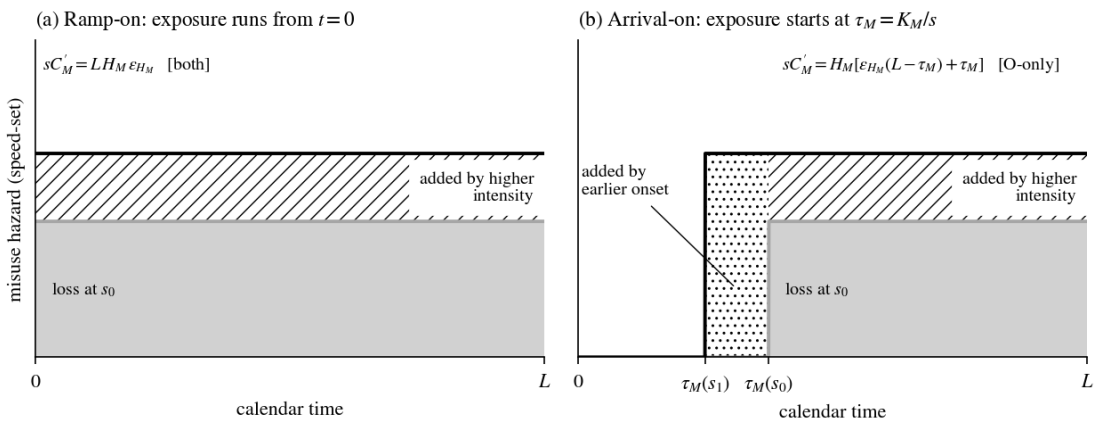


Figure 6. When the speed-set hazard starts running. (a) Under ramp-on exposure the hazard runs from date 0, so acceleration raises misuse loss only through intensity: *sC_M′ = LH_Mε_{H_M}*. (b) Under arrival-on exposure the hazard runs only once the capability is reached, at *τ_M = K_M/s*. Acceleration then also adds a block of earlier exposure: *sC_M′ = H_M[ε_{H_M}(L − τ_M) + τ_M]*. Grey is the misuse loss at s_0. Hatched is the loss added by the higher hazard at s_1. Dotted is the loss added by earlier onset. The two readings give different results throughout the model section, which is why both are carried. Schematic; axes are not to scale; areas computed by integrating the model. *Notation:* the panels and this caption use the model checks' *H_M* and *ε_{H_M}*; the text writes the speed-set hazard and its elasticity *Λ_M* and *ε_Λ* (§4a.1, note on notation).

Two axes, then, kept apart throughout. The **specification** is level-set or speed-set. The **exposure reading** is ramp-on or arrival-on, and it exists under both specifications, though it arises differently. Under speed-set it is a separate choice, because *Λ_M(s)* does not say when exposure begins. Under level-set it is a choice of **level profile**: the arrival-on reading is the profile that is zero below *K_M* and *h_M(K_M)* at it, so the hazard is gated by the capability's arrival rather than running up a ramp. Writing it that way matters for §4b.1, because a comparison **between** the specifications must hold the reading fixed; the gated level profile has ramp average *$\bar{h}$_M* = 0, and substituting it into the level-set formulas is what makes the arrival-on comparison like-for-like. The mechanism results of §4a.3 and §4a.4 are stated on the ungated level profile and on both speed-set readings, and the facility comparison of §4a.5 carries both readings under each specification. That comparison adds a third choice under level-set, the **convention** for what moving *K_M* means (Figure 26).

4a.3 R2: the plateau

Take the case in which the misuse capability is reached inside the horizon (*τ_M < L*) and there is no closure.

R2, level-set. *For a continuous, nondecreasing, nonconstant hazard,*

sC_M′ = τ_M[h_M(K_M) − $\bar{h}$_M] > 0.

Without continuity the derivative is ≥ 0, strictly positive iff h_M(K_M) > h̄_M. Figure 7 shows why: acceleration shortens the ramp and lengthens the plateau by the same calendar time. A level-set hazard does not make misuse risk speed-invariant. The derivative is zero exactly when *h_M(K_M) = h̄M; for a continuous nondecreasing hazard that requires a hazard constant on [0, K_M], but a non-monotone profile can meet it too. Discounted, sC{M,r}′ = τ_M[h_M(K_M)e^{−rτ_M} − h̄_{M,r,D}]*, which is nonnegative for a nondecreasing hazard; *h̄_{M,r,D}* is RTTR's derivative-weighted average of (C3).

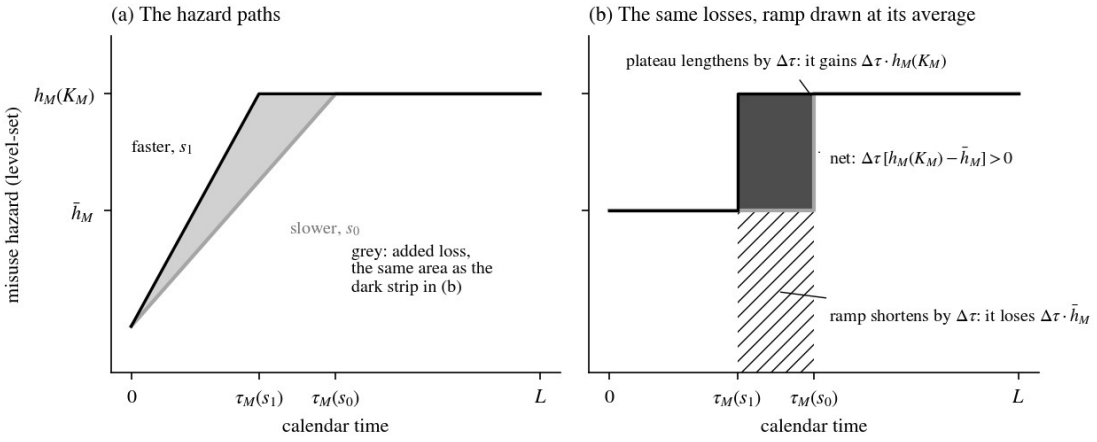


Figure 7. The plateau (R2): speed shortens the ramp and lengthens the plateau by the same calendar time. This is the level-set case with the capability reached inside the horizon and no closure. (a) Faster progress (black) climbs the same ramp sooner. (b) The same two losses, with the ramp drawn at its average hazard *h̄_M. *Speed shortens the ramp by Δτ* = τ_M(s_0) − τ_M(s_1)* and lengthens the plateau by exactly that much calendar time. The ramp loses Δτ·*h̄_M* (hatched), and the plateau gains Δτ·*h_M(K_M)* (hatched plus dark). The net is the dark strip, Δτ[*h_M(K_M)* − *h̄_M*], which is positive for any continuous, nondecreasing, nonconstant hazard. The grey area in (a) equals the dark strip in (b); in the limit this is *sC_M′ = τ_M[h_M(K_M) − h̄_M]*. On the speed-set ramp-on reading the plateau result holds more broadly, though not with a larger derivative, because it rests on a different hazard assumption, A4. Ramp-on gives *sC_M′ = LH_Mε_{H_M}*, positive whenever the hazard rises with speed, with no condition on the level profile. Schematic; axes are not to scale. *Notation: H_M* and *ε_{H_M}* are the model checks' symbols for the text's *Λ_M* and *ε_Λ* (§4a.1, note on notation).

R2, speed-set. *Under ramp-on, sC_M′ = LΛ_Mε_Λ, positive under A4 and zero at ε_Λ = 0.* Under arrival-on, *sC_M′ = Λ_M[ε_Λ(L − τ_M) + τ_M]*: positive, with a second term because earlier onset lengthens exposure, and the sign survives discounting.

Under A4 the ramp-on sign needs no monotonicity of any level profile, flat profiles included, so on the ramp-on reading the result is stronger in scope. It rests on a different hazard assumption, though, and neither assumption implies a larger derivative.

Under both specifications the plateau is RTTR's own algebra. Under level-set it is RTTR's (C2) with $\chi_c = 0$, the level-set entry of A.7, with K_M for K and the horizon in place of the adoption tail. Under speed-set, ramp-on, it is the elasticity sign of P1's adoption branch, with L in place of T_a. We claim nothing new for R2 itself. What this paper adds lies elsewhere: truncation, under level-set only (§4a.4); the moving defence boundary (§4a.5); the disagreement between the specifications (§4b.1); and the coupling of misuse with the benefit and attack channels.

R2 with the benefit (coupled, on the benefit's capability branch, $\chi_B = 1$, $t_B + \delta_T < L$, $\tau_M < L$, with delivery interactions and every other marginal term set to zero). Under level-set,[2] acceleration raises the balance iff $pyK_B > V_MK_M[h_M(K_M) - \bar{h}_M]$, a test that does not depend on s on that branch. Under speed-set the test is $pyq > V_M{\cdot}L{\cdot}\Lambda_M\varepsilon_\Lambda$ on the ramp-on reading and $pyq > V_M\Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$ on the arrival-on reading. Both right sides grow with L, and with ε_Λ holding the other quantities fixed, so on either speed-set reading the sign can flip with the horizon. Compared on one reading rather than across two, the arrival-on pair is the one Table 4b.A row 7 settles: with the level side gated the level-set test is $pyK_B > V_MK_M{\cdot}h_M(K_M)$, which under the comparison convention of §4b.1 is the speed-set arrival-on test at $\varepsilon_\Lambda = 0$. Growth with s is not claimed. Off the capability branch each test keeps χ_B: $pyK_B\chi_B$ in place of pyK_B under level-set, and $pyq\chi_B$ in place of pyq under speed-set.

4a.4 R1: truncation

Now take the case in which the capability lies beyond the horizon ($sL < K_M$).

R1, level-set. *For a nondecreasing hazard,*

$$sC_M' = L[h_M(sL) - \bar{h}_M(0, sL)] \geq 0,$$

signed by monotonicity alone. If h_M is continuous at K_M, the derivative meets R2's at s = K_M/L with no jump. R1 **has no speed-set counterpart**, so the paper's most novel result rests on one of its two co-equal specifications — and on the horizon as much as on the specification. The regime R1 is defined on, $sL < K_M$, is a joint condition on the two: at a fixed horizon it is the set of speeds below K_M/L, and as L grows that set shrinks to nothing, because with no finite date at which counting stops every positive speed reaches the capability eventually. The result therefore needs both a level-indexed hazard and an objective that stops counting. Whether a discounted objective, which has no sharp beyond-horizon region, yields an analogue is not derived here (§4a.8). It is also distinct from RTTR's level-set entry below its own kink, where the derivative is $-K\bar{h}/s^2$, negative (RTTR §4.3; Appendix A.7). The two are not in conflict: RTTR integrates the hazard over a transition that completes at K, so acceleration shortens the ramp and lowers cumulative hazard, whereas here the horizon binds and the capability is never reached, so acceleration pulls previously unreached levels into the window. The sign difference is the horizon, and that is the content of the result. Figure 8 shows the mechanism: acceleration pulls levels of the ramp that lay beyond the horizon inside it. Whether this case is the operative one is not

[2] It follows by substituting R2's derivative into the marginal form of §4a.1, with every other marginal set to zero (Appendix A4a.4(vi)).

settled by how distant the misuse capability is: *sL < K_M* is a joint condition on that distance and on the horizon chosen, so a longer horizon takes the regime away and no horizon at all leaves it empty (§4a.8). Without monotonicity the formula still holds and can be negative (§4b.1, row 2).

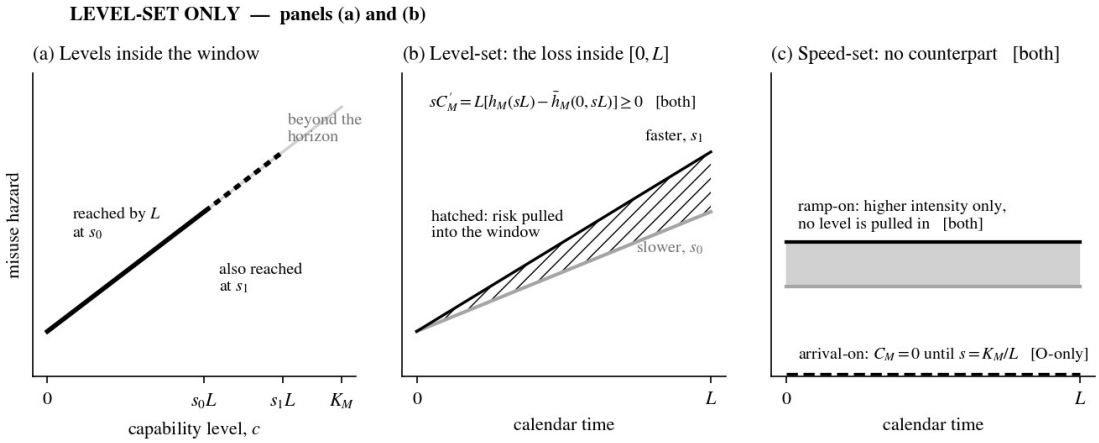


Figure 8. Truncation (R1): the capability lies beyond the horizon, and acceleration pulls level-indexed risk into the window. This holds under the level-set specification only. (a) Within the horizon L, slower progress reaches capability levels up to s_0L. Faster progress reaches levels up to s_1L, so levels that lay beyond the window are brought inside it. (b) In calendar time the faster path lies above the slower one at every date. The hatched area is the misuse loss that acceleration adds: *sC_M′ = L[h_M(sL) − h̄_M(0, sL)]* ≥ 0 for a nondecreasing hazard. (c) Under speed-set there is no counterpart, because the hazard is not indexed by the level reached, so acceleration has no level-indexed risk to pull in. With ramp-on exposure, acceleration raises loss through intensity alone (grey). With arrival-on exposure, loss is zero throughout the unreached region, and the effect appears only at the activation speed *s = K_M/L*. The paper's most novel result therefore exists under only one of its two co-equal specifications, and only where a finite horizon leaves the regime *sL < K_M* non-empty. Schematic; axes are not to scale.

Under speed-set the mechanism is absent, because it needs a hazard indexed by level. Under ramp-on, *sC_M′ = LΛ_Mε_Λ > 0*: positive for a different reason, intensity, with *τ_M* absent and no kink, and zero at *ε_Λ* = 0 even where the level-set derivative is positive. Under arrival-on, *C_M* = 0 throughout the unreached region, so *sC_M′* = 0 there; the effect appears only at the activation kink *s = K_M/L*, where the right-hand value of *sC_M′* is *Λ_ML*, the analogue of P16's horizon-activation kink.

Truncation is therefore a level-set result, and Figure 9 draws the regime it needs. Whether misuse risk scales with the level reached or with speed is unmeasured [b1, b12], and §11 lists it among the estimands.

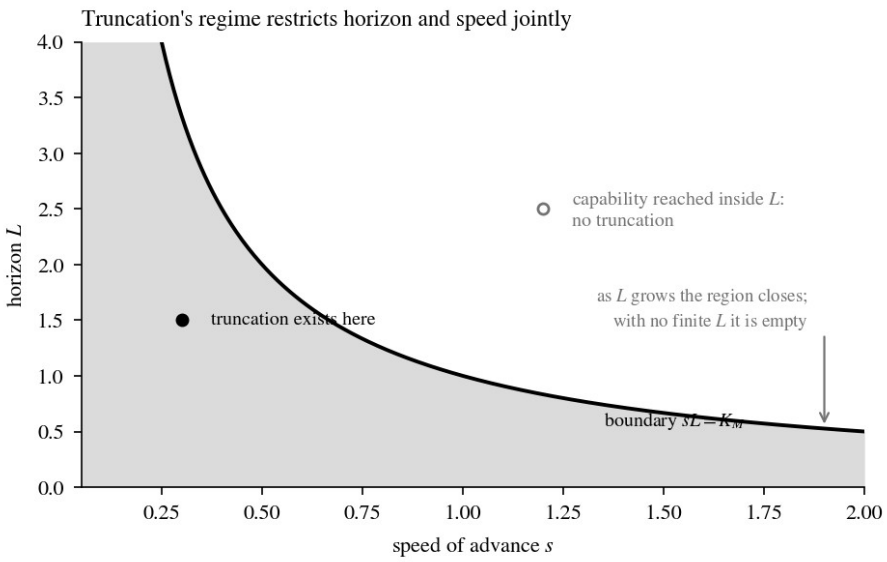


Figure 9. Truncation's regime restricts the horizon and the speed jointly. The shaded region is $sL < K_M$, where the misuse capability is not reached inside the counted window and R1 applies. The filled point lies inside it, the open point outside. The region closes as the horizon lengthens, and without a finite L it is empty: every positive speed reaches the capability eventually (§4a.4, A4a.4). Illustrative values: $K_M = 1$. The axes are the model's own; neither carries a measured quantity.

4a.5 R3: the defence clock

Take a step at t_D to the residual $h_M\text{^}post$, with $\tau_M \le t_D < L$; complete closure is $h_M\text{^}post = 0$. Every test below holds the residual fixed unless it says otherwise. Figure 10 shows the two clocks at two speeds, the hazard paths they produce, and the level-set condition as a region.

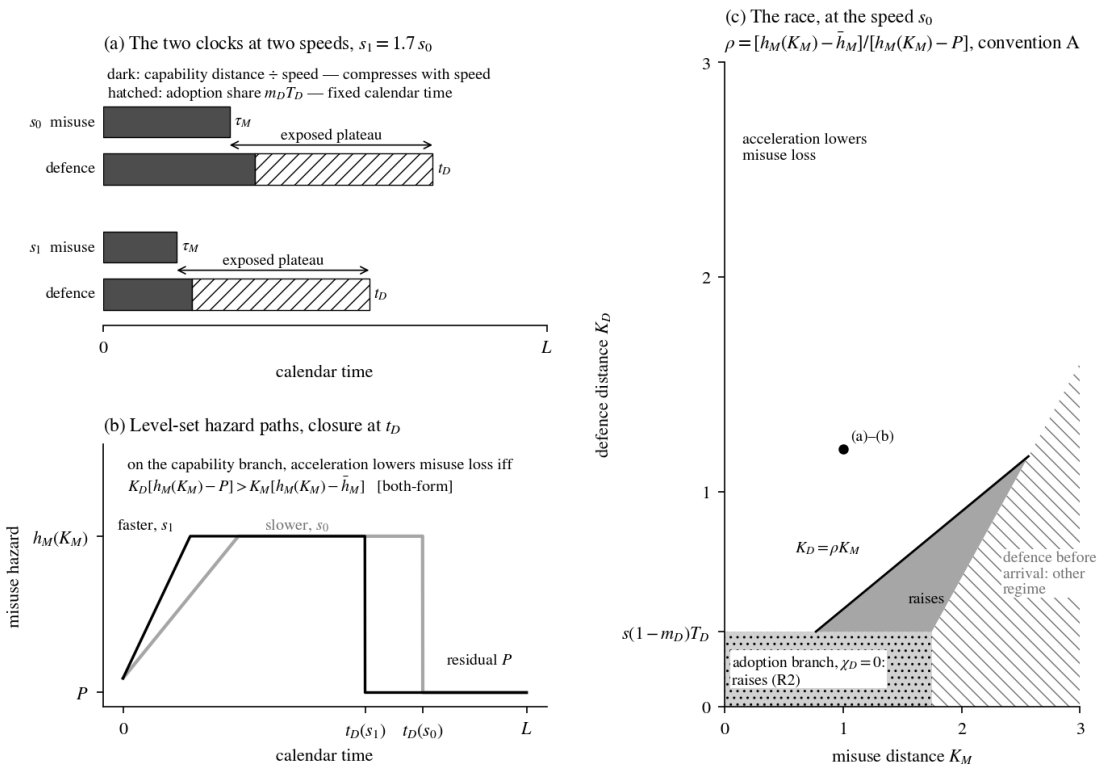


Figure 10. The defence clock (R3): a race between two distances. (a) Each clock, at two speeds. The misuse capability is reached at $\tau_M = K_M/s$, and the defence is deployed at $t_D = K_D/s + m_DT_D$. Speed compresses the capability-set intervals (dark) but not the adoption share m_DT_D (hatched), so the exposed plateau between τ_M and t_D can shrink or grow. (b) The resulting level-set hazard paths, with closure to a residual P at t_D. With the residual fixed and the defence waiting on its own distance, on its capability branch ($\chi_D = 1$), acceleration lowers misuse loss iff $K_D[h_M(K_M) - P] > K_M[h_M(K_M) - \bar{h}_M]$. (c) That condition drawn as a region at one speed. For $h_M(K_M) > P$, acceleration lowers misuse loss above the line $K_D = K_M[h_M(K_M) - \bar{h}_M]/[h_M(K_M) - P]$, on the capability branch in the stated interior regime; equality is neutrality. Below it, acceleration raises misuse loss, and it does so as well, under R2's hazard restrictions, on the adoption branch (dotted band), where the defence waits on adoption rather than capability ($\chi_D = 0$; the plateau case, R2). The lower-right corner, where the defence arrives before the capability, is a different regime and is not signed here. The panel is drawn under convention A (defined in §4a.5 below), where the hazard gaps do not vary with K_M. The two-distance form is level-set only. Under speed-set ramp-on, K_M drops out, and the test becomes $\varepsilon_\{H_M\} < \alpha_D(1 - P/H_M)$: the hazard elasticity against the capability-set share of the deployment wait. In the arrival-on form the gap $K_D - K_M$ enters. R3 signs a derivative with respect to speed, so in general it cannot sign the facility comparison of Figure 11; the one mechanical exception is at §4a.5. Schematic; axes are not to scale. *Notation:* the panels and this caption use the model checks'

symbols: P is the text's residual $h_M\text{^}post$, and H_M and $\varepsilon_\{H_M\}$ are its Λ_M and ε_Λ (§4a.1, note on notation).

R3, level-set:

$sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] + s{\cdot}t_D'{\cdot}[h_M(K_M) - h_M\text{^}post]$.

If the residual responds to speed, add $(L - t_D)s{\cdot}h_M\text{^}post'$. The branches:

- *Adoption branch* ($\chi_D = 0$, so $t_D' = 0$). This is R2: speed hurts under R2's hazard restrictions.
- *Paced by K_M*. $sC_M' = \tau_M[(1 - \chi_D)h_M(K_M) + \chi_Dh_M\text{^}post - \bar{h}_M]$, negative on the capability branch ($\chi_D = 1$) at $h_M\text{^}post = 0$ when *$\bar{h}$_M > 0*.
- *Own distance, capability branch* ($\chi_D = 1$).[3] With $s{\cdot}t_D' = -K_D/s$, acceleration lowers misuse loss **iff**

  $K_D[h_M(K_M) - h_M\text{^}post] > K_M[h_M(K_M) - \bar{h}_M]$.

  With a responsive residual, add $(L - t_D)s^2h_M\text{^}post'$ to the right side.

Under level-set, then, the sign on the capability branch turns on a race of two capability distances, drawn as a region in Figure 10(c). Gradual coverage adds terms of underived sign. The two-distance form is level-set only.

R3, speed-set. The result survives and changes character. Write $\alpha_D = -s{\cdot}t_D'/t_D$, the deployment-date elasticity. With a responsive residual every test below receives $(L - t_D)s{\cdot}h_M\text{^}post'$ (Appendix A4a.4(v)).

- *Ramp-on*. $sC_M' = \Lambda_Mt_D[\varepsilon_\Lambda - \alpha_D(1 - h_M\text{^}post/\Lambda_M)]$. Acceleration lowers misuse loss **iff** $\varepsilon_\Lambda < \alpha_D(1 - h_M\text{^}post/\Lambda_M)$; equality is neutrality. On the own-distance clock $\alpha_D = \chi_DK_D/(s{\cdot}t_D)$,[4] and on its capability branch at any fixed residual the condition is $\varepsilon_\Lambda < \tau_D(1 - h_M\text{^}post/\Lambda_M)/(\tau_D + m_DT_D)$, which at $h_M\text{^}post = 0$ is $\varepsilon_\Lambda < \tau_D/(\tau_D + m_DT_D)$. ***K_M* drops out**.
- *Paced by K_M, ramp-on, capability branch*.[5] Acceleration helps iff $\varepsilon_\Lambda < (\tau_M/t_D)(1 - h_M\text{^}post/\Lambda_M)$.

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[3] It follows by substituting $s{\cdot}t_D' = -K_D/s$, the capability-branch value, into the level-set R3 derivative and multiplying by s (Appendix A4a.3); the threshold $\bar{\alpha}_\ell$ of §4b.1 gives the same condition.

[4] It follows by substituting the own-distance clock's speed derivative, $s{\cdot}t_D' = -\chi_DK_D/s$ (Appendix A0), into the definition $\alpha_D = -s{\cdot}t_D'/t_D$. It is a clock identity: it holds under either specification and either exposure reading, needs no convention, and has no hazard content. It requires the own-distance clock with K_D, m_D and T_D independent of s, and it does not hold at the clock kink $K_D/s = (1 - m_D)T_D$, where χ_D is undefined and only one-sided derivatives exist.

- *Arrival-on, own distance, capability branch*. With positive exposure, $t_D - \tau_M = \tau_D - \tau_M + m_DT_D > 0$, acceleration lowers loss iff $\varepsilon_\Lambda < [\tau_D(1 - h_M\text{^}post/\Lambda_M) - \tau_M]/(\tau_D - \tau_M + m_DT_D)$; at $h_M\text{^}post = 0$ this needs $K_D > K_M$. At zero exposure the divided threshold is undefined. For the own-distance clock this boundary is generally a kink: the positive-exposure-side limit of sC_M' is $\Lambda_M\tau_M - \tau_D(\Lambda_M - h_M\text{^}post)$, while the defence-before-arrival-side expression, on the convention that exposure then starts at τ_M at the residual hazard (Appendix A0), is $h_M\text{^}post \cdot \tau_M$. Use one-sided derivatives. When $K_D = K_M$ and $m_DT_D = 0$, the speed derivative is $h_M\text{^}post \cdot \tau_M$.
- *Paced by K_M, arrival-on*. On the capability branch $sC_M' = \varepsilon_\Lambda\Lambda_Mm_DT_D + h_M\text{^}post \cdot \tau_M \geq 0$, strictly positive iff $m_DT_D > 0$ or $h_M\text{^}post > 0$; on the adoption branch $sC_M' = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M]$.
- *Adoption branch* ($\alpha_D = 0$): speed hurts under A4, on either reading.
- *Parallel defence timing* ($m_D = 0$), on the capability branch with $h_M\text{^}post = 0$: the ramp-on condition is exactly P1's $\varepsilon_\Lambda < 1$.[6] With $h_M\text{^}post > 0$, or off that branch, it is not.

These conditions sign the effect of speed; they are not advice about the distances. Under level-set, on the capability branch with $h_M(K_M) > h_M\text{^}post$, a longer K_D lowers sC_M': $\partial(sC_M')/\partial K_D = -[h_M(K_M) - h_M\text{^}post]/s$. It leaves more of the defence wait for speed to compress (Figure 10(a)), so acceleration is more likely to lower misuse loss. Under speed-set, ramp-on, a longer K_D raises the elasticity threshold $\alpha_D(1 - h_M\text{^}post/\Lambda_M)$ when $m_DT_D > 0$ and $\Lambda_M > h_M\text{^}post$, but it lowers the signed marginal if and only if also $\varepsilon_\Lambda < 1 - h_M\text{^}post/\Lambda_M$.[7] Whether a shorter K_D lowers misuse loss at a given speed is a different question, taken up next. No measure of ε_Λ was located [b1, b12]; the evidence on K_D and T_D is in §7.

The facility comparison at a fixed speed

Sections 2 and 9 ask how the changes a laboratory makes to the defence distance K_D and the misuse distance K_M together move misuse loss at a fixed speed. **R3 cannot sign that**

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[5] It follows by substituting $K_D = K_M$ on the capability branch, so that $\alpha_D = \tau_M/t_D$, into the ramp-on condition (Appendix A4a.4(iv), A4b).

[6] It follows by setting $m_D = 0$ and $h_M\text{^}post = 0$ in the own-distance ramp-on condition (Appendix A4a.4(iv)).

[7] On the own-distance capability branch, substituting $t_D = \tau_D + m_DT_D$ and $s \cdot t_D' = -\tau_D$ into the speed-set ramp-on derivative gives $\Xi_s = \Lambda_M\varepsilon_\Lambda m_DT_D + (K_D/s)[\Lambda_M(\varepsilon_\Lambda - 1) + h_M\text{^}post]$, which is affine in K_D at fixed s (Appendix A4a.4(iv)). Both statements are read off that expression and the capability-branch share $\tau_D/(\tau_D + m_DT_D)$. The conditions are: speed-set, ramp-on only; $\chi_D = 1$ at both values of K_D; $t_D < L$ after the change; a fixed residual; and s, m_D, T_D fixed. The arrival-on analogue has the same K_D coefficient, $[\Lambda_M(\varepsilon_\Lambda - 1) + h_M\text{^}post]/s$, by differentiating the arrival-on R3 derivative in K_D on the own-distance capability branch (Appendix A4a.4(iv)).

comparison. R3 signs the derivative of misuse loss with respect to speed, and its inequality compares distance levels. The facility question compares distance changes with speed held fixed. In general neither sign tells you the other; the one exception, a cut of both distances in proportion under convention A or, with an added intensity term, under speed-set, is in the footnote.[8] Under speed-set the independence is immediate: Λ_M does not depend on the capability level, so a facility's effect through the misuse distance at a fixed speed is $-\Lambda_M/s$ under arrival-on and zero under ramp-on (A4a.7), neither carrying ε_Λ, while R3's speed sign carries ε_Λ freely; varying ε_Λ alone therefore moves one sign and leaves the other where it is. Under level-set each direction needs a counterexample, and both are two points on one profile. Take convention A with $\hat{h}_M(x) = x^2$ on [0, 1], so $\hat{h}_M(1) = 1$ and $*\bar{h}_M* = 1/3$; set $K_M = 1$, $h_M^{post} = 0.2$, $L = 6$, $m_{DT_D} = 1$ and $s = 0.5$, which puts $\tau_M = 2$ and $t_D = 2K_D + 1$ inside $[\tau_M, L)$ throughout. *R3's sign does not determine the facility comparison.* A facility that shortens the defence distance alone ($\Delta K_M = 0$, $\Delta K_D = -0.05$) lowers misuse loss by 0.080 at every K_D below, since it moves t_D along the plateau and the plateau hazard exceeds the residual. R3 meanwhile changes sign as K_D crosses the two-distance boundary $K_D[h_M(K_M) - h_M^{post}] = K_M[h_M(K_M) - \bar{h}_M]$, here $K_D = 0.8333$: C_M' is +0.747 at $K_D = 0.6$, exactly 0 at the boundary, and −1.813 at $K_D = 1.4$. One sign moves and the other stays. *The facility comparison does not determine R3's sign either.* At $K_D = 1$, where $C_M' = -0.533$, two facility moves of the same size go opposite ways: $\Delta K_D = -0.05$ lowers misuse loss by 0.080 while $\Delta K_M = -0.05$ raises it by 0.067, because under convention A compressing the profile raises the hazard at every level below K_M. *Numerical check:* the losses are built from the set-up and the derivative is a central difference of them, which also reproduces R3's printed zero at the boundary exactly.

Everything in this subsection holds under these conditions: fixed s; $\tau_M < t_D < L$; a step at t_D to a constant residual; one severity class; V_M, the hazard and the rest of the schedule fixed except as stated; the same s converting both distances into dates; undiscounted; away from kinks. The shared speed is a substantive restriction, not a convenience: a laboratory directed at biology may advance biology-relevant capability at a different rate from general capability, and that case is not derived here (§4a.8). Write h_M^{plat} for the hazard on the plateau, $h_M(K_M)$ under level-set and Λ_M under speed-set, and ΔK_D, ΔK_M for the changes a facility makes, negative when it shortens a distance. The algebra is in Appendix A4a.7. Figure 11 shows, case by case, which pairs of cuts lower misuse loss, and where that rule stops applying.

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[8] Under level-set convention A, R3's derivative equals the fixed-speed derivative taken along a proportional cut of both distances: $s\partial C_M/\partial s = -K_M\partial C_M/\partial K_M - K_D\partial C_M/\partial K_D$. Under speed-set the identity needs an added intensity term: $s\partial C_M/\partial s = \Lambda_M\varepsilon_\Lambda\ell_M - K_M\partial C_M/\partial K_M - K_D\partial C_M/\partial K_D$, where $\ell_M = t_D$ under ramp-on and $\ell_M = t_D - \tau_M$ under arrival-on; the added term is the change in hazard intensity. Under convention B the identity fails in general (it holds, for instance, for a constant profile). Each form needs a fixed residual and m_{DT_D} held fixed, and is proved in Appendix A4a.7(vii) from the degree-zero homogeneity of the loss (Lemma H). The numerical check tests the identity in these corrected forms, the speed-set one with the intensity term, and confirms all three statements [NUMCHECK]. It is a property of the convention, not a relation the paper uses, and it does not make speed a facility.

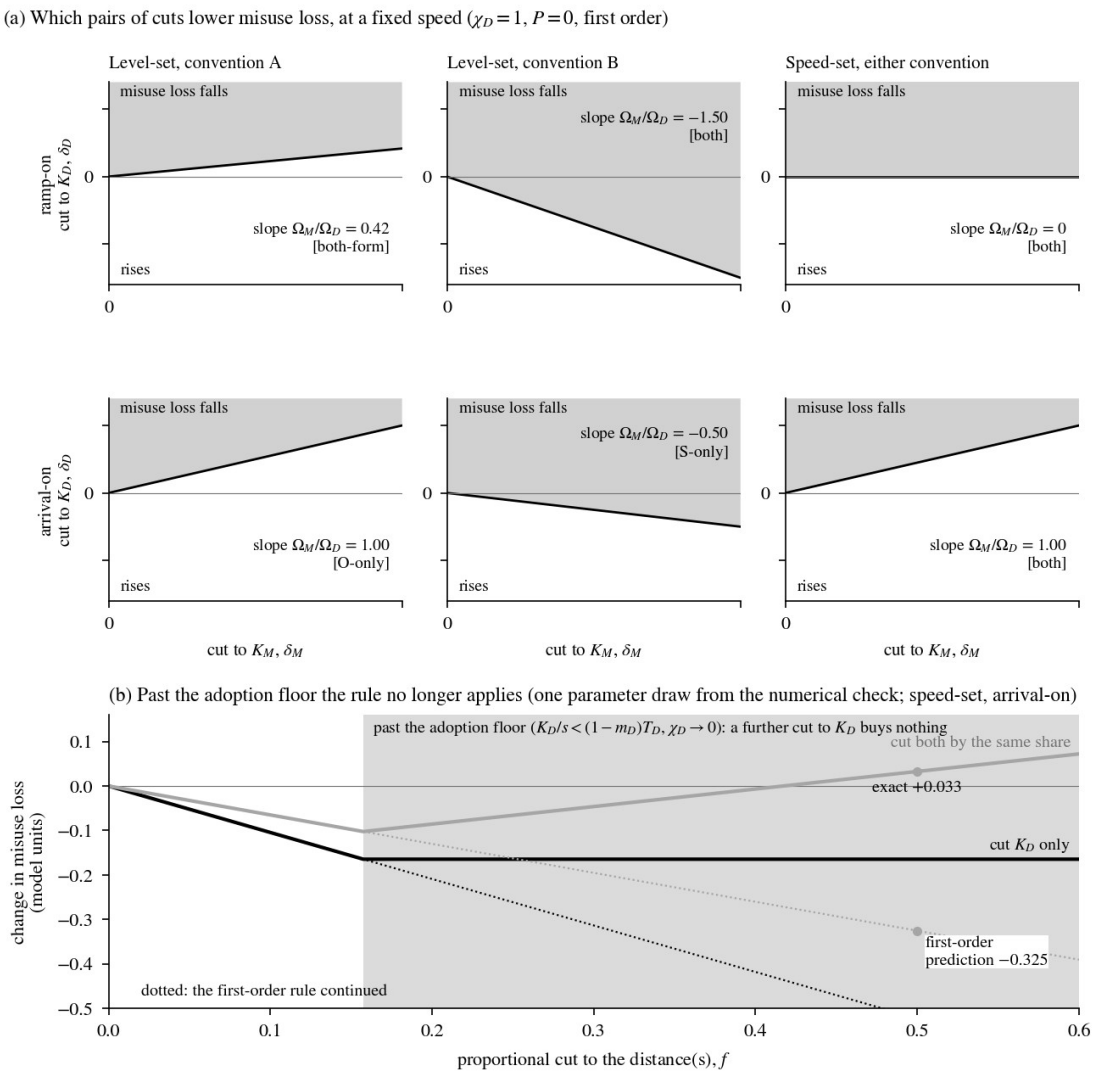


Figure 11. The facility comparison at a fixed speed, with its weights. (a) A facility may cut the misuse distance by δ_M and the defence distance by δ_D. To first order, misuse loss falls iff $\Omega_D\delta_D > \Omega_M\delta_M$. The boundary is therefore a line through the origin with slope Ω_M/Ω_D, and grey marks the pairs of cuts that lower misuse loss. The slope depends on the specification, on the exposure reading and, under level-set, on the convention: - **Speed-set ramp-on:** K_M carries no weight (slope 0). - **Speed-set arrival-on, at $P = 0$:** the rule is unweighted, $\delta_D > \delta_M$. - **Level-set, convention A:** a genuine trade-off. Ramp-on slope $(h - \bar{h})/(h - P)$; arrival-on slope $h/(h - P)$, which needs convention A. - **Level-set, convention B:** under ramp-on exposure, cutting K_M weakly lowers loss for a nondecreasing hazard, and the slope is nonpositive when defence is effective. Under arrival-on exposure the slope can have either sign; its weight is $h - s(t_D - \tau_M)h_M'(K_M)$, negative at the parameter set drawn.

The panels are drawn at $P = 0$ and $\chi_D = 1$, at one illustrative parameter set. (b) The rule holds only inside one regime. This panel is at one numerical-check draw:

speed-set, arrival-on. Cutting K_D alone lowers misuse loss linearly until K_D/s falls below the adoption term $(1 - m_D)T_D$. Past that adoption floor (shaded), the defence waits on adoption ($\chi_D \to 0$), and a further cut to K_D buys nothing (black, flat). Cutting both distances by the same share (grey) turns a predicted fall of 0.325 into an actual rise of 0.033 at a 50 per cent cut, because the K_M cut keeps raising loss after the K_D cut has stopped helping. Dotted lines continue the first-order rule past where it applies. This is the misuse component only, with everything else held fixed by assumption; the whole balance is unsigned. Model units at illustrative values; not calibrated. Panel (a) is schematic; axes are not to scale. Panel (b) is schematic in that no axis carries a measured quantity. *Notation:* the panels and this caption use the model check's symbols: δ_D and δ_M are the shortenings $-\Delta K_D$ and $-\Delta K_M$; P is the residual $h_M\text{^}post$; h is the plateau hazard $h_M(K_M)$; Ω_D and Ω_M are the weights on the two cuts, unrelated to the evidence set Ω_E of §4b.2.

The defence distance. In all four cases, two specifications by two readings, $\partial(V_MC_M)/\partial K_D = V_M(\chi_D/s)(h_M\text{^}plat - h_M\text{^}post)$. Shortening K_D lowers misuse loss iff $\chi_D = 1$ and $h_M\text{^}plat > h_M\text{^}post$. On the adoption branch it does nothing. So a cut to K_D helps only while the defence waits on the capability; past the adoption floor drawn in Figure 11(b), where its own deployment time binds, further cuts buy nothing [NUMCHECK]. That is a change of regime, not a failure of the condition. **What the agreement across cases is.** It is an agreement of form, with each case's own plateau hazard substituted, and not of magnitude: at one illustrative parameter set the derivative is 0.750 under convention A, 0.250 under B and 0.715 under speed-set, in model units — the three differ because $\hat{h}_M(1)$, $h_M(K_M)$ and Λ_M take different values there, not because the choices reach the derivative's form. What is free of the three choices is the sign condition, and it is free of them because K_D enters the loss at one point, the deployment date, while the specification sets the plateau level, the exposure reading sets where counting starts — which K_D does not move, t_D being interior to both windows — and the convention sets how the profile responds to K_M, which this derivative holds fixed. The contrast is inside this subsection: the K_M derivative below, which the conventions do govern, flips sign between them. Written as a shift in time the derivative also extends further than the step at t_D: the same form holds under gradual coverage that translates with t_D and completes before L, and can fail when coverage does not complete before L (A4a.7(i)) [NUMCHECK].

Speed-set, with the facility leaving Λ_M unchanged. Ramp-on, misuse loss falls iff $\chi_D(\Lambda_M - h_M\text{^}post)(-\Delta K_D) > 0$: K_M carries zero weight. Arrival-on, it falls iff $\chi_D(\Lambda_M - h_M\text{^}post)(-\Delta K_D) > \Lambda_M(-\Delta K_M)$. Only at $h_M\text{^}post = 0$ and $\chi_D = 1$ does this reduce to $-\Delta K_D > -\Delta K_M$, a comparison of date changes, or of capability changes under the shared-speed assumption above. It is the only unweighted comparison of the two distances the paper states.

Level-set: two conventions, co-equal. Under level-set the sign on K_M depends on what "advancing K_M" means, and no source we read fixes it. We carry two conventions co-equal, as we carry the two specifications, and give every level-set facility statement under both. Figure 26 (§9) draws what each does to the hazard path.

- **Convention A, shape-preserving.** The whole profile is compressed with the threshold, $h_M(c; K_M) = \hat{h}_M(c/K_M)$, so misuse reaches the same plateau hazard at a lower capability level.
- **Convention B, fixed profile**, the model's literal definition. The hazard at each capability level is unchanged, and the plateau hazard is h_M at the new, lower K_M.

Ramp-on, the misuse-distance derivative is:

- under A, $\partial C_M/\partial K_M = -[h_M(K_M) - \bar{h}_M]/s$.[9] Shortening K_M raises misuse loss when *h_M(K_M) > h̄_M*, and the two distances trade off: misuse loss falls iff* χ_Dh_M(K_M) − h_M^post > h_M(K_M) − h̄_M*;
- under B, $\partial C_M/\partial K_M = (t_D - \tau_M)h_M'(K_M)$. A nondecreasing hazard therefore makes a cut to K_M weakly beneficial under ramp-on exposure. Both cuts are weakly beneficial when defence also satisfies $h_M^{post} \le h_M(K_M)$; if the residual exceeds the plateau hazard, shortening a capability-bound K_D raises loss.

The sign on K_M flips between the conventions. Arrival-on, $\partial C_M/\partial K_M$ is $-h_M(K_M)/s$ under A, and $(t_D - \tau_M)h_M'(K_M) - h_M(K_M)/s$ under B, which is unsigned from monotonicity alone; Appendix A4a.7(iii) gives the condition that decides its sign.

Finite moves. Under A, and under speed-set, misuse loss is affine in the two distances within one regime, so these conditions hold for real moves and not only marginal ones, and the numerical check confirms it under A and under speed-set for moves of up to half of each distance [NUMCHECK]. Under B misuse loss is not affine, and a finite change has to be integrated.

What the check covers. The numerical check's derivative tests matched the facility distance-derivative formulas, wherever their conditions hold, with a maximum normalised error below 2.1×10^{-8}, against the check's acceptance threshold of 10^{-6}, under both conventions and both readings [NUMCHECK]. Finite-move comparisons use a separate error measure. That facility check did not test gradual coverage, discounting, more than one severity class or the whole balance $J_I3 - J_0$, and nothing here rests on it for those. Coverage that moves rigidly with the deployment date is tested by the separate translation check reported in A4a.7(i), over 1,440 case-draws. The condition signs the misuse component only. The whole difference $J_I3 - J_0$ also carries the changes in benefit, facility events, occupational loss and remediation, and it stays unsigned. Which distances a facility moves, and by how much, no source read records [a2, a3]. Figures 12 to 15 draw this subsection's four pictures: what speed can compress in the deployment wait, the K_D derivative and its floor, the counterexample behind the non-determination just stated, and the speed-set condition as a race between two ratios.

[9] It follows, with the weighted condition, by substituting $h_M(c; K_M) = \hat{h}_M(c/K_M)$ into the ramp-on loss, whose ramp term *τ_Mh̄_M* then has *h̄_M* fixed in K_M (Appendix Lemma 1(c), A4a.7(ii) and (iv)).

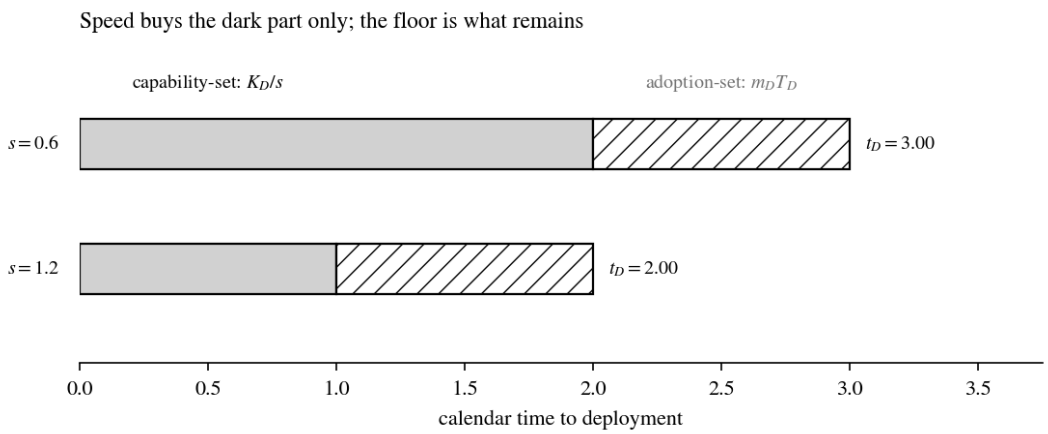


Figure 12. The deployment wait, split, and what speed can compress. The countermeasure is deployed at *t_D = K_D/s + m_DT_D* on the capability branch. Speed compresses the capability-set part (solid) in proportion to itself and leaves the adoption-set part (hatched) exactly where it is. The division is this paper's, and the floor that follows from it is why cuts to *K_D* stop buying anything once the clock is adoption-bound. Illustrative values: *K_D* = 1.2, *m_DT_D* = 1, speeds 0.6 and 1.2.

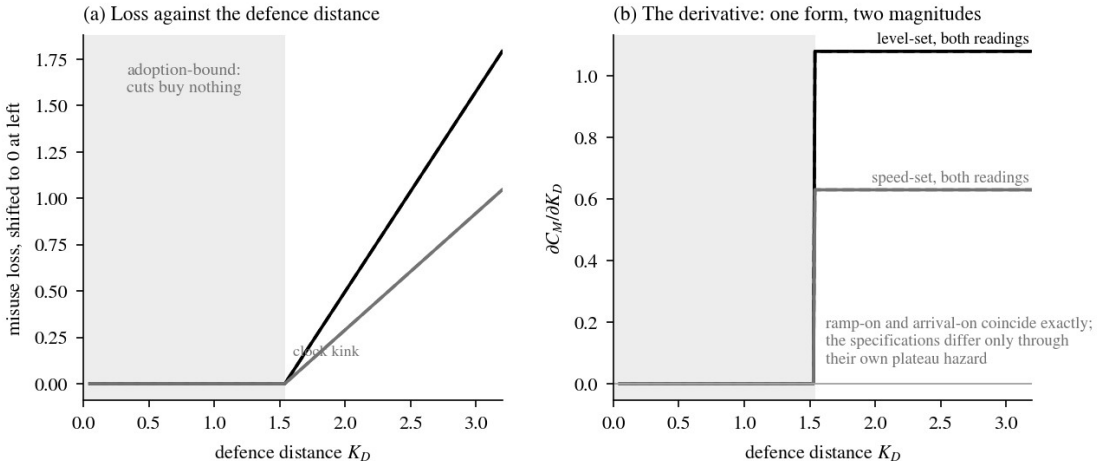


Figure 13. The *K_D* derivative: one form, and a floor. (a) Misuse loss against the defence distance in all four cases — two specifications by two exposure readings — each shifted to zero at the left edge. Below the clock kink the wait is adoption-bound and cuts to *K_D* buy nothing. (b) The derivative. Above the kink it equals *(χ_D/s)(h_M^plat − h_M^post)* with each case's own plateau hazard, so the two exposure readings coincide exactly and the two specifications differ only through that hazard; below the kink it is exactly zero. The curves are drawn from the model's own loss function, and the figure checks both statements before it is saved (§4a.5, A4a.7(i)).

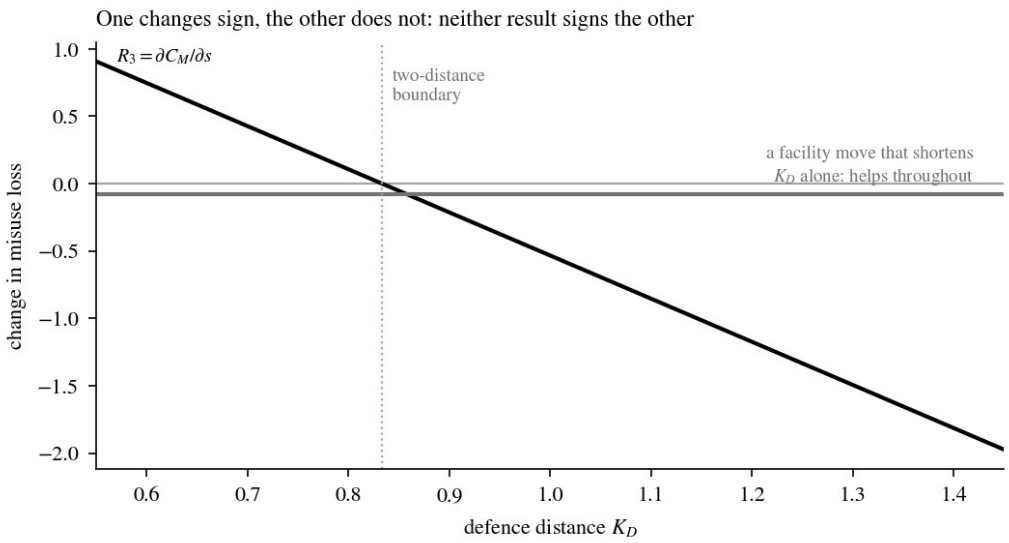


Figure 14. Neither R3 nor the facility comparison signs the other. Level-set, convention A, ramp-on, with $\hat{h}(x) = x^2$, *K_M* = 1, *h_M^post* = 0.2, *s* = 0.5, *L* = 6 and *m_DT_D* = 1. As *K_D* crosses the two-distance boundary, R3 changes sign while a facility move that shortens *K_D* alone keeps lowering misuse loss by the same amount. So knowing how misuse loss responds to speed does not tell you whether the facility helps, and the counterexample runs in both directions (§4a.5). The curves are computed by the same script that supplies the counterexample, so figure and proof cannot drift apart.

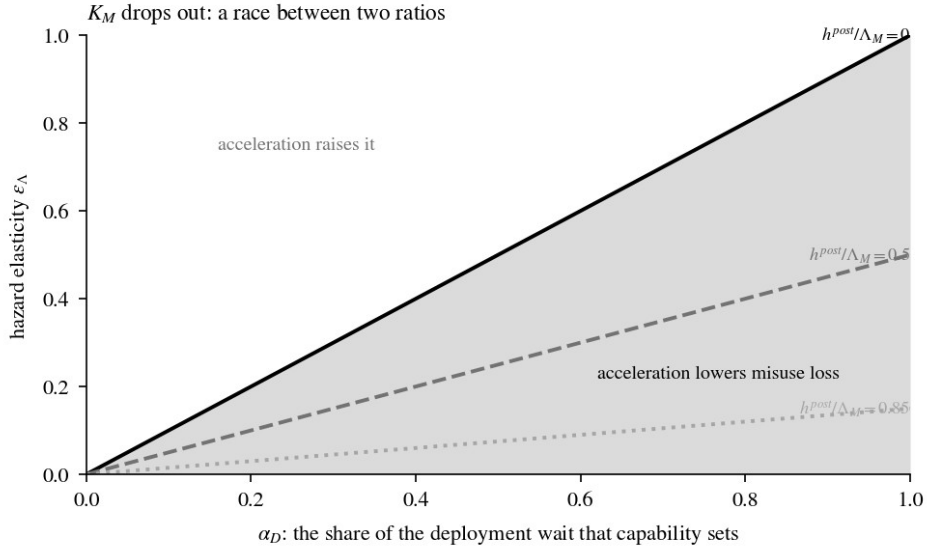


Figure 15. Under speed-set, the defence clock is a race between two ratios. Acceleration lowers misuse loss exactly when *ε_Λ < α_D(1 − h_M^post/Λ_M)*: the hazard's elasticity in speed against the share of the deployment wait that capability sets, discounted by the residual. The misuse distance *K_M* does not appear at all, which is the structural difference from the level-set two-distance

form. Three residual ratios are drawn. Schematic; the axes are the model's own ratios and neither carries a measured quantity.

4a.6 The mixed case

Mixed defence timing and a mixed hazard are different objects.

Mixed timing with a pure speed-set hazard gives RTTR's share test, $\varepsilon_\Lambda < \alpha_D$, under ramp-on exposure with complete closure.[10] On the capability branch the share is $(K_D/s)/(K_D/s + m_D T_D)$, which exceeds $1 - m_D$ whenever $m_D > 0$ and equals it at $m_D = 0$.

A genuinely mixed hazard has both a level component and a speed component. A share condition of RTTR's kind then holds only under conditions: it needs a closing clock whose post-capability segment does not lengthen with speed.

- *Additive form.* With ramp-on exposure, closure paced by K_M and $h_M^{post} = 0$, acceleration lowers loss iff $\alpha_H \varepsilon_\Lambda < \sigma_M$, where α_H is the speed component's share of misuse loss and σ_M the capability-set share, the loss accrued on $[0, \tau_M]$ over total loss. At $\alpha_H = 1$ this is RTTR's $\varepsilon_\Lambda < \tau_M/t_D$; at $\alpha_H = 0$ it is the level-set result. It is the general additive form (Appendix A4a.5) evaluated on that branch.
- *Separable multiplicative form.* With ramp-on exposure and complete closure at $\tau_M \leq t_D < L$, loss falls iff ε_Λ is below an exposure-response threshold $*\bar{\varepsilon}_g*$ (Appendix A4a.5). On the K_M-paced capability branch $*\bar{\varepsilon}_g*$ is the capability-ramp share of accumulated loss, the hazard-weighted share. That share equals the calendar-time share $\alpha_D = \tau_M/t_D$ exactly when the level component's value at K_M equals its average along the ramp, or when $t_D = \tau_M$; a constant level component is sufficient, not necessary.
- *On the plateau, no share condition exists*, on either exposure reading (Appendix A4a.5). Under a fixed horizon, $K_M > 0$ and a nondecreasing level component the threshold is nonpositive, so A4 raises loss.

4a.7 The benefit side, stated discounted

The balance is undiscounted, as a marginal approximation, for comparability with RTTR on the risk side [DIS 9, g10]; the benefit-side result is stated discounted. Take $B_r = p(y/r)e^{-rt_B}$, or its horizon form $p(y/r)(e^{-rt_B} - e^{-rL})^+$.

Result B. *With $p > 0$, $\chi_B = 1$ and, in the horizon form, $t_B < L$: the timing margin is $s\partial B_r/\partial s = pyq\chi_B e^{-rt_B}$ in both forms; the probability margin is $\partial B_r/\partial p = (y/r)e^{-rt_B}$ in the perpetual case and $(y/r)(e^{-rt_B} - e^{-rL})$ with a horizon. Both fall with r, and the probability margin falls faster in proportion. The ratio of the proportional margins, $(s\partial B_r/\partial s)/(p\partial B_r/\partial p)$, is $rq\chi_B$, or $rq\chi_B/[1 - e^{-r(L - t_B)}]$ with a horizon, and it rises in*

[10] The capability-branch share at complete closure is RTTR's mixed-timing condition. The general form $\varepsilon_\Lambda < \alpha_D$ is the ramp-on R3 condition at $h_M^{post} = 0$ (Appendix A4a.4(iv)). The conditions are: speed-set; ramp-on only; complete closure, held fixed; $t_D < L$; away from the clock kink; $\Lambda_M > 0$. No convention is involved.

r. Off the capability branch ($\chi_B = 0$) the timing margin and the ratio are zero; after the horizon ($t_B \geq L$) both margins vanish locally.

So discounting favours timing relative to the probability channel (Figure 16(a)). p multiplies the timing margin, so the two channels are complements: a higher success probability raises the return to speed. And probability is itself an argument for speed whenever $sp' > 0$, adding $(y/r)e^{-rt_B} \cdot sp'$ in the perpetual case and $(y/r)(e^{-rt_B} - e^{-rL}) \cdot sp'$ with a horizon. The comparison is between proportional parameter changes, not between equally costly interventions.

Corollary. *For a benefit of fixed duration ℓ_B after arrival, holding p, y and ℓ_B fixed and counting the whole duration with no binding cutoff, the timing margin is*

$$pyq\chi_B e^{-rt_B}(1 - e^{-r\ell_B}),$$

which is zero at r = 0. Earlier arrival then moves the end of the benefit forward as well as its start. This is why the benefit side is stated discounted; Figure 16(b) shows the margin starting from zero. If the horizon cuts the benefit off ($t_B < L < t_B + \ell_B$), earlier arrival raises counted benefit, which is P16(i). At small enough r the timing margin rises with r; Appendix A4a.6 gives the interval exactly.

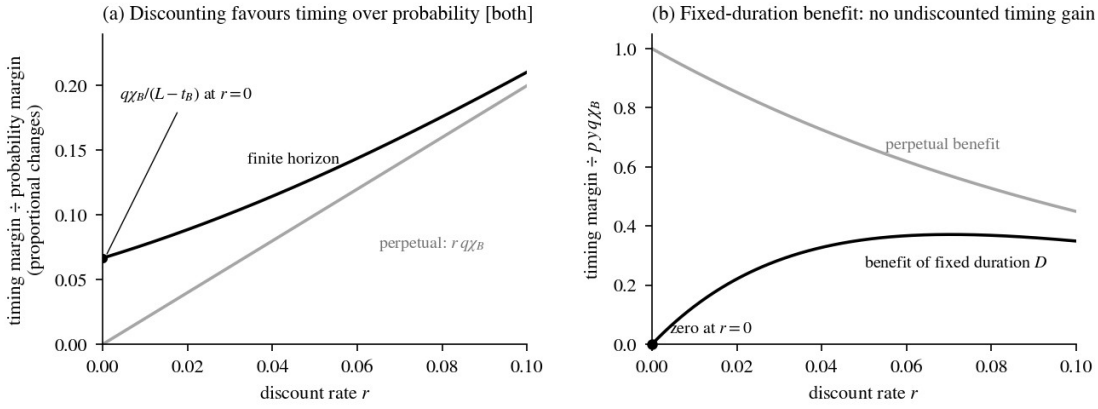


Figure 16. The benefit side, stated discounted. (a) For proportional changes, the ratio of the timing margin to the probability margin rises with the discount rate r. It is $rq\chi_B$ for a perpetual benefit and $rq\chi_B/[1 - e^{-r(L - t_B)}]$ with a finite horizon. Discounting therefore favours the timing channel relative to the probability channel, not the reverse, and the two are complements. (b) Take a benefit of fixed duration D, counted in full with no binding cutoff. Its timing margin, $pyq\chi_B e^{-rt_B}(1 - e^{-rD})$, is zero at $r = 0$. Undiscounted, arriving earlier only moves the same benefit earlier and adds nothing; this is why the paper states the benefit side discounted. The margin then rises with r at small r. Grey shows a perpetual benefit for comparison. The formulas are exact, but the parameter values are illustrative ($q = 2$, $t_B = 8$, $D = 15$, $L - t_B = 30$, in years), not calibrated. Schematic in that no axis carries a measured quantity. *Notation: D* is the model checks' symbol for the fixed benefit duration, which the text writes ℓ_B (§4a.1, note on notation).

One observation, not a result: under these conditions the timing argument for haste on health grounds relies on discounting the future, and it sits beside a case for weighing long-run catastrophe heavily [J]. The paper does not weigh the two, and it offers no decision rule for acting while the balance is unidentified. That is a declination, not an omission: a rule would convert a non-identification result into a recommendation the evidence will not carry. Nor would the obvious candidate work on this structure. That candidate is maximin. Rawls gives three conditions under which it is defensible — a knowledge of likelihoods impossible or at best extremely insecure; a chooser who cares very little for what might be gained above the minimum he can be sure of; and rejected alternatives whose outcomes one can hardly accept — and calls the rule itself a maxim, a rule of thumb that comes into its own in special circumstances, rather than a general guide to choice under uncertainty (Rawls 1971, §26, pp. 153–154). Gardiner adopts those same three as the criteria for a core precautionary principle, and states that they pick out sufficient conditions for precaution rather than necessary ones (Gardiner 2006, pp. 47–48). The first condition this paper arguably meets. The second and third it does not, and for reasons internal to the model rather than gaps in the search. Acting as though the worse reading held needs a fixed referent for "worse", and the signs flip across the rows of §4b.1 — which is the paper's own finding — so such a rule would have to be applied cell by cell and could select an action that is optimal under no single reading. It also needs bounds, and the documented search supplies none: V_M multiplies both readings' misuse terms alike, so the stakes per event are the same under each, and no bound on the health flow y or the success probability p was located [g7, g2] — so no finite worst case can be formed from what was read. Non-identification is itself symmetric between the two errors; what need not be symmetric is the loss from acting on the wrong one, and nothing read bounds that either, so the paper assumes no asymmetry and supplies none.

4a.8 Derivations still open

These remain open, and nothing above rests on them: life-years accounting; which invariances of expected loss, beyond invariance of every argument to speed, make the misuse derivative zero; a full characterisation of how the region in which speed helps changes once misuse is added; the signs of Φ' by clock branch beyond the case in which both benefit and harm decouple from general capability; §4b's parameter-level construction under each specification (§4b.2); the relative response of benefit to p and to s when attacks delay arrival; the comparison of I3 with I4 (§9); gradual coverage in R3; speed-set truncation at the activation kink as a non-local result; truncation under a discounted or otherwise untruncated objective, which has no sharp beyond-horizon region (§4a.4); R3 discounted under speed-set; §4b's specification-level result as a stated theorem; and the whole set of results with V_M a function of the capability level reached rather than a constant; and the benefit side under a speed-set reading of p, the paper having taken a level-set reading of it (§4a.1) while carrying both specifications for the misuse side. For the facility comparison: gradual coverage beyond the K_D derivative of A4a.7(i), discounting, more than one severity class, the whole balance $J_I3 - J_0$, and a facility that changes the local speed of the biology clock, which is outside this comparison. Two of these open items arise only in this paper: whether the right side of the speed-set test in R2

with the benefit is monotone in *s*; and which convention, A or B, describes a real facility, which no source read records. Appendix A-open lists what the appendix does not prove.

4b. The model: non-identification

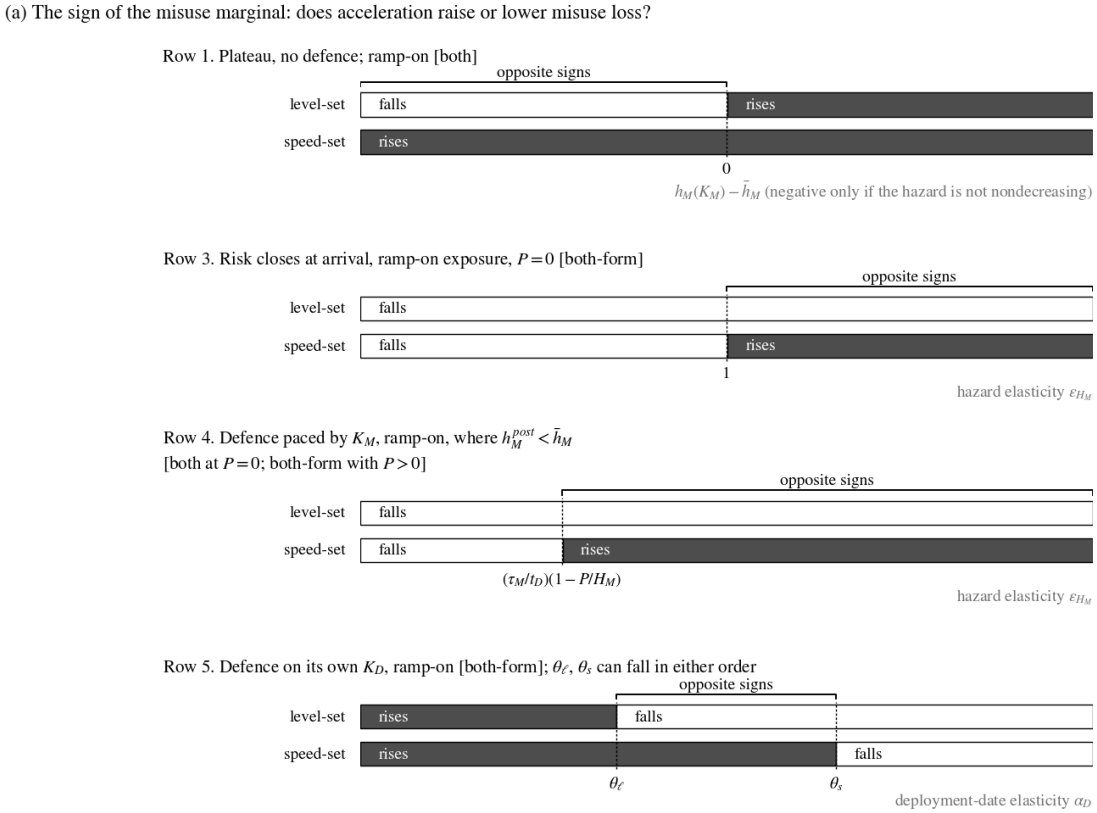
Section 4a signs the misuse marginal under each specification. This section asks what those signs leave unidentified: first what the choice of specification leaves open, then what the unmeasured parameters leave open within one.

4b.1 Specification-level: four conditional opposite signs

Level-set and speed-set are both consistent with the documented evidence, and no source read settles which holds for the misuse term [b12]. On the ramp-on exposure reading, the two give opposite strict signs for the misuse marginal sC_M' in four cases, rows 1, 3, 4 and 5 of Table 4b.A (appendix material), each under the restrictions its row states. Each row's two signs are proved in the appendix, rows 3 and 5 by substitution into the formulas of §4a.[11] The rows are conditional examples, not an enumeration over every level profile: row 2 also diverges once its monotonicity is dropped (below). The count of four rests on the ramp-on reading alone. **On the arrival-on reading the comparison must gate both sides**, since under level-set the arrival-on reading is the level profile that is zero below K_M (§4a.2). Substituting it sets the ramp average $*\bar{h}_M*$ to zero, and rows 1 and 4 then cannot take opposite strict signs at all: their divergence conditions become $h_M(K_M) < 0$ and $h_M\text{^}post < 0$, and a hazard is not negative. **Only row 5 diverges on the arrival-on reading**. Earlier drafts of this paper compared the ungated level derivative with the gated speed-set derivative and reported three arrival-on divergences; that pairing changes the exposure reading and the specification at once, which is the pairing §4b declines for the thresholds, and two of the three were artefacts of it. Row 5 survives for a reason worth stating: its two tests are not two evaluations of one quantity. The level-set test compares **distances** — the defence distance against the misuse distance, each weighted by the hazard gap it closes — while the speed-set test compares an **elasticity** against the capability-set share of the deployment wait. The two are not independent objects: under the comparison convention the first is the second at $\varepsilon_\Lambda = 0$, as row 7 sets out. What makes row 5 survive is that the gap between them, unlike row 1's and row 4's, is not signed by the sign of a hazard, so a positive elasticity can carry the speed-set marginal across zero while the level-set one stays below it. A divergence means opposite strict signs; neutrality against a sign does not count. A4 ($\Lambda_M' > 0$) is imposed throughout, the residual $h_M\text{^}post$ is held

[11] Row 3 follows by substituting closure at the capability's arrival ($t_D = \tau_M$, $h_M\text{^}post = 0$, capability branch) into the K_M-paced formulas of §4a, level-set and speed-set ramp-on (Appendix A4a.1(iii), A4b). Row 5's level-set side is the two-distance inequality of §4a.5 and its speed-set side the own-distance elasticity condition (Appendix A4a.3, A4a.4(iv)).

fixed, and the defence rows use the capability branch of the clock ($\chi_D = 1$).[12] Figure 17 draws each of the four rows along the quantity that decides it.



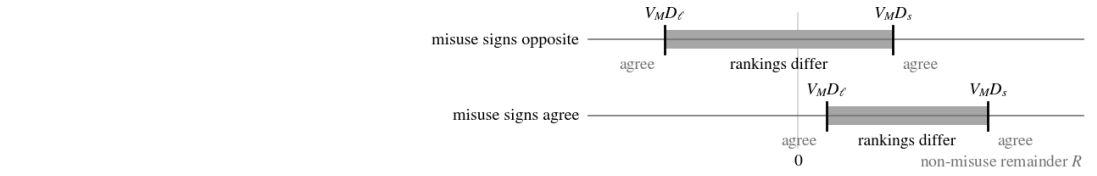


Figure 17. Where the two specifications disagree, and where that matters. (a) The sign of the misuse marginal under each specification: whether acceleration raises (dark) or lowers (open) misuse loss. It is shown for the four rows of the disagreement table that have conditional opposite signs, each along the quantity that decides it. Equality is neutrality. - **Row 1, plateau without defence.** Level-set follows the sign of *h_M(K_M) − h̄_M*, while speed-set rises*

[12] Symbols are §4a's (§4a.1, notation note): *Λ_M*, *ε_Λ* and *h_M^post* for the model checks' *H_M*, *ε_{H_M}* and *P*; *Ξ_ℓ*, *Ξ_s* for the two misuse derivatives; *Υ* for the non-misuse remainder; *ᾱ_ℓ*, *ᾱ_s*, *ᾱ_s^arr* for the thresholds; *Ω_E* for the evidence set. Each replaces a symbol that collides with RTTR's.

whenever the hazard rises with speed. They diverge only when the hazard at the level reached is below its average along the path, h_M(K_M) < h̄*M, which no nondecreasing hazard allows. -* ***Row 3, risk closing at arrival, with ramp-on exposure and P = 0.*** *Level-set always lowers. Speed-set lowers only while* ε{H_M}* < 1, a strict reversal above unity. - **Row 4, defence paced by *K_M*, where *h_M^post < h̄_M*. **Level-set lowers. Speed-set ramp-on rises once *ε_{H_M}* exceeds (*τ_M/t_D*)(1 − *P/H_M*). -** Row 5, defence on its own distance.** Each specification lowers misuse loss once the deployment-date elasticity *α_D* exceeds its own threshold, θ_ℓ or θ_s. The two diverge between the thresholds, which can fall in either order.

The arrival-on versions are not drawn. Compared like with like, with the level-set side gated, only row 5 diverges on that reading (§4b.1). (b) These are signs of the misuse term only, not policy directions. Let *R* be the sum of every non-misuse derivative. The whole balance then moves as *R* − *V_MD_i* under specification *i*. So the two specifications rank speed differently only when *R* lies strictly between *V_MD_ℓ* and *V_MD_s*, with *V_M* fixed, *R* held common to both specifications by assumption, and both endpoints taken on one exposure reading. Opposite misuse signs therefore need not mean opposite rankings, and agreeing misuse signs need not mean agreeing rankings. Threshold positions are schematic; axes are not to scale. *Notation:* the panels and this caption use the model checks' symbols: *H_M*, *ε_{H_M}* and *P* are the text's *Λ_M*, *ε_Λ* and *h_M^post*; θ_ℓ and θ_s are *ᾱ_ℓ* and *ᾱ_s*; *R* is *Υ*; and *D_ℓ*, *D_s* are *Ξ_ℓ*, *Ξ_s* (§4a.1, note on notation).

Row 1: plateau, no defence. The two derivatives are in Table 4b.A. They agree that speed hurts whenever *h_M(K_M) > h̄_M*, as for any continuous, nondecreasing, nonconstant level profile, and they diverge if and only if* h_M(K_M) < h̄_M*, a profile higher on average along the path than at the level reached; then level-set says acceleration lowers misuse loss. The deciding quantity,* h_M(K_M) − h̄_M*, is unmeasured: nothing read estimates misuse risk per unit of capability [b1], and the hazard's form was not located as an estimate [b11]. A measure of the hazard at the level reached against its average over the levels traversed would settle the row. On the arrival-on reading, with the level-set side gated (§4a.2), the row cannot diverge:* h̄_M* = 0 makes its condition *h_M(K_M) < 0*.

Row 2: truncation, and its restriction. Row 2's agreement needs a nondecreasing *h_M* on *[0, sL]*. Without it, R1's level-set derivative *L[h_M(sL) − h̄_M(0, sL)]* can be negative while speed-set ramp-on stays positive under A4, so row 2 then diverges iff *h_M(sL) < h̄_M(0, sL), as row 1 does.*[13] *On the arrival-on reading, with the level-set side gated, the row cannot*

[13] It follows by substituting a hazard that is not nondecreasing on *[0, sL]* into the level-set truncation derivative *sC_M′ = L[h_M(sL) − h̄_M(0, sL)]*, which holds without monotonicity (the "≥ 0" needs a nondecreasing hazard; Appendix A4a.1(ii)), and comparing with the speed-set ramp-on truncation derivative *LΛ_Mε_Λ > 0* under A4 (Appendix A4a.4(iii)). For example, *h_M(c) = 2 − c*, *K_M = L = 1*, *s = 1/2* gives *−0.25*. The conditions are: ramp-on only, the arrival-on speed-set truncation derivative being zero, so that a negative level-set sign against zero is not a divergence; the truncation regime *sL < K_M*, away from the activation kink; A4; and *h_M ≥ 0*. No convention is involved.

diverge for a different reason: the path never reaches K_M* in this regime, so the gated hazard is zero throughout and both derivatives are exactly zero.

Row 3: risk closes when the capability arrives (ramp-on exposure, *t_D = τ_M* on the capability branch, *h_M^post* = 0, *h̄_M > 0*). Level-set gives* −τ_M h̄_M < 0*: acceleration helps. Speed-set helps if and only if* ε_Λ < 1* and is neutral at 1, so the reversal is strict if and only if *ε_Λ > 1*. The deciding quantity, *ε_Λ* against unity, has no located measure [b1, b12]; the response of the misuse hazard to speed at a fixed capability level would settle it. The row is defined on ramp-on exposure, and we state no arrival-on version.

Row 4: defence paced by *K_M*. The two conditions and the divergence inequality are in Table 4b.A. The divergence runs in two directions: *h_M^post < h̄_M* with *ε_Λ* above the speed-set threshold, where level-set favours speed and speed-set opposes it; and *h_M^post > h̄_M* with *ε_Λ* below it, where the signs swap. Equality in either factor is neutrality. The second direction can occur with a continuous increasing level hazard and effective protection under both specifications. At *h_M^post* = 0 and *h̄_M > 0* the first is a strict reversal under A4 itself: level-set gives *−τ_M h̄_M < 0*, while speed-set loss rises whenever* ε_Λ > τ_M/t_D*. The deciding quantities,* h_M^post − h̄_M* and *ε_Λ* against the capability-set share of the closure date, are unmeasured [b1, b12]; the residual hazard after the step against the average before it, and the elasticity against that share, would settle them. On the arrival-on reading, with the level-set side gated, the row cannot diverge: its level-set condition becomes *h_M^post < 0*, and the gated level derivative *h_M^post·τ_M* carries the same weak sign as the speed-set one.

Row 5: defence on its own distance *K_D*. The two tests are in Table 4b.A. On the ramp-on reading the row diverges when they give opposite strict signs, either way round, and reversals can occur with *ε_Λ* below unity. On the gated arrival-on reading, under the comparison convention stated before Table 4b.A or at *h_M^post* = 0, the direction is fixed: where the two differ it is level-set that favours speed. Under the convention the reason is row 7's, the gated level-set marginal being the speed-set one at *ε_Λ* = 0; at *h_M^post* = 0 without the convention the same direction holds, although the two marginals need not coincide at *ε_Λ* = 0. Without that normalisation the direction is not fixed — rescaling *Λ_M* alone reverses it in part of the admissible region (A4b). A parameter point, on the gated arrival-on reading: *h_M(K_M)* = 0.5, *K_M* = 0.6, *K_D* = 0.7, *h_M^post* = 0, *m_DT_D* = 0.4, *ε_Λ* = 0.6 and *s* = 0.6 give a gated level-set marginal of −0.0833 and, with *Λ_M = h_M(K_M)* = 0.5 as the comparison convention requires at the speed in question, a speed-set one of +0.0867: opposite strict signs under A4. An earlier draft of this sentence took *Λ_M* = 0.5*s*^{*ε_Λ*} = 0.368 here, which fixes the two hazards against each other at *s* = 1 rather than at the speed in question, and printed +0.0638; the divergence survives either way. On the ramp-on reading this point has *α_D* = 0.745 > *ε_Λ* with a zero residual, so the speed-set marginal there is negative for every *Λ_M*, which is why the example belongs to the arrival-on reading. A larger *K_D* does not move the two tests alike: under level-set it lowers the misuse marginal, given *h_M(K_M) > h_M^post*; under speed-set, ramp-on, it raises the elasticity threshold when *m_DT_D > 0* and *Λ_M > h_M^post*, and lowers the signed marginal if and only if also *ε_Λ < 1 − h_M^post/Λ_M* (§4a.5). The deciding quantities are the two-distance inequality and the elasticity threshold; no measure of *K_D* was located (§11), nor of *ε_Λ* [b1, b12]. A dated record placing a countermeasure's capability requirement against the

corresponding misuse requirement, with the elasticity, would settle the row. The arrival-on condition is in Table 4b.A.

Rows 6 and 7. Row 6, defence on the adoption branch, agrees on every reading: gated, the level-set derivative is $h_M(K_M)\cdot\tau_M$, positive, against a speed-set derivative also positive under A4.

Row 7 is of a different kind from the rest of the table, and an earlier draft of this section mis-stated it twice. It compares *thresholds* on a coupled health benefit, not signs of the misuse marginal, so it needs the comparison convention stated before Table 4b.A, and it is not a divergence in this section's sense at all. Under that convention the two tests are nested rather than incommensurable: the gated level-set test is the speed-set arrival-on test at $\varepsilon_\Lambda = 0$, because a gated profile holds the hazard constant after arrival, which is what a constant Λ_M means. The thresholds differ by $V_M\Lambda_M\varepsilon_\Lambda(L - \tau_M)$, which A4 makes positive whenever $\tau_M < L$. So a benefit between them passes the level-set test and fails the speed-set one, never the reverse — the disagreement has a direction, which a divergence of signs does not. Nor does the row widen anything: it is the whole-balance interval rule of this section with $Y = pyq$ and every other term at zero, and that rule holds on either reading, gated or not. What the row does show is that the choice of specification moves the threshold the benefit must clear, not only the sign of the misuse marginal, and that the quantities deciding it are the same unmeasured ones, ε_Λ and the hazard at the level reached [b1, b12].

On the ramp-on reading the two tests do not depend on the same parameters. The ungated level-set test, $pyK_B > V_MK_M[h_M(K_M) - \bar{h}_M]$, is free of L, ε_Λ and s. The speed-set test carries L and ε_Λ on both readings: $pyq > V_M\cdot L\cdot\Lambda_M\varepsilon_\Lambda$ under ramp-on and $pyq > V_M\Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$ under arrival-on (A4a.4(vi)). What the two share is only the shape of the test, a benefit against V_M times the misuse marginal, which is how the test is defined. On the arrival-on reading with the level side gated the level-set test is $pyK_B > V_MK_M\cdot h_M(K_M)$, whose parameters are a subset of the speed-set arrival-on test's — which is the nesting above. We therefore make no claim that the two tests take "the same form" on the ramp-on reading, and do not use that phrase as a comparison between them.

The threshold rule.[14] On the defence rows, each specification has a threshold on the deployment-date elasticity, $\bar{\alpha}_\ell$ and $\bar{\alpha}_s$ (appendix material), and favours acceleration exactly when α_D exceeds it. Below both thresholds both oppose acceleration; above both, both favour it; between them they diverge; at a threshold that specification is neutral. The displayed comparison holds the residual fixed and pairs the ungated level-set threshold

[14] It follows by writing $s\cdot t_D' = -\alpha_D t_D$ in the level-set R3 derivative and in the speed-set ramp-on derivative, and dividing by $t_D(h_M(K_M) - h_M^{post}) > 0$ and $t_D(\Lambda_M - h_M^{post}) > 0$ (Appendix A4b). The conditions are: the speed-set side ramp-on only, the arrival-on threshold $\bar{\alpha}_s^{arr}$ being stated separately (Appendix A4b); a fixed residual for the thresholds as displayed, since with $h_M^{post\prime} \neq 0$ the term $(L - t_D)s\cdot h_M^{post\prime}$ enters and each threshold must be shifted by that term over t_D times its hazard gap (Appendix A4b) before it signs the derivative; $h_M(K_M) > h_M^{post}$ and $\Lambda_M > h_M^{post}$; and $\tau_M \leq t_D < L$, away from kinks.

with the speed-set ramp-on threshold. The speed-set arrival-on threshold $\bar{\alpha}_s\hat{}arr$ is stated separately (appendix material); it needs positive arrival-on exposure and $\Lambda_M > h_M\hat{}post$, and no same-reading arrival-on threshold comparison is established here.

What the four rows do and do not say. They are signs of the misuse marginal, not opposite directions for the whole policy. Write Ξ_ℓ and Ξ_s for the two misuse derivatives sC_M', and Υ for the sum of every non-misuse derivative in $sW_L\hat{}X'$. If — and this is an assumption of the result, not a property derived from it — Υ takes the same value under both specifications, each specification's whole-balance marginal is $\Upsilon - V_M\Xi_i$. The assumption is substantive: Υ contains the facility term $-V_M{\cdot}s{\cdot}C_F'$, and a facility hazard indexed to capability would make Υ itself specification-dependent. The specifications rank speed oppositely if and only if Υ lies strictly between $V_M\Xi_\ell$ and $V_M\Xi_s$ [b12].[15] They give the same strict sign when Υ lies strictly below both endpoints or strictly above both; at a distinct endpoint one specification is neutral, and if the endpoints coincide the interval is empty. Agreeing misuse signs need not mean agreeing rankings, and opposite misuse signs need not mean opposite rankings (Figure 17(b)). Υ carries the benefit and hospital terms and V_M scales the misuse term; none is measured [g7, d4, b1], so whether Υ falls inside the interval is itself unidentified.

Each row names the quantity that decides it and the observation that would fix it. That is our answer to the objection that this is a known sensitivity to modelling choice, and it is judgment, not a derived result. One further reading is ours and not a finding: since these divergences cannot be dismissed as free parameters flipping a sign, they may be a stronger core than the construction that follows. Figures 19 to 21 set out the seven rows on both readings, the nesting of row 7's two tests, and the interval rule that keeps opposite misuse signs from being opposite rankings of speed.

[15] It follows from the additive whole-balance marginal of §4a.1, in which misuse enters only as $-V_M{\cdot}sC_M'$, by evaluating it under each specification with the non-misuse remainder Υ held common, which is an assumption of the result rather than a derived property, and reading off the sign of the product of the two marginals (Appendix A4b). V_M is held independent of speed in this derivative. Across different fixed values of V_M, the interval criterion still applies pointwise, but Υ must be recomputed because it contains $-V_M{\cdot}s{\cdot}C_F'$; it cannot be held constant while only the endpoints move. The interval's own algebra involves no exposure reading and no convention, but its two endpoints must be taken on the same exposure reading: the ungated Ξ_ℓ with the ramp-on Ξ_s, or the gated Ξ_ℓ (§4a.2) with the arrival-on Ξ_s.

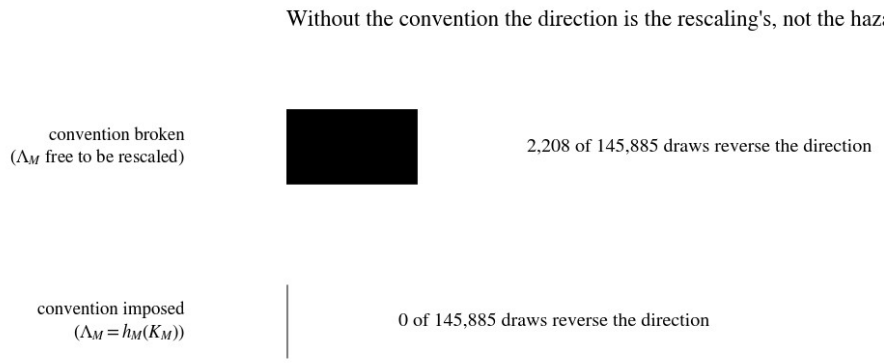


Figure 18. What the comparison convention is doing. Over 145,885 draws inside row 5's stated case, the two specifications' directions reverse in 2,208 of them when *Λ_M* is left free to be rescaled, and in none when the convention *Λ_M = h_M(K_M)* is imposed. A direction that a rescaling can reverse is the rescaling's, not the hazard's. The counts are from the project's own numerical check over the paper's row-5 tests, not from any source.

Four divergences on one reading; one survives the like-for-like comparison

	ramp-on	arrival-on, level-set side gated
1 plateau, no defence	diverges	cannot: would need a negative hazard
2 truncation	level-set only	level-set only
3 risk closes at arrival (ramp-on only)	diverges	no arrival-on version
4 defence paced by K_M	diverges	cannot: would need a negative hazard
5 defence on its own K_D	diverges	diverges
6 defence on the adoption branch	agree	agree
7 R2 with a coupled benefit	thresholds	nested at $\varepsilon_\Lambda = 0$

Figure 19. The seven rows on both readings, compared like with like. On the ramp-on reading four rows give conditional opposite signs for the misuse marginal. On the arrival-on reading, with the level-set side gated as §4a.2 defines it, only row 5 — defence on its own distance — still diverges: rows 1 and 4 would need a negative hazard to, row 3 is defined on ramp-on exposure and has no arrival-on version, row 2's two sides are both exactly zero, since the path never reaches *K_M*, row 6's two sides agree, and row 7 compares thresholds rather than signs. The table states the paper's own results (§4b.1, Table 4b.A).

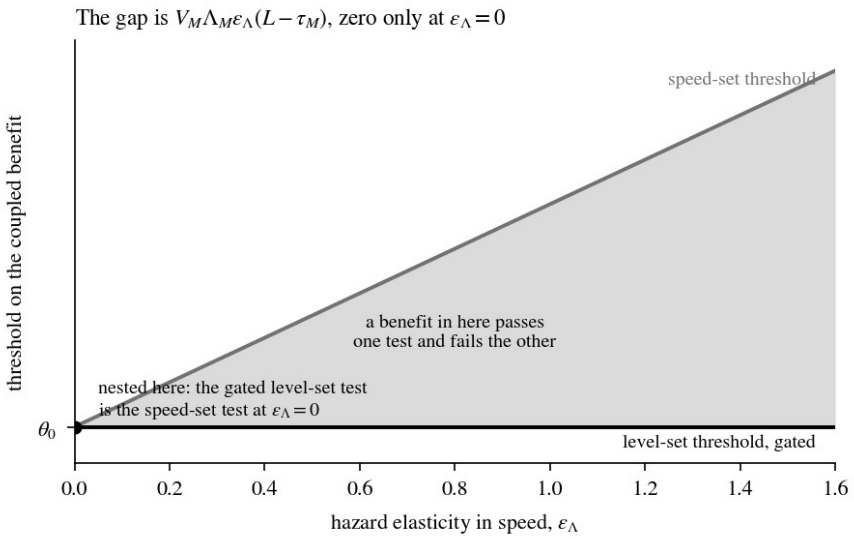


Figure 20. Row 7's two tests are nested, not incommensurable. On a common arrival-on reading, with the gated level profile and the stated normalisation, the level-set test on a coupled health benefit is the speed-set test at $\varepsilon_\Lambda = 0$. The thresholds differ by $V_M\Lambda_M\varepsilon_\Lambda(L - \tau_M)$, which is positive under A4 when $\tau_M < L$ and $V_M > 0$, so the speed-set threshold is the higher one always and a benefit between them passes one test and fails the other. Illustrative values: $\Lambda_M = 0.8$, $L = 6$, $\tau_M = 1.4$, $V_M = 1$. This is the correction made on 27 October (§4b.1).

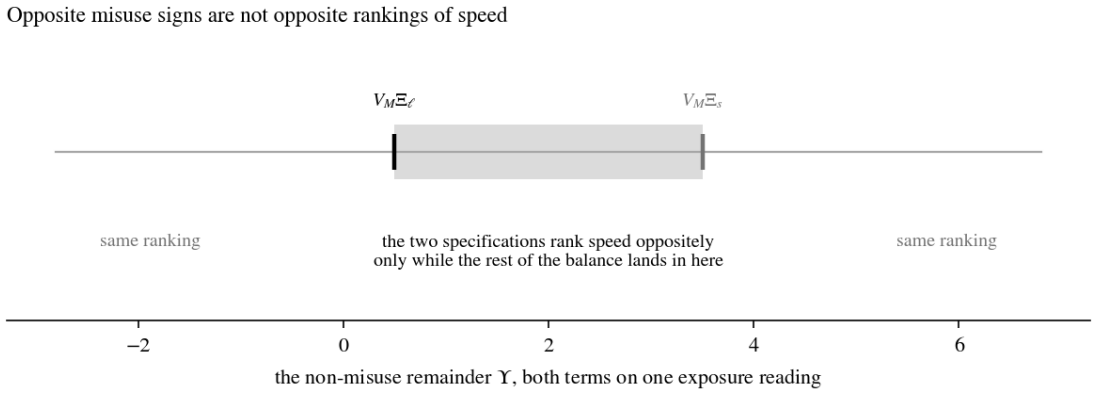


Figure 21. Opposite misuse signs are not opposite rankings of speed. With V_M fixed and the non-misuse remainder Υ assumed common to both specifications, the whole balance ranks speed oppositely if and only if Υ lies strictly between the two specifications' misuse marginals, both taken on the same exposure reading. The two marks are computed from the model rather than placed by hand. Outside the interval the two specifications agree about speed however far apart their misuse terms are (§4b.1).

4b.2 Parameter-level: can the unmeasured terms flip the sign?

Fix a specification and take §4a's misuse signs as given. Can assignments of y, V_M, p and the hospital DALY conversion, each a point of an evidence set Ω_E [d1, d3, d4; b2, b2a; b1; g7; g1; g6], flip the sign of $\Delta = W_L\text{^}X(s_1) - W_L\text{^}X(s_0)$, the change in the balance between two speeds, evaluated at that point? Answering in full takes a construction of several parts. One part is done: a bound on the hospital channel, and the margin condition it implies. The rest is not, and it is set out after the bound.

The hospital bound. Two separate checks computed it blind from the same inputs. It covers one component of the hospital term, the one the causal design measures: direct in-hospital mortality among patients already admitted when an attack began. Write V_c^mort for its expected value per attack, the death component of $V_c = \mathrm{E}[B_e]$. Only the working-paper version enters, because the two versions are never mixed and the published full text was not read [d1]. Excess deaths per attack are $d = 0.568–0.905$, the authors' own count of 42–67 divided by the working paper's 74 attacks; that is our arithmetic, WEAK, and a range across reported estimates, not a confidence interval [d3]. No conversion from an excess death to DALYs was located [d4], so the bound is stated as a function of the conversion ℓ, the DALYs lost per excess death, with V_c^mort $= d\ell$, at three anchors.

What fixes ℓ, and what does not. Two separate choices do, and only one of them is the paper's unit. The first is the **counterfactual**: if the attack does not occur, does this patient die later, and how much later? The second is the **valuation** of whatever interval that leaves: the reference-table standard loss at the age of death, which is WHO's rule, or the cohort's actual expectancy, which is not. The unit settles the second. It does not settle the first, and an earlier draft of this subsection said that it did.

The reason the counterfactual matters so much here is that the ledger is marginal. It differences two states of the world, and WHO books a death at the standard loss for the age at which it occurs in both of them. A death moved forward by an interval δ therefore contributes $SE(a) - SE(a + \delta)$, the difference of two standard losses, and not $SE(a)$. Since $dSE/da = \mu(a)SE(a) - 1 > -1$ for any life table with positive mortality, that difference is strictly less than δ; on a reference table with the low mortality that gives $SE(72) = 21.62$ it is a large fraction of δ, between about two-thirds and seven-eighths of it across plausible hazard shapes (*numerical check:* 65 to 87 per cent at Gompertz slopes from 0.06 to 0.12). $SE(a)$ itself is the contribution of a death that is **averted**, not displaced, with the decedent then facing the reference table. The two counterfactuals are not two readings of one number; they are two different quantities, and the ledger needs whichever one is true.

This is the problem that air-pollution epidemiology calls mortality displacement, or harvesting, and has a developed treatment of (Zeger, Dominici and Samet 1999; Schwartz 2000 — cited for the existence of that treatment; existence verified by DOI, neither read). We have not applied their methods, and the study's design does not let us; we name the problem so that the labelling below is read as the open empirical question it is.

- *Displaced death, standard valuation.* If each excess death brought forward by at most thirty days a death that would otherwise have occurred, which is our reading of the

30-day null and not the authors' [d2], then the marginal quantity is $SE(a) - SE(a + \delta)$ with $\delta \leq 30/365.25$, so $\ell < 30/365.25$ years and V_c^mort < 0.047–0.074 DALY per attack (0.568–0.905 × 30/365.25). The inequality is strict by the bound above, so these figures are a ceiling for this counterfactual and not its value. This anchor uses WHO's valuation throughout; it is as consistent with the paper's unit as the next one. The null is consistent with the displacement reading without establishing it, so the anchor is not a floor the data enforce.

- *Averted death, standard valuation.* If the attack caused a death that would not otherwise have occurred within the window at all, the decedent then facing the reference table, each excess death loses the WHO Global Health Estimates standard loss for persons aged 70–74, 21.62 years (WHO 2024, Table 2.1; read in part), which gives 12.3–19.6 DALYs per attack (0.568–0.905 × 21.62). This anchor also uses WHO's valuation — the years lost are the standard loss at the age of death, whatever the decedent's own prognosis — and it differs from the one above only in its counterfactual. The 30-day null bears on this counterfactual, since a death that would not otherwise have occurred within the window should raise 30-day mortality as well as in-hospital mortality — but only as far as that test had power to detect it, which the inputs do not establish (below). As with the anchor above, the data neither enforce this one nor rule it out.
- *Averted death, actual-expectancy valuation.* This is the one anchor that departs from the unit. It answers a different question, how much life these particular patients lost, and substituting the cohort's actual expectancy changes the valuation partway through the ledger. At ℓ = 12 years, assumed and not sourced, the anchor is 6.8–10.9 DALYs per attack (0.568–0.905 × 12); both checks reached for that figure separately. A sourced comorbidity-adjusted expectancy for this population would replace the assumption; none was located.

The age group is set by the mean age of the study's inpatients, 74.21 [d1; working paper Table 1]. That is the mean age of all inpatients, not of the excess decedents, who were probably older and sicker, and the inputs do not supply the decedents' age. Older decedents would lower both high anchors, sicker ones only the alternative, since the standard loss ignores prognosis. At fixed d, and with WHO's valuation held throughout, the **counterfactual alone** spans a factor of about 260 between the displaced-death ceiling and the averted-death anchor (21.62 × 365.25/30). So the spread in the bound is not the data's, and it is not the unit's either: it is the unanswered question of whether these deaths were caused or advanced. **The horizon convention, and which way it leans.** Each loss is counted once (A18; §4a.1) and booked undiscounted at the date of death, with the life-years it represents not cut off at the horizon L. The benefit is treated differently: it flows at y a year only to L (§4a.1; Figure 4). So in an undiscounted ledger a year of life gained after L is worth nothing and a year of life lost after L is worth one, and the asymmetry favours the harm channels — including the misuse channel, whose per-event loss V_M is on the same untruncated footing, which the paper had not said.

The convention is not arbitrary, and the alternative is not free; Figure 23 draws the asymmetry it leaves. Truncating the harm side would make each per-event loss a function of the event's date, $\min(\text{standard loss}, L - t)$ rather than a constant, and A20's fixed V_M is

exactly what lets the misuse term be written as the product V_MC_M and differentiated with V_M outside. Every marginal in §4a would have to be retaken with the per-event value inside the integral and a boundary term at L. Truncating the benefit side instead, by counting it past L, would remove the horizon from the benefit results and with it the result that an earlier benefit of fixed duration counted in full is worth nothing undiscounted. We keep the convention, state it, and say what it costs: the undiscounted comparisons in this paper are not neutral between a gain and a loss that fall beyond the horizon, and the horizon is a choice (§11). Note that this is a different use of the word from R1's *truncation*, which is the capability path running past the horizon and is a regime of the model rather than a booking rule. The population is US Medicare fee-for-service inpatients in 2016–2021; an unmeasured $\kappa \geq 1$ scales the figures to all payers and to the paper's population. The high anchors are the top of one component, not an upper bound on hospital harm, which also includes the losses §6 lists as omitted [d3]. One asymmetry in what the null can do: a 30-day null is the signature displacement would leave, but it is also what a real effect too small to detect at thirty days would leave, since the 30-day denominator and the 30-day baseline rate are both larger than the in-hospital ones and the same absolute excess is a smaller relative effect against a noisier background. The inputs record no displacement analysis by the authors, and we did not compute one; the published full text was not read [d1], so this is an absence in what we read and not a claim about the study. So the null discriminates between the two counterfactuals only as far as the 30-day test had power to, and nothing we read establishes that power. Neither counterfactual is established and neither is ruled out. The bound is therefore carried as a bracket between them rather than resolved, and the paper leans on neither end: a reader who judges the displacement reading the likelier one can take the low end from what is written here, and so can a reader who judges the other way, but the choice is not ours to make on this evidence.

The margin condition. For two options compared on one branch of §4a's results, extend Δ to any quantity, as its value under option 1 minus its value under option 0, and write Δ_other for the change in every channel except deliberate misuse and the bounded mortality component: benefit, foregone benefit Φ, facility events, occupational harm, natural-outbreak protection, remediation, and any hospital harm outside the mortality calculation. Option 1 then has the lower whole balance exactly when

$$\Delta(V_MC_M) > \Delta_other - \Delta(V_c\text{^}mort\ C_c).$$

Where misuse and the hospital channel pull opposite ways and everything else is held fixed, the facility term included because V_M also scales it (§4b.1), the bound turns this into a bracket that needs no conversion. Misuse decides the comparison, whatever the conversion between the anchors, if $V_M|\Delta C_M|$ exceeds the upper hospital bound times $|\Delta C_c|$. The hospital channel decides it, whatever the conversion between the anchors, if $V_M|\Delta C_M|$ falls below the lower bound times $|\Delta C_c|$. **One rule at both ends.** The bracket's two ends must be read off the same rule, and an earlier draft of this subsection did not do that: it took the upper end from a 95 per cent limit of one control-group specification and the lower end from an interval of the other, while §1.1 quoted the authors' point estimates. The three rules give three different brackets, and each is stated here complete. All figures are DALYs per unit change in the expected number of hospital attacks of the kind the study

counts, before the unmeasured all-payer scaling κ, with the lower end at the displaced-death ceiling and the upper end at the averted-death standard loss.

Rule	Lower end	Upper end	Lower region
The authors' point estimate, main specification: 42 deaths over 74 attacks (d = 0.568)	0.047κ	12.3κ	not empty
The authors' point estimate, the more adverse specification: 67 deaths over 74 attacks (d = 0.905)	0.074κ	19.6κ	not empty
95 per cent limits, main specification (p = 0.07)	0	25.4κ	**empty**, since its interval −0.0005 to 0.0159 includes zero
95 per cent limits, the more adverse specification	0.019κ	34.2κ	not empty, since its interval 0.0031 to 0.0215 excludes zero

So whether the lower region is empty — whether any change in misuse loss is small enough for the hospital channel to decide the comparison — is not a fact about the data but a consequence of which rule and which specification are used, and the earlier draft's pairing happened to pick the one combination that emptied it. Within a rule, the conversion decides where in the bracket the boundary falls, and nothing measured settles that. Under the actual-expectancy valuation the point-estimate upper ends are 6.8κ and 10.9κ instead of 12.3κ and 19.6κ. Each row takes both of its ends from one value of d, which is the same discipline the rule above states: an earlier version of this table paired the lower end of d = 0.568 with the upper end of d = 0.905. Arithmetic for every cell is in PROOFCHECK/bracket.py and recorded in S.6; the implied exposed patients, 5,455 and 5,447, come from 42 ÷ 0.0077 and 67 ÷ 0.0123 and reproduce the authors' own per-attack counts. Both checks derived the condition, in complementary forms. What is unmeasured in it is V_M, ΔC_M, ΔC_c (how speed or an intervention moves the attack count, which nothing read links to capability [d10]), κ and ℓ itself; what is measured is d, the cohort's mean age and the 30-day coefficients. V_M and ΔC_M enter only as a product, which is why the paper can write the term and must not calibrate it [b1]. The hospital bound is common to both misuse specifications and the misuse side is not, so the margin is evaluated once per specification, with ramp-on and arrival-on as a separate axis, and each evaluation carries the conditions of its own misuse sign.

What the bound does not do. It is graded WEAK, and it cannot be graded above its weakest necessary input: the conversion is ours, and the per-attack death rate is our arithmetic [d3]. It bounds one summand of Y. It does not calibrate V_M, does not sign the balance and ranks no intervention. Whether $V_M|\Delta C_M|$ lies above, inside or below the bracket is unidentified, so the non-identification claim of §4b.1 and of this section is unchanged; both checks reached that conclusion separately. The claim is sharper, not weaker: the hospital channel's ignorance is now one named quantity, the conversion, anchored at two values, beside V_M, y and p, which nothing read bounds. A computed bound on one channel reads like progress on the whole balance, and it is not.

What remains open. A parameter-level argument needs five further parts. None is done, and together they are the paper's largest open item. *A target*: a population, horizon L, comparator and two speeds on one branch of §4a's local results, with P16's primitives held common and its favourable speed test holding so that any flip comes from the new channels, and with the kinks between branches accounted for (under level-set the join of truncation and plateau at $s = K_M/L$; under speed-set arrival-on the activation kink there). *The evidence set written as constraints*: Ω_E is so far a list, in which the hospital rows, one developer-reported programme for q [g1] and narrow-tool gains that cannot enter y as a function of speed [g7, g6] constrain little, no row bounds p [g5] or V_M [b1], every deciding quantity of §4b.1 is free [b1, b12], and the one controlled comparison enters wide, an insensitive null rather than a tight zero (§5.2–§5.4) [b2, b2a]. *Witnesses*: two assignments in Ω_E with opposite signs of Δ, each tested against the bracket above; none has been written, a pair differing only in V_M would be vacuous because no row bounds it, and in the level-set divergence regions of rows 3 and 4 a witness against speed must take its flip from the hospital term or foregone benefit unless the facility derivative or another added channel's is left free. *A restriction*: a defined and ordered family of restrictions, the minimal member that restores one sign, and a showing that the evidence does not imply it and that it is not a preference, with candidates that differ by specification (ε_Λ under speed-set; the level profile or the two distances under level-set). *Information*: the construction assumes fixed information and an exogenous path; RTTR's model has neither learning nor irreversibility, which is where the strongest economics objection to speed parts from it, and the learning comparison that would answer it has not been made [g11, DIS 10]. Appendix Table 4b.B sets out each part by specification.

Where this leaves non-identification. Section 4b.1's four divergences are derived on the ramp-on reading; compared like with like on the arrival-on reading, one survives. Section 4b.2 computes one bound, on one channel, graded WEAK, and leaves the rest of its construction unexecuted. Every quantity either part turns on, other than the hospital study's own estimates, is unmeasured and listed in §11 as an estimand. That is the paper's limitation, and the content of its non-identification claim.

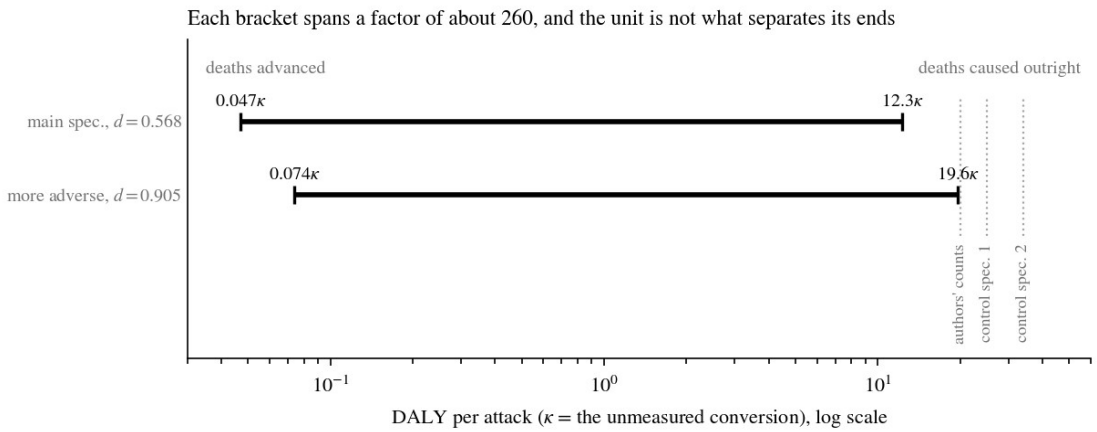


Figure 22. The one measured channel, and what its bracket does not settle. Each rule's bracket on the mortality part of the hospital harm, on a log scale, as a multiple of the unmeasured conversion κ from deaths to healthy life. Both ends use WHO's own valuation; what separates them by a factor of about 260 is whether an attack caused a death or advanced one, which the study was not designed to answer. The dotted lines are the misuse-side thresholds the text computes. This is our own arithmetic over the study's counts, graded WEAK (§4b.2) [d1, d3, d4].

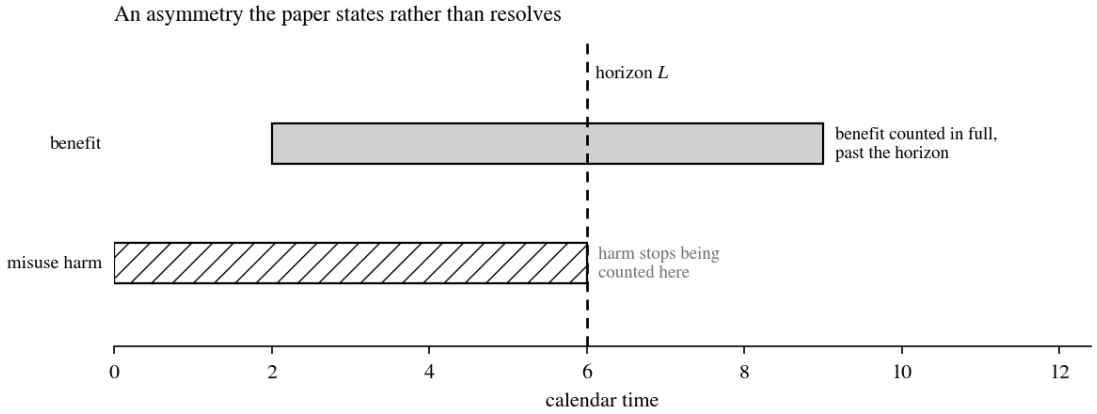


Figure 23. The two sides are booked on different horizons. A benefit of fixed duration is counted in full even where it runs past the counted window, while misuse harm stops being counted at L. The paper states this convention rather than resolving it: truncating the harm side would make each per-event loss depend on its date, which breaks A20's fixed V_M and the product form V_MC_M (§4b.2). Schematic; the durations are illustrative.

5. Evidence I: misuse

The misuse terms are h_M (level-set), Λ_M (speed-set) and V_M, the loss per event.

5.1 No usable estimate was located

No source read estimates the change in attack probability, casualties or DALYs per unit of general capability [b1]. The expert-consensus report we read says the field has no standard for the quantity: there are not yet standardised benchmarks for uplift, and real-world impact remains unclear because of secrecy, testing prohibitions and immature evaluation [b1]. That is a search result, not a claim that none exists, and the gap is contingent on those conditions rather than logical. The paper may write the terms; it may not calibrate them.

One record of attempted misuse was located, and it supplies no quantity. The company reported in September 2026 that it had disrupted several potential plots that year by scientists whose research with its models could have helped develop biological weapons, and said it could not determine whether that research served a legitimate or a nefarious purpose, erring on the side of caution in the face of that uncertainty [b14]. Three limits travel with it. We read a press account of that report, via syndication and with one paragraph of the account itself not read; the company's report was not located. The number is of plots the company says it detected and chose to report, with no denominator of attempts and no outcome for any of them, and it comes from an interested party on the risk-asserting side. And it speaks to the fourth link of the chain rather than the fifth: evidence that attempts occur is not evidence that harm followed. We grade it WEAK, and it moves none of h_M, Λ_M or V_M.

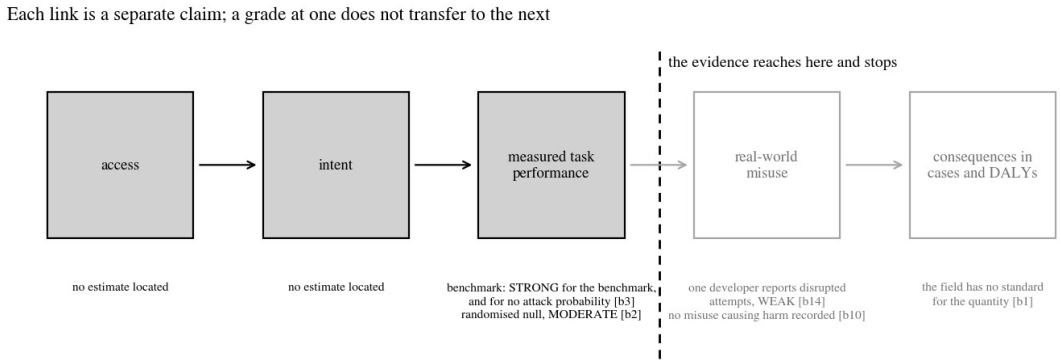


Figure 24. The chain from a model to a health loss, and where the evidence stops. Each link is a separate claim, and a grade at one does not transfer to the next. The evidence reaches measured task performance and stops: a purpose-built benchmark, STRONG for the benchmark and support for no attack probability [b3]; and the one randomised comparison, a null from a design its authors say could detect only a change about five times the one observed [b2]. Beyond the cut: one developer's report of disrupted attempts, WEAK [b14]; no misuse causing real-world harm recorded in what was read [b10]; and no standard in the field for the quantity the last link needs [b1]. Nothing operational appears.

5.2 The one controlled comparison

One study uses a randomised no-AI arm [b2]. Twelve cells of three researchers each were randomised across four scenario strata: four internet-only, four with one model, four with a second. Each produced one plan, scored on a nine-point viability scale by eight adjudicators, four from biology and four from security, under a Delphi protocol. It is the only randomised-arm design located, and not a developer's evaluation of its own product.

The result is a null: model access associated with 0.22 points lower adjudicated viability, t-statistic −0.49, p-value 0.64 [b2]. The load-bearing number is the authors' own sensitivity figure. Their design could have detected about **1.0 point**, roughly five times what it observed, stated under the assumption that all other variables remained constant — an assumption their own report undercuts where it says that cell composition likely was a much greater source of variability than model access [b2]. Figure 25 sets the observed difference beside the detectable one, on the study's own scale.

It prints estimates, t- and p-values, and no interval, standard error or inter-rater statistic anywhere; the significance tables exclude three non-randomised cells [b2]. Every plan in every arm, control included, scored between the rubric's "untenable" and "problematic" labels, so the outcome was floor-bound in the control arm too [b2]. We grade the study MODERATE: design transparency raises it; twelve cells, that variance attribution and the missing interval hold it below STRONG.

The authors' limits travel with the result: adjudicated viability of a written plan, one rubric, planning phase, the guardrailed models of mid-2023, none fine-tuned. They decline to extend it, writing that their results "do not and could not disprove the existence of risk" [b2]. Access, intent, task performance, realised misuse and consequences are separate links; this sits at the third.

5.3 Our reconstruction

Because no interval is printed, we reconstructed one from the printed estimates and t-values: implied SE = difference ÷ t ≈ 0.45 — the printed estimate and t-statistic are each rounded to the nearest hundredth, so the implied standard error is itself only determined to about ±0.02 — giving an approximate 95 per cent envelope of about **−1.5 to +1.0 viability points**, stated to the precision the printed inputs support [b2a]. We did it twice, separately; neither could recover a unique interval from what is printed, and we carry the wider of the two envelopes they found. Figure 25's third row draws it.

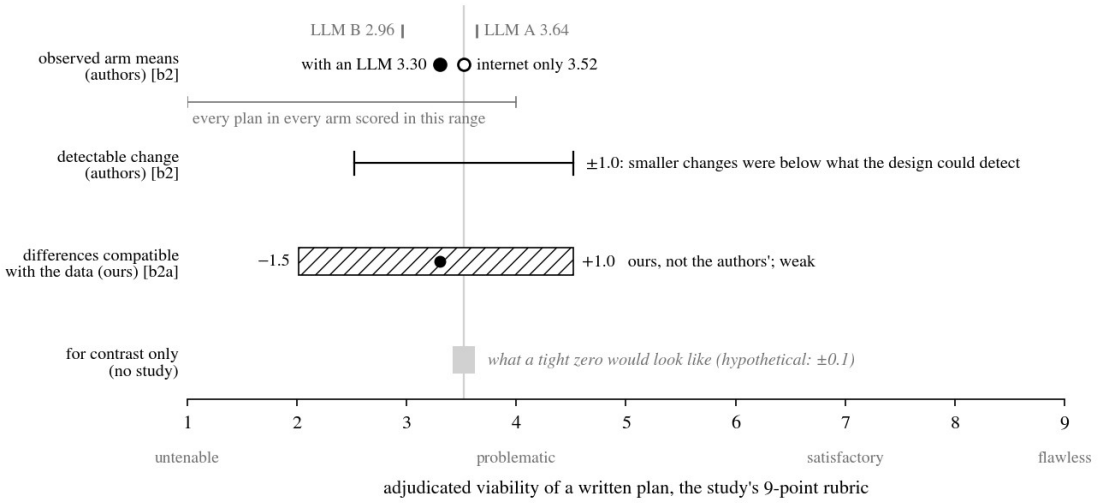


Figure 25. The one controlled misuse comparison, on its own 9-point scale. Mouton et al. (2024) compared written plans from red cells that had internet access only against cells that also had one of two large language models (summer-2023, guardrailed). Top row: the arm means as printed. Internet only scored 3.52 and with an LLM 3.30 (LLM A 3.64, LLM B 2.96), a difference of −0.22 points (p-value 0.64). Every plan in every arm scored between "untenable" and "problematic" [b2]. Rows two to four are differences from the internet-only mean, drawn about that mean (the vertical line). Second row: the authors' own sensitivity statement, that the design could have detected a change of about 1.0 point, roughly five times the difference observed, stated under the assumption that all other variables remained constant, which the report itself undercuts (§5.2) [b2]. Third row: our arithmetic reconstruction of the differences compatible with the data, about −1.5 to +1.0 points, stated to the precision the printed inputs support (§5.3). The report prints no interval, standard error or degrees of freedom, so this envelope is ours, not the authors'. It is weak, resting on five stated assumptions [b2a]. Bottom row, for contrast only: what a tight zero would look like (a hypothetical ±0.1; no study). The data are compatible with a moderate decrease and with an increase of about one scale point, so this null is not a tight zero. All of this concerns adjudicated viability of a written plan under this rubric, in the planning phase only. The authors decline to extend it to attack probability or casualties, and write that their results "do not and could not disprove the existence of risk". Published statistics and our reconstruction; drawn to scale.

It is ours, not the authors': they print no interval, standard error or degrees of freedom anywhere, and do not invite the arithmetic. It needs five assumptions, named — that the printed estimate and t are the same zero-null mean contrast; ordinary two-sample inference on twelve cell means, the relevant endnote fixing neither pairing, pooling nor tail count; a valid two-sided Student-t reference distribution; nearest-hundredth rounding; and unprinted degrees of freedom, taken in the range compatible with the printed p-values [b2a]. The sensitivity range we searched is 3 to 10 degrees of freedom, read continuously;

enforcing compatibility with the printed statistics under nearest-hundredth rounding puts the binding lower limit at about 4.27, and the envelope above is the union of the conditional intervals across that range, not a uniquely recovered confidence interval of the authors' and not a bound on any biological-risk parameter [b2a]. Neither p nor n identifies the actual test. We grade it **WEAK as an interval**, and MODERATE only for the qualitative point that the data do not bound the effect near zero. The design as §5.2 describes it is stratified, twelve cells randomised across four scenario strata, and the inversion assumes it is not: the envelope is for an unstratified comparison the authors did not run. It therefore constrains no element of Ω_E, and we carry it only as a caution against reading the null as a tight zero. Section 4b refers here rather than restating these figures.

5.4 An insensitive null is not a tight zero

The design was not sensitive enough to establish an effect of the size it estimated — a minimum detectable effect is a design quantity under an assumed power, not a prohibition on smaller effects being real — and on our reconstruction, a moderate decrease and an increase of about one scale point are both compatible with what was printed [b2a]. Neither licenses a direction. The constraint entering Ω_E is **wide, not tight**, in both directions, and narrows neither h_M, nor Λ_M, nor V_M. Zero uplift is compatible with this comparison; it is not established by it, and the comparison supplies no quantitative constraint on the misuse hazard or on the loss per event. The tempting error is to read the null as evidence of safety. The study's authors do not make it. Narrowing the constraint needs a design much larger than twelve cells; none was located [b1, b2].

5.5 The specification is unsettled

Nothing read decides whether misuse risk is set by the capability level reached or by the speed of approach [b1, b12]. The state-risk reading behind the level-set form appears in two theoretical works RTTR already cites, neither estimating anything; the speed-set form is that of RTTR's main results, and RTTR carries the level-set form as a scoped alternative. The quantity that would settle it is an estimand with no located measure (§11). The paper carries both, and §4b's specification-level non-identification follows.

6. Evidence II: health systems

This section supplies B_e, the baseline care loss per attack: harm to the care patients were already receiving [ACC (2)]. With foregone benefit carried separately (§4a.1, Figure 3), $V_c = E[B_e]$ [ACC]. The units are the studies' own.

One causal study, two versions. One study identifies the effect of ransomware attacks on hospital patients. As the working paper describes it, the design is a stacked event study and an admission-level difference-in-differences on Medicare fee-for-service claims: 74 attacks on 163 hospitals, 2016–2021, standard errors clustered by attack, funded by the NIHCM Foundation [d1]. Attacks were found through news and federal reports, so undisclosed ones are missed [d1].

The published version: we hold the abstract only, because the published body was closed to us [d1; §11]. The abstract reports hospital volume down 17–24 per cent in the attack week and in-hospital mortality up 34–38 per cent among patients already admitted [d1].

The working paper, which we read in full, reports absolute effects with their baselines: in-hospital mortality up 0.77 percentage points on 3.72 per cent in the main specification (p-value 0.07), and up 1.23 points on 3.38 per cent with later-attacked hospitals as controls [d1].

The versions differ in specification and magnitude. The working paper's absolute effects are its own estimates and are not conversions of the published percentages [d1; DIS 4]. The record is STRONG on direction and MODERATE on magnitude [d1].

Deaths. The count is unresolved. The authors' own 2023 account, from the working-paper version, puts the deaths at 42–67 Medicare patients over 2016–2021 [d3]. A systematic review reports 69–76 on its reading of the published text, which we were not able to read [d3; §11]. Divided by the working paper's 74 attacks, its count gives roughly 0.6–0.9 excess in-hospital Medicare deaths per attack; we do not divide the review's figure, because we could not read the published version's own attack count. The rate is our arithmetic and WEAK [d3]. Either count is a lower bound for all payers, omitting morbidity, deaths after discharge and deaths among displaced patients [d3].

The nulls. The working paper finds no effect on 30-day mortality or readmissions, and none among patients already admitted to neighbouring hospitals [d2]. The first null is the most informative thing the record supplies about the conversion §11 lists as missing: an in-hospital mortality effect with no 30-day counterpart is consistent with deaths brought forward rather than life-years lost at the decedents' full remaining expectancy, which would make the DALY loss per excess death substantially smaller than a life-expectancy conversion implies. That reading is ours, not the authors' [d2], and it is why the hospital margin of §4b.2 must state the conversion it assumes rather than take a life-expectancy one for granted. Single-incident studies, with cruder outcomes and shorter outages, report no detectable mortality difference or no reported harm [d5]. They can be reconciled with an average effect across 74 attacks, and they show that harm is not visible in every attack: loss per event is a distribution [d5; DIS 4].

Spillover and other channels. In all-payer California data, emergency visits fell 16.21 per cent at attacked hospitals in week 2 and rose up to 7.10 per cent at nearby hospitals in week 3, recovering within eight weeks [d6]. The volume result is STRONG; outcome evidence for spillover rests on one event with tiny counts and is WEAK, so displacement is the better-supported account [d6; DIS 5]. One national attack cut trial referrals by 85 per cent and recruitment by 55 per cent, on ten survey responses held at abstract level [d7]; in the model that lengthens arrival through δ_T rather than cutting y. Hospital quality measures worsened after data breaches; the authors propose remediation as the cause and did not observe it [d8]. Any such loss is assigned once, to V_cC_rem or to B_e [ACC (6)].

What it does not supply. Ransomware attacks on US health care delivery organisations rose from 43 to 91 a year over 2016–2021, 44.4 per cent of them disrupting care; no later

count of disruptive attacks was located [d9]. No primary-source report of an AI-enabled attack on a named hospital was located, so the capability term is not linked to this channel empirically [d10]. No conversion of any hospital result into DALYs, QALYs or a value of life was located; converting deaths needs the decedents' remaining healthy life expectancy, which no source read gives [d4]. *V_c* is unmeasured in the paper's units; §4b.2 bounds its mortality component only under conversions we assume and label (§11).

7. Evidence III: defence

R3 (§4a.5) needs a recognition date *t_R*, a coverage schedule *u(t, s)*, a completion date *t_D(s)* built from the deployment time *T_D* and the capability distance *K_D*, and a residual hazard *h_M^post*.[16]

Deployment. For one vaccine programme, 66 days passed from sequence release to a first trial and 326 days to rollout, against a funder's 100-day target from recognition of a threat to a product ready for initial authorisation; the fast part rested on years of prior platform work [e5]. These are dates for one disease, one platform and one programme, STRONG as dates [e5]. They identify timings, not a branch. Nothing in them puts countermeasures on the benefit's adoption branch, and RTTR §4.7 says sequential timing carries no such cutoff, so the paper carries sequential, mixed and parallel timing (§4a.5–§4a.6) [e5]. The funder's target presupposes that the platform exists [e5].

Coverage. Uptake does most of the work once a countermeasure arrives. One model estimates 19.8 million deaths averted in the first year of COVID-19 vaccination (95% CrI 19.1–20.4), 79 per cent through direct protection; a reported-deaths baseline gives more than 14 million [e6]. The figure is in deaths, not DALYs, and bears mainly on *u* and on *G_nat*, credited once [e6; ACC].

Recognition and residual. In the one dated case, recognition can be placed at the sequence release from which the 66 days are counted [e5]; no general distribution of *t_R* was located. Complete closure at *t_D* is an assumption (A23), not a finding [e4, e7]. One framework argues that the cyber structure may not transfer because a software patch can fully resolve a vulnerability within weeks, while some biological threats still lack any countermeasure; it is conceptual and carries no data [e4]. One model gives a criterion for control through isolation alone under its own idealised assumptions [e7]. That constrains one intervention in one model. It is not a floor on defence, and nothing read establishes one [e7].

The capability distance. No measure of *K_D* was located in the documented search, and no dated record placing a countermeasure's capability requirement against the corresponding misuse requirement (§11). The offence–defence literature does not close the gap. Its one direct study finds four of five asymmetries favour attackers, excludes non-pharmaceutical defences by construction, and states that it has done no quantitative analysis [e1, e2].

[16] We use §4a's renames: *Λ_M* for the speed-set misuse hazard (the model checks' *H_M*), *ε_Λ* for its elasticity, and *h_M^post* for the residual (the checks' *P*).

Formal theory makes the balance a function of investment level and leaves biology unmodelled [e3].

What follows for R3. Every form of the condition (§4a.5, Figure 10) needs K_D, and the speed-set forms need ε_Λ too; neither has a located measure (§11). Our own assessment, a judgment about the result rather than a sourced finding, is that R3 cannot be tested on the evidence this search located, under either specification.

8. Evidence IV: benefit

The benefit term needs $q = K_B/s$, the capability-set part of the wait; T_b, the adoption time; m_B, the share of the benefit's adoption that waits for capability (RTTR's m_B); p, the probability that an AI-originated candidate reaches patients; and y, the flow once it does.

Clinical durations: STRONG [g3]. An analysis of 406,038 trial entries, 2000–2015, gives median durations of 1.6, 2.9 and 3.8 years for Phases 1, 2 and 3, median time in the clinic of 5.9–7.2 years outside oncology and 13.1 in oncology, and an overall probability of success of 13.8 per cent; a separate estimate puts the clinical phase at about 95 months against 31 non-clinical [g3]. Both are pre-AI. On this row most of the wait is clinical, consistent with RTTR's reading that q is a minority of it; the row does not measure m_B [g3]. The familiar 17-year lag from research to practice carries its reviewer's warning that lag studies measure different things, MODERATE [g4].

Discovery-stage speed-ups: WEAK [g1]. One programme, reported by its developer, took about 18 months from target discovery to preclinical candidate and under 30 months to the end of phase 0/1 [g1]. A review's 30–70 per cent reductions in early-stage timelines trace to no checkable study [g1]. One programme is not a parameter, and no cross-drug estimate of AI's effect on total development time was located [g1]. Downstream clinical timelines are reported as comparable to conventional development [DIS 8]. The best-known external evidence for AI-driven acceleration was withdrawn by its own institution, and it concerned materials, not health [g8]. Nothing here shows AI shortening the time from target to approval [g1, g2].

Probability of success: MODERATE [g2, g5]. At abstract level, AI-discovered candidates show Phase I success of 80–90 per cent, above historic averages, and Phase II success of about 40 per cent on a small sample, comparable to them [g2]. The higher Phase I rate is consistent with a safety filter rather than a gain in efficacy [g2]. A developer-side author stated that none had been approved as of December 2024; we have not established the position since [g2]. Three sources from different directions suggest that choosing the right compound matters more than speed, which is why the model carries p as a level lever: expected benefit scales as $p{\cdot}y$, and p responds to capability on a margin other than K_B/s [g5].

Health gained [g7]. No DALY or population QALY figure for health gained from AI was located in the documented search. What has been measured is for narrow tools, small and modelled: an interquartile range of incremental QALY gains of 0.001–0.019 across 48 economic evaluations, from a review held at abstract level that flags systematic optimism; a second review finds most such studies rest on decision-analytic models rather than

observed outcomes [g7]. Scaling either to a population needs prevalence and uptake, which no source read supplies. These rows bound the smallness of what has been measured and give no value for y [g7].

The narrow-tool limit [g6]. Realised health-relevant output comes mainly from task-specific tools. General models are evidenced to in-vitro or vignette level; in one randomised trial, 50 physicians given a 2023 model as a diagnostic aid did not reason significantly better (adjusted difference 2 points, 95% CI −4 to 8) [g6]. No source read measures what share of realised benefit comes from general rather than narrow models [g6]. On this record, benefits shown by narrow tools are not evidence for general capability: no source read measures how much of them general models supply [g6]. For A16 the rows are consistent with RTTR's decoupling of benefit from general capability and do not establish it [g6].

9. Facility and ownership

One question to ask of a model developer's own laboratory is whether it is safer or less safe than a public one. The public argument our search located does not ask it; it is about legitimacy (§1) [a8, DIS 11]. No accident rate differentiating facilities by operator type appears in the documented search [c9], and two searches run separately found no comparison of safety outcomes across privately and publicly owned life-science facilities [f6].

The model poses a different question. A facility is an intervention on distances: I3, the developer-owned laboratory taken as reported, and I4, its public or academic equivalent, are defined in §2, and each may move K_D, the distance to the defensive capability, and K_M, the distance to the misuse capability. The question is how the changes a laboratory makes to K_D and K_M together move misuse loss at a fixed speed: a weighted comparison of two changes in distance, derived on its own at §4a.5, whose weights depend on the specification, the exposure reading and, under level-set, the convention (Figure 11). It is not what R3 signs, and, as §4a.5 shows, R3 cannot sign it in general — proved there for speed-set and shown by counterexample for level-set (§2; Figure 10; the one mechanical exception is footnoted at §4a.5). The comparison also holds one speed fixed for both distances, and that is a substantive restriction: a facility that moved the speed of the biology clock rather than a distance would bring R3 back into the question, and that case is not derived (§4a.5, §4a.8).

What the convention means. Under level-set, "the facility brings the misuse capability closer" can mean two things, and they are not the same intervention. Figure 26 shows what the two conventions do to the hazard path: under **convention A** the same path arrives at a lower threshold; under **convention B** the threshold moves while the hazard at each capability level stays as it was, so, with a rising hazard, a nearer threshold is also a less dangerous one. The paper carries the two co-equal, neither as base, as it carries the two hazard specifications; every level-set statement below appears under both, and on the ramp-on reading the sign flips between them. Nothing read chooses between them; the choice is a modelling decision, not a result.

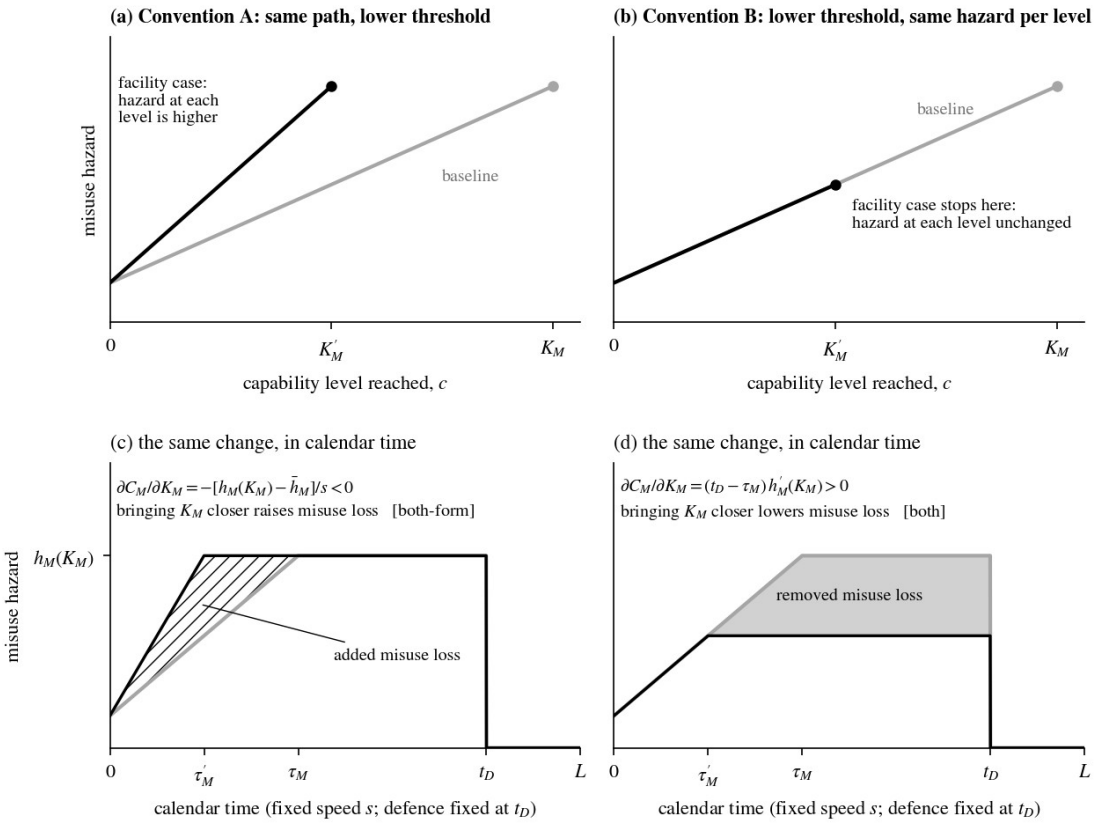


Figure 26. Two readings of "a facility brings misuse capability closer" (level-set specification, ramp-on exposure; fixed speed *s*; defence date *t_D* fixed). Under convention A (left), the facility moves the same hazard path to a lower threshold *K_M′*. The path is compressed, so the hazard at every capability level is higher (a). The plateau *h_M(K_M)* is reached sooner and held longer, and misuse loss rises (c, hatched): *∂C_M/∂K_M = −[h_M(K_M) − h̄_M]/s*. Under convention B (right), the hazard at each capability level is unchanged and only the threshold is lower, so the path stops at the lower hazard *h_M(K_M′)* (b). With a rising hazard the plateau is lower, and misuse loss falls (d, grey): *∂C_M/∂K_M = (t_D − τ_M)h_M′(K_M)*. The same verbal intervention therefore carries opposite signs. The paper's two model checks first reported these opposite signs, with no algebra error, because they had evaluated different conventions. The paper carries both conventions co-equally, and every level-set facility statement names its convention. On the arrival-on reading the convention-A derivative keeps its sign and the convention-B derivative is unsigned from monotonicity alone (§4a.5). Under the speed-set specification, the facility comparison is the same under either convention (Figure 11). Schematic; axes are not to scale; areas computed by integrating the model's loss for a linear level profile.

The defence side is the same in every case: with the residual hazard *h_M^post* fixed, shortening *K_D* lowers misuse loss exactly when *χ_D* = 1 and the plateau hazard *h_M^plat* exceeds *h_M^post*, does nothing on the adoption branch, and raises it if the residual

exceeds the plateau hazard; the formula and the full list of conditions it holds under are at §4a.5.

The misuse side, under level-set, under both conventions. On the ramp-on reading, Figure 26(c) and (d) show the two signs, and §4a.5 gives the derivatives. Under A, advancing K_M raises misuse loss whenever the terminal hazard exceeds the ramp average, so the two distances trade off: whatever a facility does to the defence distance must be weighed against what it does to the misuse distance. Which distance this laboratory moves, if either, is not known (§2). Under B, where the hazard is nondecreasing, bringing the misuse threshold closer weakly lowers ramp-on misuse loss, because the plateau it installs is a lower one; where defence is also effective, $h_M\text{^}post \le h_M(K_M)$, neither cut raises loss and there is no trade-off at all. On the arrival-on reading A's form keeps its sign and B's is unsigned from monotonicity alone (§4a.5).

Under speed-set the facility condition is cleaner: on the ramp-on reading K_M carries **zero** weight, and on the arrival-on reading the comparison is weighted and reduces to the bare comparison of the two shortenings only at $h_M\text{^}post = 0$ and $\chi_D = 1$; §4a.5 gives the conditions and the finite-move result, and Figure 11(a) draws the weights in every case.

The evidence for either distance is thin. The company describes the laboratory's purpose as fundamental biology, not drug discovery specifically, and says it runs no clinical work for now; no independent record of what it does was located [a2, a3]. We take no position on which distance it moves.

Oversight, at the precision the evidence supports. US law reaches a laboratory through two separate triggers: possession of a listed agent, and receipt of federal research funding [c4]. A privately funded laboratory at modest containment holding no listed agent falls outside the two research-oversight regimes those triggers define. That is not a claim that no regulatory floor applies: general occupational-safety and recombinant-DNA guidance remains, which the source describes as lacking enforceable penalties [c4]. Whether any particular facility falls inside or outside turns on facts about funding source and holdings that we could not establish [c4, c5]. Canada licenses the activity regardless of funding source [c5]. The paper makes no present-tense claim about the status of synthesis screening; the texts we hold leave it unresolved [c10].

The whole balance. Formally, I3 raises the balance if and only if the complete difference $J_I3 - J_0$ is positive. That difference sums the change in the benefit product, including the withholding loss ω; the change in the facility term, taken pathway by pathway with each consequence counted once; and any change in misuse loss, whether through the two distances of §4a.5 or through the automation route of §2 [b8]. The distance condition of §4a.5 signs the misuse change alone, and a condition in the facility terms alone holds only once every other difference is set to zero. The comparison of I3 against I4 remains open (§4a.8). The economics of ownership gives the question its shape without supplying an answer: the case for in-house provision strengthens as the adverse consequences of non-contractible cost cutting on non-contractible quality grow, and a public operator can be kept replaceable where a private one can invest in being costly to replace [f1]. Those are two answerable questions — how much of biosafety is non-contractible, and how

replaceable the operator is — rather than a ranking. Public ownership does not remove insider risk [f2].

10. Levers, conditionally

Two further interventions can be read through the defence clock, R3, under each specification. Neither is ranked against the other or against speed; the model supplies no common resource variable, and the quantities a ranking needs are unmeasured (§11).

I5, the countermeasure programme. It moves T_D, the countermeasure deployment time, m_D, the share of deployment that waits for capability, and K_D. Two things are true of K_D at once, and they are easily confused. At a fixed speed, shortening K_D lowers misuse loss whenever $\chi_D = 1$ and the pre-deployment hazard exceeds the residual (§9) — that is what the programme is for. But under level-set with the residual fixed, a *longer* K_D makes acceleration more likely to lower misuse loss, since $\partial(sC_{M'})/\partial K_D = -[h_M(K_M) - h_M^{post}]/s < 0$ on the capability branch (§4a.5). Under level-set, then, a programme that shortens K_D lowers misuse loss at a given speed and weakens the case that speed helps; under speed-set, ramp-on, the second effect runs through the share α_D (below). There is no contradiction between the two; they are derivatives of different arguments.

Figure 11(b) draws the point past which shortening the countermeasure distance stops helping. A large enough cut takes the defence out of the regime in which the condition was derived: the clock becomes adoption-bound, χ_D falls to zero, and further capability buys nothing. In the numerical check's test of the unweighted rule (§4a.5), all 46 sign disagreements with that rule, across speed-set arrival-on and the two tested level-set convention-A arrival-on cases at zero residual, followed a move onto the adoption branch [NUMCHECK]. The check's separate finite-move test also shows that linear predictions can fail when a move leaves its regime.

What the programme would have to do differs by specification. Under speed-set on the ramp-on reading, the condition is an elasticity against a window share (§4a.5), so I5 acts by raising α_D — a larger capability-set share of the deployment wait — or by lowering the residual. On the capability branch $\alpha_D = \chi_D K_D/(s \cdot t_D)$, so K_D still matters, but it enters through a share rather than as a distance raced against K_M. Under the arrival-on reading the two-distance form returns, and at zero residual the condition requires $K_D > K_M$. A shorter T_D weakly lowers misuse loss and raises the natural-disease benefit only given a specified monotone effect of deployment on the coverage schedule, which dated milestones alone do not supply [e5]. The dates we have are for one programme, one platform, one disease, and they identify timings rather than a branch [e5].

I6, oversight and interface screening. It moves the facility event rate or the misuse hazard [c4, c5, c10]. Its effect on the distances is not derived. What the structure does say is where it could enter: closing the funding-source gap changes which facilities a regime reaches [c4, c5], and screening at the digital-to-physical interface is a distinct instrument from regulating a facility [c10]. Under the intervention mapping we assume (§4a.1), neither moves K_D, so neither changes the defence-clock condition through that channel; both would act on the facility term or on the hazard level, and the magnitude of either is

unmeasured. The international layer imposes purpose-based prohibitions and domestic prevention duties and contains no facility-level verification system; that is a statement about the text, and nothing read measures what those duties achieve [c6].

The levers come out as conditions, each with a named deciding quantity and each unmeasured. The paper does not order them.

11. Limits

This paper rests on a documented search of seven literatures, run on 26 October 2026: 173 source records and 162 distinct works. A work read in two strands of the search counts twice as a record and once as a work, and the *International AI Safety Report*, recorded under two keys, counts once. Of the records, 22 were read in full, 113 in part, 17 at abstract only, and 21 were not obtained. A work we could not obtain appears only as an item we tried and failed to obtain, never as evidence. Findings resting on abstracts or secondary accounts are labelled so.

What we excluded ourselves. On safety grounds we did not collect, read for, or record technical detail about how biological harm could be caused, and on the cyber side we excluded exploit detail. The paper can therefore say what a study measured, how well, who funded it and what the field concludes, and it cannot say which agents or methods any capability bears on. No part of the risk term rests on such material. One source was kept as a safe excerpt only — its design, its authors' stated limits, and its governance proposals. The one controlled comparison was read the same way: design, sample, scoring rubric and statistics were taken, and its model-output section was never opened. What we excluded was operational detail, not risk estimates.

What a safety classifier stopped. Four steps were stopped. One, summarising our reviewers' methodological criticism of how incident, containment and evaluation statistics had been used, was stopped four times across two models, including after rephrasing. We did not re-derive it. The reports stayed on disk and were applied by reference to their numbered findings. A second step was stopped once and resumed at the structural level only. A third step, drafting the misuse-evidence section, was stopped once when the statistics-only extraction from the controlled comparison was held in the same turn as our summary of the misuse evidence. That section was then drafted from our summary alone, and separately from §6–§8. The division is ours and deliberate, and it cost the paper no content. A fourth step was stopped once while planning this section and §9, in a turn that read across the misuse and the facility material at once; no draft text existed yet, the files stayed on disk, and both sections were then written at the level of governance, oversight, study design and model structure only.

Ordinary gaps. The published text of the one causal hospital-mortality study was closed to us, so its abstract and its working paper are reported separately and never spliced [d1]. The body of the one controlled comparison was obtained and read for its statistics alone; it prints no confidence interval, standard error or inter-rater agreement statistic, which is why the interval carried beside it in §5 is our own arithmetic reconstruction and is WEAK as an interval [b2, b2a]. Incident updates from one national health service were blocked to

automated fetching. A laboratory-safety manual and the current US synthesis-screening rules could not be confirmed from primary documents [c10]. Two benefit studies were paywalled, and the 2026 edition of the international safety report was not read, so the absence recorded at [b1] speaks only to the editions and sources in the search logs.

What no source read supplies. The health gain from AI in population units [g7]; the health cost per attack in those units [d4]; the risk per unit of capability [b1]; a company-against-public facility safety comparison [c9, f6]; and whether misuse risk scales with the capability level reached or with speed [b12]. These are statements about our search, not about the world. They head the estimand list below, and they are the reason the paper's results are structural rather than calibrated. Throughout, "unmeasured" means that no measure was located in the documented search.

Estimands. Figure 27 maps the main ones to what each would settle. The results need these quantities, and the documented search supplies none of them in the paper's units: the health flow y [g7]; the success probability p [g2]; the capability-set wait q [g1]; the adoption time T_b and the share m_B that waits for capability [g3, g4]; the withholding loss ω [a9]; the baseline care loss per attack B_e [d1–d4]; the foregone-benefit terms Φ_e, δ_T and A_Φ [d7]; the attack rate [d9]; the misuse hazard and its distance, h_M and K_M [b1, b10]; the loss per misuse event V_M [b1, e8]; the defensive schedule t_R, u, $h_M\text{\^{}}post$, T_D and m_D [e5]; the defensive distance K_D, for which no measure was located; the facility terms λ_F, n_F and V_O [c9, c1, c2]; countermeasure gains against natural outbreaks G_nat [e6]; the hazard elasticity ε_Λ; and whether misuse risk scales with the capability level reached or with speed [b1, b12]. The last two, with the level profile, the residual hazard and K_D, decide the rows of §4b.1. The observations that would move them are a sensitive bound on misuse-relevant assistance, from a design much larger than the one controlled comparison [b2]; a within-programme estimate of development time under differing availability of general capability [g1, g6]; a DALY translation of the hospital result [d4]; and, for R3, a dated record placing a countermeasure's capability requirement against the corresponding misuse requirement. One quantity the results turn on is not a search result at all: the horizon L is a choice, and a consequential one. Truncation exists only where $sL < K_M$, and on either speed-set reading the coupled benefit test can change sign with L (§4a.3). Nothing in the documented search fixes a horizon for a health balance of this kind, and the results are stated for a horizon taken as given.

Tools. The drafting, the literature search and the model checks were run with large language models; the arrangement is stated at the end of the paper.

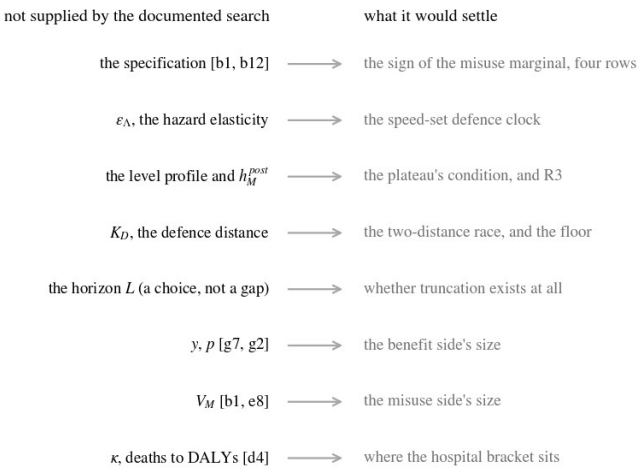


Figure 27. What would settle what. Eight of the quantities §11 lists as unsupplied by the documented search, against the part of the argument each would decide. The horizon is marked as what it is: a choice the paper makes, not a gap a longer search would close. The map states the paper's own dependencies; it supplies none of the quantities.

12. Conclusion

Four results survive every choice the paper refuses to make, each within its stated restrictions. R3 cannot sign the facility comparison in general, and the facility comparison cannot sign R3 (§4a.5) — proved for speed-set, shown by counterexample for level-set. Once the misuse capability has been reached, faster progress raises misuse loss under each specification's own hazard restriction. On the adoption branch both specifications agree that speed hurts. On the benefit side, discounting favours timing over the probability channel, and the undiscounted timing margin for a fixed-duration benefit counted in full, with no binding cutoff, is zero. A fifth survives in the same form, and the sameness is structural rather than three independent confirmations. Under both specifications, both exposure readings and both conventions, shortening the defence distance at a fixed speed lowers misuse loss exactly while the countermeasure still waits on capability and the residual hazard lies below the plateau hazard — $\partial C_M/\partial K_D = (\chi_D/s)(h_M^{plat} - h_M^{post})$ — and buys nothing once the clock becomes adoption-bound. The invariance is of that form and of the sign condition, not of the magnitude, which carries each case's own plateau hazard, and it holds because K_D enters the loss only at the deployment date while each of the three choices governs another object: in the same subsection the K_M derivative, which the conventions do govern, flips sign between them. The derivative is elementary, a shift in time; what is this paper's is the division of the wait into capability-set and adoption-set parts, the adoption floor that follows, and the conditions the invariance needs: a deployment date interior to the counted window ($\tau_M < t_D < L$), a fixed residual, and, under gradual coverage, a schedule that translates rigidly and completes before the horizon (A4a.7(i)).

The rest is conditional on a choice. Under the level-set specification there are three mechanism results: the plateau, which is RTTR's own algebra carried to the misuse term; truncation; and the defence clock, whose two-distance form follows by substitution into the general level-set derivative. Under the speed-set specification on the ramp-on reading, the plateau holds with no condition on the hazard's level profile, though not with a larger derivative, truncation has no counterpart at all, and the defence clock survives as an elasticity-against-share condition in which K_M drops out. The arrival-on reading of the speed-set specification yields a plateau result, a kink at activation, a two-distance defence condition and a coupled benefit test. Each of these is proved in the appendix. The rest of the facility comparison is conditional: the speed-set condition under both exposure readings, and the level-set condition under each of the two co-equal conventions, with the sign flipping between them on the ramp-on reading when the terminal hazard exceeds its ramp average and its endpoint slope is positive; on the arrival-on reading convention A's form keeps its sign and convention B's is unsigned in general (§4a.5, §9).

Truncation is the paper's most novel result, and it rests on two of the paper's conventions rather than one: it exists under only one of the two co-equal specifications, and only under an objective that stops counting at a finite horizon. That there is no speed-set counterpart is a statement about mechanism rather than a sign result: A4a.4(iii) proves the speed-set derivatives in that regime, and under ramp-on misuse loss rises with speed there too. The second dependence is the regime's own condition: $sL < K_M$ restricts the horizon and the speed jointly, and without a finite L every positive speed reaches the capability eventually, so the regime is empty (§4a.4). We state this as a limit of the result, not as a reason to prefer the specification that produces it.

What is not derived is larger. No calibrated sign of the whole balance follows, anywhere. No ordering of the interventions follows. No decision rule follows either, and §4a.7 declines one rather than leaving a reader to supply it: any such rule would need a fixed referent for the worse reading, which the sign flips deny it, and bounds on magnitudes the documented search did not supply. Nor does anything here settle a severity channel: the loss per misuse event is held independent of the capability level reached (A20), so under level-set the capability indexes how often events occur and not how bad they are, and V_M varying with the level reached is listed open (A-open). Nor is the measured part of the balance a guide to its sign: the one channel with a measured causal magnitude, attacks on hospitals, comes to the order of one excess in-hospital death per attack on a rate we grade WEAK, while for the misuse channel the paper is about no estimate of any kind was located, so a reassuring reading would be an artefact of which channel can be measured (§1.1). The parameter-level construction over the unmeasured quantities is set out in §4b.2. One part is computed, a bound on the hospital channel graded WEAK that leaves the non-identification claim unchanged, and the rest is not done. On the ramp-on exposure reading, two specifications, each consistent with the documented evidence, give conditional opposite signs for the misuse marginal in four of the seven cases of Table 4b.A, each under the restrictions its row states. On the arrival-on reading, with the level-set side gated so that the comparison holds the reading fixed (§4a.2), **only one of the four survives**, and it is the two-distance row. Rows 1 and 4 cannot diverge there, because a gated profile would need a negative hazard for them to, and row 3 is defined on ramp-on exposure with no arrival-on version. A

seventh row compares thresholds on a coupled health benefit rather than signs; under the comparison convention §4b.1 states, its two tests are nested, and it is an instance of the whole-balance interval rule below rather than a further non-identification. With V_M fixed and the non-misuse remainder Υ assumed common to both specifications, the whole balance ranks speed oppositely if and only if Υ lies strictly between $V_M\Xi_\ell$ and $V_M\Xi_s$, both taken on the same exposure reading.

What turns on the specification is therefore the sign of the misuse marginal in those four cases, and the existence of truncation. What would settle it is a measurement: whether misuse risk scales with the capability level reached or with the speed of approach, and then each row's own deciding quantity, which is the hazard elasticity ε_Λ under speed-set and the level profile, the residual hazard or the two distances under level-set (§4b.1). None has been measured in anything we read [b1, b12]. Until one of them is, the choice between the specifications decides the sign of the misuse term in those cases. Whether it also decides the direction of policy turns on where the rest of the balance lies, and that is unmeasured too. What the paper adds is the list of quantities that would settle each case.

Acknowledgements

The literature search, the drafting and the model checks were run with large language models under a division of labour set out here, so that a reader can see what was machine-produced and how it was checked.

Seven search tasks were run separately — six literatures, and one verification of the factual record the paper's case rests on — each producing a source table with reading status, study design, funder, stated uncertainty and the main limitation of every core source. A documented search log records database, query, date, hits screened and what was kept, and the paper writes "not located in this search" rather than "none exist" wherever a search returned nothing. Every evidential claim carries a key to the source-table row behind it, and statements that are our own judgment are labelled as such. Keys [DIS n] point to the project's register of unresolved disagreements between sources, and [ACC] to its accounting note; both are available with the source table.

Each mathematical result was first put to two models working blind and separately, in three rounds: one on the level-set specification, one on the speed-set specification, and one on the facility comparison at a fixed speed. A fourth pair, run the same way, computed the hospital bound of §4b.2. Agreement between two models working separately is not a warrant, since nothing establishes that their errors are uncorrelated, and no result in the paper rests on it. Each model result is instead proved in the appendix, which lists in its closing section what it does not prove and reports beside each proof the worst error of any numerical check of it against the loss computed directly from the model; the supplement records which proofs have also had a line-by-line check, and how the results were first derived. Two further blind reads, one of the prose and claims and one of the mathematics, were run on the sections as drafted, before assembly, and their findings were applied section by section. A third blind read was then run on the assembled paper: an outside reading of the whole text, a contrarian attempt to break it, a check of the figures against the code that draws them, a re-derivation of the mathematics run separately, and a separate

fairness reading on a model the company discussed did not build. Most of its findings were applied, a few were declined, and a number of decisions were left to the author.

A safety policy was in force throughout: no technical or operational detail about how biological harm could be caused was collected, read for or recorded, whatever its source, and the same rule applied to cyber exploit detail. Section 11 records where a safety classifier stopped a step and what that cost the paper. The author is responsible for every claim.

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Appendix

This appendix proves the results of §4a and §4b. Each proof starts from the set-up in A0 and uses only the lemmas stated there and the assumptions it lists. Where a result has also been checked numerically, one clause says so and gives the worst error. A numerical check is evidence of a different kind from a proof, because it integrates the model's definition directly and shares no step with the algebra, so it sits beside the proof and never replaces it. Twenty-three results — the project's inventory of sixteen and the seven rows added after it, which together are not the whole of this appendix — carry a third warrant: they are formalised in a proof assistant, which checks the deduction itself rather than sampling it. The Supplement's S.4 names the artefact, lists the results that carry no formal warrant, and states what the checking does and does not certify. A-open, at the end, lists what is not proved. The project's record of how each result was first derived is in the Supplement.

A0. Set-up and lemmas

Set-up. Horizon L. One speed $s > 0$ converts every capability distance into a date. Misuse capability K_M is reached at $\tau_M = K_M/s$. A countermeasure is deployed at $t_D = \max\{K_D/s, (1 - m_D)T_D\} + m_D T_D$. Away from the clock kink this gives $s \cdot t_D' = -\chi_D \tau_D$, with $\tau_D = K_D/s$, and $\partial t_D/\partial K_D = \chi_D/s$. t_D does not depend on K_M unless the clock is paced by K_M ($K_D = K_M$). The misuse hazard is either **level-set**, $h_M(c(t))$ with $c(t) = \min(st, K_M)$, or **speed-set**, $\Lambda_M(s)$, with $\varepsilon_\Lambda = s\Lambda_M'/\Lambda_M$. Exposure is either **ramp-on** (the hazard runs from date 0) or **arrival-on** (it runs only from τ_M). After t_D the hazard steps to a residual h_M^{post}, which is constant in calendar time. The residual is fixed unless a result says it responds to speed, $h_M^{post}(s)$. Misuse loss is $C_M = \int_0^L$ (hazard) dt. All results are local derivatives away from kinks and undiscounted unless stated, with $V_M > 0$ fixed. K_M, K_D, L, m_D and T_D are held fixed when s moves.

Three further statements complete the set-up.

- **Level-set arrival-on** means that the ramp carries no hazard: exposure starts at τ_M, at the plateau hazard. This is the reading that the arrival-on formula $C_M = (t_D - \tau_M)h_M^{plat} + (L - t_D)h_M^{post}$ encodes.
- **Conventions.** Under **A**, $h_M(c; K_M) = \hat{h}_M(c/K_M)$. Substituting $x = c/K_M$ shows that the plateau hazard $h_M^{plat} = \hat{h}_M(1)$ and the ramp average $^*\bar{h}_M = \int_0^1 \hat{h}_M{}^*$ do not depend on K_M, and under A "$h_M(K_M)$" means $\hat{h}_M(1)$. Under **B**, $h_M(c)$ is a fixed function, so $h_M^{plat} = h_M(K_M)$ and $dh_M^{plat}/dK_M = h_M'(K_M)$ wherever h_M is differentiable at K_M.
- **Defence before arrival, speed-set arrival-on** ($t_D < \tau_M$). We assume, as a modelling choice and not as a result, that exposure still starts at τ_M and that it starts at the residual because the defence is already in place. The hazard is therefore 0 on $[0, \tau_M)$ and h_M^{post} on $[\tau_M, L]$. The one-sided limit in A4a.4(iv) and the responsive-residual term in that case rest on this choice. Under level-set arrival-on the same case is left undefined, and no result is claimed for it (A-open).

Leibniz's rule for an integral with moving limits is *d/ds ∫_{u(s)}^{v(s)} f(t, s)dt = f(v, s)v′ − f(u, s)u′ + ∫_u^v ∂_sf dt. *Below, a* boundary term* is one of the first two terms and the *interior term* is the third.

Lemma 0 (speed-set step loss). Let exposure start at t_0, with $t_0 = \tau_M$ under arrival-on and $t_0 = 0$ under ramp-on, and suppose $t_0 \le t_D \le L$ on a neighbourhood of s. Then $C_M = \int_{t_0}^{t_D} \Lambda_M\, dt + \int_{t_D}^{L} h_M^{post}\, dt$, and

arrival-on: $sC_M' = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M] + s \cdot t_D'(\Lambda_M - h_M^{post}) + (L - t_D)s \cdot h_M^{post}{}'$, (0.1) ramp-on: $sC_M' = \Lambda_M t_D \varepsilon_\Lambda + s \cdot t_D'(\Lambda_M - h_M^{post}) + (L - t_D)s \cdot h_M^{post}{}'$. (0.2)

Proof. Apply Leibniz's rule to each piece. For the first piece, the interior term is $\Lambda_M'(t_D - t_0)$ and the boundary terms are $+\Lambda_M t_D'$ and $-\Lambda_M t_0'$. For the second piece, the upper limit is fixed, the boundary term is $-h_M^{post} t_D'$, and the interior term is $(L - t_D)h_M^{post}{}'$.

Sum the terms, multiply by *s*, and use $s\Lambda_M' = \varepsilon_{\Lambda\Lambda_M}$, together with $s\cdot t_0' = -K_M/s = -\tau_M$ under arrival-on and $s\cdot t_0' = 0$ under ramp-on. ■

Assumptions. Λ_M, t_D and h_M^{post} are differentiable at *s*, and the ordering holds on a neighbourhood of *s*, so the point is away from $t_D = \tau_M$ under arrival-on and away from $t_D = L$. No sign is claimed. With no closure inside the horizon, set $t_D \equiv L$; the second piece is then empty. The term $s\cdot t_D'(\Lambda_M - h_M^{post})$ is the jump in the integrand across the moving boundary at t_D, times the speed of that boundary. The last term is Leibniz's interior term, not a boundary term.

Lemma L (level-set, the speed derivative). Under level-set, on the ungated path (ramp-on), with $\tau_M \le t_D \le L$,

$$C_M = \int_0^{\tau_M} h_M(st)\,dt + (t_D - \tau_M)h_M(K_M) + \int_{t_D}^{L} h_M^{post}\,dt,$$

$$sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] + s\cdot t_D'[h_M(K_M) - h_M^{post}] + (L - t_D)s\cdot h_M^{post\prime}. \quad \text{(L.1)}$$

Proof. Take the three pieces in turn. (i) Changing variables to $c = st$ gives $\int_0^{\tau_M} h_M(st)\,dt = s^{-1}\int_0^{K_M} h_M(c)\,dc = \tau_M\bar{h}_M$, *and* s times its s-derivative is $-\tau_M\bar{h}_M$. *(ii)* $s\cdot d/ds[(t_D - \tau_M)h_M(K_M)] = (s\cdot t_D' + \tau_M)h_M(K_M)$, *because* $s\cdot\tau_M' = -\tau_M$. *(iii) The third piece contributes* $-s\cdot t_D' h_M^{post} + (L - t_D)s\cdot h_M^{post\prime}$, as in Lemma 0. Sum the three. ■

Assumptions. h_M is integrable on $[0, K_M]$; step (i) needs nothing more. t_D and h_M^{post} are differentiable at *s*, and $t_D < L$ holds on a neighbourhood of *s*. No monotonicity is used. With no closure, $t_D \equiv L$, and the derivative is $\tau_M[h_M(K_M) - \bar{h}_M]$.

Lemma 1 (level-set, the distance derivatives, step after arrival). Fix *s*, and let $\tau_M < t_D < L$.

arrival-on: $C_M = \int_{K_M/s}^{t_D} h_M^{plat}(K_M)\,dt + \int_{t_D}^{L} h_M^{post}\,dt$; (1.1) ramp-on: $C_M = \int_0^{K_M/s} h_M(st; K_M)\,dt + \int_{K_M/s}^{t_D} h_M^{plat}(K_M)\,dt + \int_{t_D}^{L} h_M^{post}\,dt.$ (1.2)

- **(a) The K_M derivative, arrival-on.** In (1.1), the lower limit K_M/s moves at rate $1/s$, which gives the boundary term $-h_M^{plat}/s$. The integrand depends on K_M, which gives the interior term $(t_D - \tau_M)dh_M^{plat}/dK_M$. The upper limit t_D does not move with K_M. So

  $$\partial C_M/\partial K_M = -h_M^{plat}/s + (t_D - \tau_M)\cdot dh_M^{plat}/dK_M. \quad \text{(1.3)}$$

- **(b) The K_M derivative, ramp-on, convention B.** In (1.2) the first integrand $h_M(st)$ contains no K_M. Its upper limit moves at $1/s$, which gives $+h_M(K_M)/s$. The second integral gives the two terms of (1.3), with $h_M^{plat} = h_M(K_M)$. The two boundary terms cancel, because the path $c(t)$ is continuous at τ_M and h_M is continuous at K_M, so the integrand on each side of the moving boundary tends to $h_M(K_M)$. Continuity at K_M is implied by differentiability of h_M at K_M, which (1.4) needs anyway; with a jump at K_M the boundary terms differ by $[h_M(K_M^-) - h_M(K_M)]/s$. Hence

$\partial C_M/\partial K_M = (t_D - \tau_M)h_M'(K_M)$. (1.4)

- **(c) The *K_M* derivative, ramp-on, convention A.** Substituting $x = st/K_M$ in the first integral of (1.2) gives $*(K_M/s)\int_0^1 \hat{h}_M = \tau_M\bar{h}_M*$, so

  $C_M = \tau_M[\bar{h}_M - \hat{h}_M(1)] + t_D[\hat{h}_M(1) - h_M\text{^}post] + L \cdot h_M\text{^}post$. (1.5)

  Since $*\bar{h}_M*$ and $\hat{h}_M(1)$ are constant in K_M, $\partial C_M/\partial K_M = -[\hat{h}_M(1) - \bar{h}_M]/s$. This route needs no differentiability of $\hat{h}_M$. (Differentiating the ramp integral under the integral sign gives $-(1/s)\int_0^1 x\hat{h}_M'(x)dx$, after integration by parts, for $\hat{h}_M$ absolutely continuous.)

- **(d) The *K_D* derivative, both readings and both conventions.** K_D enters only through t_D, which is the boundary between the plateau and the residual. Leibniz's rule on the two adjacent integrals gives $\partial C_M/\partial t_D = h_M\text{^}plat - h_M\text{^}post$, so

  $\partial C_M/\partial K_D = (\chi_D/s)(h_M\text{^}plat - h_M\text{^}post)$. (1.6)

- **(e) Speed-set.** With $h_M\text{^}plat = \Lambda_M$ fixed by s, arrival-on $C_M = \Lambda_M(t_D - K_M/s) + h_M\text{^}post(L - t_D)$ gives $\partial C_M/\partial K_M = -\Lambda_M/s$. Ramp-on $C_M = \Lambda_Mt_D + h_M\text{^}post(L - t_D)$ gives $\partial C_M/\partial K_M = 0$. Both readings satisfy (1.6) with $h_M\text{^}plat = \Lambda_M$.

Assumptions. $\tau_M < t_D < L$. t_D does not depend on K_M. The residual is constant. The point is away from the clock kink. Under B, h_M is differentiable at K_M wherever $h_M'(K_M)$ appears.

Lemma 2 (level-set, step before arrival, ramp-on). If $t_D < \tau_M$, the path never reaches K_M before the step. For $t_D < L$,

$C_M = s^{-1}\int_0\text{^}\{st_D\}h_M(c; K_M)dc + (L - t_D)h_M\text{^}post$. (2.1)

If $t_D \geq L$, the upper limit is sL and the residual term vanishes. *Proof.* Under ramp-on the hazard runs from date 0 at the level reached, $h_M(st; K_M)$, and $t_D < \tau_M$ means $st < K_M$ for every $t < t_D$, so no part of the pre-step path is at or above K_M and the plateau is never entered before the step. Splitting $\int_0\text{^}L$ at t_D and changing variables to $c = st$ on the first piece, as in Lemma 0(i), gives the integral term; the second piece is the constant residual over $[t_D, L]$, which is $(L - t_D)h_M\text{^}post$. If $t_D \geq L$ the second piece is empty and the first runs to sL. ■

Lemma H (homogeneity). Under level-set convention A, on either reading and with a fixed residual, $C_M(\lambda s, \lambda K_M, \lambda K_D) = C_M(s, K_M, K_D)$ for every $\lambda > 0$. Hence, wherever C_M is differentiable,

$s\partial C_M/\partial s + K_M\partial C_M/\partial K_M + K_D\partial C_M/\partial K_D = 0.$ (H.1)

Proof. Scaling s, K_M and K_D by λ leaves τ_M and K_D/s unchanged, and so leaves t_D unchanged on either branch of the clock. The hazard at date t before the step is $\hat{h}_M(\min(st, K_M)/K_M) = \hat{h}_M(\min(t/\tau_M, 1))$, which is unchanged. The arrival gate at τ_M and the residual are unchanged too. So the integrand is the same at every date, and so is the loss.

Differentiating in λ at $\lambda = 1$ gives (H.1), which is Euler's theorem for a function homogeneous of degree zero. ■

Under speed-set, write $C_M = F(\Lambda_M(s); s, K_M, K_D)$, where $F(\Lambda; \cdot)$ is the loss with the intensity held at the number Λ. By the same argument, F is homogeneous of degree zero in (s, K_M, K_D). Also $\partial F/\partial \Lambda = \ell_M$, the exposure length at full intensity inside the horizon, which is t_D under ramp-on and $t_D - \tau_M$ under arrival-on (for $\tau_M \le t_D \le L$; it is 0 when the defence comes first; with the step at or beyond the horizon, t_D is replaced by L, for $\tau_M \le L$). Because Λ_M does not depend on the distances,

$$s\partial C_M/\partial s = \Lambda_M \varepsilon_\Lambda \ell_M - K_M \partial C_M/\partial K_M - K_D \partial C_M/\partial K_D. \quad \text{(H.2)}$$

A4a. Proofs for §4a

A4a.0 The cyber term inside a common horizon. P16's marginal $V_{cH}(\varepsilon_{HT} - \chi_c \tau)$ needs $L \ge \sup_s T(s)$, every counted step loss inside L, and A2's fixed step. If $T > L$ is possible, the exposure inside the horizon is $C_{c,L} = H \cdot \min(T, L) + S_L$. *Proof.* Away from $T = L$, $d/ds[H \cdot \min(T, L)] = H' \min(T, L) + H \cdot 1\{T < L\} T'$. Multiply by s, and use $sH' = \varepsilon_{HH}$ and P16's clock, $s \cdot T' = -\chi_c \tau$, which is what produces P16's marginal. This gives $sC_{c,L}' = H[\varepsilon_H \cdot \min(T, L) - 1\{T < L\} \chi_c \tau] + sS_L'$. ■ At $T = L$, $\min(T, L)$ has a kink.

A4a.1 Level-set plateau, truncation, and closure at arrival.

(i) R2, level-set. With no closure and $\tau_M < L$, Lemma L with $t_D \equiv L$ gives $sC_M' = \tau_M[h_M(K_M) - \bar{h}_M]$. *Sign.* For nondecreasing h_M, $h_M(c) \le h_M(K_M)$ on $[0, K_M]$, so $*\bar{h}_M \le h_M(K_M)$, *with equality iff* $h_M = h_M(K_M)^*$ almost everywhere on $[0, K_M]$. For a nondecreasing function, that forces h_M to be constant on $(0, K_M]$: if $h_M(c_0) < h_M(K_M)$ for some $c_0 > 0$, the inequality holds on all of $[0, c_0]$. Adding continuity at 0 forces h_M constant on $[0, K_M]$. So a continuous, nondecreasing, nonconstant h_M gives $sC_M' > 0$. Without continuity, $sC_M' \ge 0$, strictly iff $*h_M(K_M) > \bar{h}_M$. *The derivative is zero exactly when* $h_M(K_M) = \bar{h}_M^*$, which a non-monotone profile can also meet.

(ii) R1, truncation. For $sL < K_M$, $*C_M = \int_0^{Lh_M(st)dt} = s^{-1}\int_0^{\{sL\}} h_M(c)dc = L\bar{h}_M(0, sL)$. Proof.* Differentiate: $C_M' = -s^{-2}\int_0^{\{sL\}} h_M + s^{-1} L h_M(sL)$. So $sC_M' = Lh_M(sL) - C_M = L[h_M(sL) - \bar{h}_M(0, sL)]$. ■ This uses continuity of h_M at sL; otherwise the derivative is one-sided. The formula uses no monotonicity. For nondecreasing h_M, $h_M(sL) \ge h_M$ on $[0, sL]$, so the derivative is ≥ 0. As $s \uparrow K_M/L$, $\tau_M \to L$ and the expression tends to $L[h_M(K_M^-) - \bar{h}_M]$. It therefore meets (i) at $s = K_M/L$ exactly when h_M is continuous at K_M from the left. *Worked instance:* $h_M(c) = 0.1 + c^2$, $K_M = 4$, $L = 10$, $s = 0.3$, so $sL = 3 < 4$. $C_M = 0.1L + s^2L^3/3 = 1 + 30 = 31$, and $sC_M' = 2s^2L^3/3 = 60$. The formula gives $10[9.1 - 3.1] = 60$.

(iii) Closure at arrival. Complete closure at τ_M is Lemma L on the K_M-paced clock with $\chi_D = 1$, $m_{DT_D} = 0$ and $h_M^{post} = 0$, so that $s \cdot t_D' = -\tau_M$. This gives $*sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] - \tau_M h_M(K_M) = -\tau_M \bar{h}_M$. *Directly,* $C_M = \tau_M \bar{h}_M^*$. This is the level-set entry of row 3 of Table 4b.A.

A4a.2 Level-set, discounted. With $\tau_M < L$, $C_{M,r} = s^{-1}\int_0^{\{K_M\}} e^{\{-rc/s\}} h_M(c)dc + h_M(K_M)(e^{\{-r\tau_M\}} - e^{\{-rL\}})/r$. *Proof.* s times the s-derivative of the first term is $-s^{-1}\int_0^{\{K_M\}} e^{\{-rc/s\}}(1 - rc/s) h_M(c)dc = -\tau_M \bar{h}\{M,r,D\}$, *where* $\bar{h}\{M,r,D\} = K_M^{-1}\int_0^{\{K_M\}}$

$e^{-rc/s}(1 - rc/s)h_M(c)dc$. s times the s-derivative of the tail is $h_M(K_M)e^{-r\tau_M}\tau_M$. So $sC_{M,r}' = \tau_M[h_M(K_M)e^{-r\tau_M} - \bar{h}_{M,r,D}]$. ■ Because $e^{-rc/s}(1 - rc/s) = d/dc[c\cdot e^{-rc/s}]$, a constant hazard h_0 gives $\bar{h}_{M,r,D} = h_0 e^{-r\tau_M}$ and a zero derivative. For a differentiable hazard, integrating by parts gives $sC_{M,r}' = s^{-1}\int_0^{K_M} c\cdot e^{-rc/s}h_M'(c)dc = \int_0^{\tau_M} e^{-rt}st\cdot h_M'(st)dt$. The same computation below the kink gives the upper limit L, so the integral form holds with $\min(L, \tau_M)$ and is nonnegative for a nondecreasing hazard. That integral form is for a differentiable hazard; for one merely nondecreasing the conclusion follows from the Stieltjes form $\int_0^{\min(L, \tau_M)} st\cdot e^{-rt}\, dh_M(st) \geq 0$, whose integrator is a nonnegative measure. The undiscounted bound $\bar{h}_M \leq h_M(K_M)$ needs neither, only monotonicity and continuity (A4a.1(i)).

A4a.3 Level-set defence clock. *R3, level-set*, with the residual fixed, is (L.1) without its last term: $sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] + s\cdot t_D'[h_M(K_M) - h_M^{post}]$, for $\tau_M \leq t_D < L$. On the mixed clock $s\cdot t_D' = -\chi_D\tau_D$. The branches are as follows.

- *Adoption branch* ($\chi_D = 0$): $t_D' = 0$, which gives A4a.1(i).
- *Paced by K_M* ($\tau_D = \tau_M$): $sC_M' = \tau_M[(1 - \chi_D)h_M(K_M) + \chi_D h_M^{post} - \bar{h}_M]$.
- *Own distance, capability branch* ($\chi_D = 1$): $s\cdot t_D' = -K_D/s$. So $sC_M' < 0$ iff $(K_D/s)[h_M(K_M) - h_M^{post}] > \tau_M[h_M(K_M) - \bar{h}_M]$. Multiplying by s gives the two-distance inequality $K_D[h_M(K_M) - h_M^{post}] > K_M[h_M(K_M) - \bar{h}_M]$. On this branch, $sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] - (K_D/s)[h_M(K_M) - h_M^{post}]$, so $\partial(sC_M')/\partial K_D = -[h_M(K_M) - h_M^{post}]/s$.

Responsive residual. (L.1)'s last term, $(L - t_D)s\cdot h_M^{post}'$, is added to the derivative. In the two-distance inequality, the same substitution followed by multiplication by s gives

$$K_D[h_M(K_M) - h_M^{post}] > K_M[h_M(K_M) - \bar{h}_M] + (L - t_D)s^2 h_M^{post}'.$$

No division by a hazard gap occurs, so $h_M(K_M) > h_M^{post}$ is not needed here. *Sign.* If $h_M^{post}' > 0$, the right side grows, so acceleration is less likely to help, and conversely. *Assumptions.* $\tau_M \leq t_D < L$, $\chi_D = 1$, $h_M^{post}(s)$ differentiable and constant in t, h_M integrable on $[0, K_M]$, and K_M, K_D fixed. *Worked instance.* $h_M(c) = 0.01 + 0.0025c$ on $[0, 4]$, so $h_M(K_M) = 0.02$ and $\bar{h}_M = 0.015$. *Take* $K_M^* = 4$, $K_D = 6$, $m_{DT_D} = 2$, $L = 10$ and $s = 1$, so $t_D = 8$, with $h_M^{post}(s) = 0.005 + \kappa(s - 1)$. The left side is $6\cdot 0.015 = 0.09$. - At $\kappa = 0.01$, the right side is $4\cdot 0.005 + 2\cdot 0.01 = 0.04 < 0.09$, so acceleration helps. Directly, $sC_M' = 0.02 - 0.09 + 0.02 = -0.05$. - At $\kappa = 0.04$, the right side is $0.02 + 0.08 = 0.10 > 0.09$, so acceleration hurts. Directly, $sC_M' = 0.02 - 0.09 + 0.08 = +0.01$.

A4a.4 Speed-set. With $C_M = \Lambda_M\ell_M$, the exposure lengths are read off the integrals of Lemma 0:

Case	Ramp-on ℓ_M (and residual)	Arrival-on ℓ_M (and residual)
Plateau, no closure	L	$L - \tau_M$
Truncation, $sL < K_M$	L	0
Step at t_D	t_D, plus $h_M^{post}(L - t_D)$	$t_D - \tau_M$, plus $h_M^{post}(L -$

Case	Ramp-on ℓ_M (and residual)	Arrival-on ℓ_M (and residual)

t_D)

(i) Plateau. Under ramp-on, (0.2) with $t_D \equiv L$ gives $sC_M' = L\Lambda_M\varepsilon_\Lambda$. Under arrival-on, with $\tau_M < L$,

$sC_M' = \Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$.

Proof. *C_M = ∫_{K_M/s}^{L}Λ_M dt = Λ_M(L − K_M/s). *By the product rule,* C_M′ = Λ_M′(L − K_M/s) + Λ_MK_M/s². *Multiplying by* s* turns the first term into $\varepsilon_\Lambda\Lambda_M(L - \tau_M)$ and the second into $\Lambda_M\tau_M$. ■ This is (0.1) with $t_D \equiv L$. The exposure-length elasticity follows: $s\ell_M' = \tau_M$, so $\varepsilon_{\ell_M} = \tau_M/(L - \tau_M)$. *Sign.* With $\Lambda_M > 0$, the derivative is positive iff $\varepsilon_\Lambda > -\tau_M/(L - \tau_M)$. A4 is therefore sufficient but stronger than needed. $\varepsilon_\Lambda \geq 0$ suffices, because the onset term $\Lambda_M\tau_M$ is positive when $K_M > 0$. The ramp-on derivative, by contrast, is zero at $\varepsilon_\Lambda = 0$. *Worked instance.* $\Lambda_M(s) = 0.02s^{1/2}$ (so $\varepsilon_\Lambda = 0.5$), $K_M = 4$, $L = 10$ and $s = 1$ give $0.02[0.5\cdot6 + 4] = 0.14$. Directly, $C_M(1.01) = 0.02009975124\cdot6.039603960 = 0.1213945372$ and $C_M(0.99) = 0.01989974874\cdot5.959595960 = 0.1185944622$. The central difference is 0.14000375. *Numerical check:* this matched central finite differences of the loss, built exactly from the set-up, to a worst relative error of 4.0×10^{-9} (348 cases).

(ii) Plateau, discounted, arrival-on. $sC_{M,r}' = \Lambda_M[\varepsilon_\Lambda(e^{-r\tau_M} - e^{-rL})/r + \tau_M e^{-r\tau_M}] > 0$. *Proof.* $C_{M,r} = \Lambda_M(e^{-r\tau_M} - e^{-rL})/r$. The interior term is $\Lambda_M'(e^{-r\tau_M} - e^{-rL})/r$. The lower-boundary term is $-\Lambda_M e^{-r\tau_M}\tau_M' = +\Lambda_M e^{-r\tau_M}\tau_M/s$. Multiply by s. ■ As $r \to 0$, the expression returns (i). For $r > 0$ and $\tau_M < L$, both bracketed terms are positive, so $\varepsilon_\Lambda \geq 0$ is again enough for the sign. *Worked instance.* (i)'s values with $r = 0.05$: $(0.8187307531 - 0.6065306597)/0.05 = 4.2440018673$ and $4\cdot0.8187307531 = 3.2749230123$. So $sC_{M,r}' = 0.02[0.5\cdot4.2440018673 + 3.2749230123] = 0.1079384789$. *Numerical check:* worst relative error 1.2×10^{-9} (96 cases).

(iii) Truncation. Under ramp-on, $C_M = \Lambda_M L$ whether or not the capability is reached, and τ_M never enters. Under arrival-on with no closure, the exposure set is $\{t \in [0, L] : t \geq K_M/s\}$, so $C_M = \Lambda_M(L - K_M/s)^+$ exactly, for every s. *Proof.* Below $s = K_M/L$, the exposure set is empty, so $C_M \equiv 0$ and $sC_M' = 0$ on the whole interval. At $s = K_M/L$, $C_M = 0$ and is continuous. Its left derivative is 0. Its right derivative is $\Lambda_M\cdot K_M/s^2$, because $L - K_M/s$ vanishes there with slope K_M/s^2. Multiplying by s, and using $K_M/s = L$, gives $\Lambda_M L$. (i) has the same limit from above. ■ This needs Λ_M finite and right-continuous at K_M/L, and no sign assumption on ε_Λ. The exact form also gives finite moves of misuse loss across the activation kink: $C_M(s_1) - C_M(s_0) = \Lambda_M(s_1)(L - K_M/s_1)$ for $s_0 \leq K_M/L < s_1$. A finite move of the whole balance across the kink is still open (§4b.2). *Worked instance.* $K_M = 4$ and $L = 10$ give a kink at $s = 0.4$. There $\Lambda_M = 0.02\cdot0.6324555320 = 0.0126491106$, the right derivative is $0.0126491106\cdot4/0.16 = 0.3162277660$, and s times that is $0.1264911064 = \Lambda_M L$. *Numerical check:* C_M and sC_M' are exactly 0 throughout the unreached region (22 cases).

(iv) Step, R3. Ramp-on, with a fixed residual, (0.2) gives $sC_M' = \Lambda_M t_D\varepsilon_\Lambda + s\cdot t_D'(\Lambda_M - h_M^{post}) = \Lambda_M t_D[\varepsilon_\Lambda - \alpha_D(1 - h_M^{post}/\Lambda_M)]$, with $\alpha_D = -s\cdot t_D'/t_D$. On the own-distance capability branch, $\alpha_D = \tau_D/(\tau_D + m_D T_D)$, and at $h_M^{post} = 0$ the condition is

$\varepsilon_\Lambda < \tau_D/(\tau_D + m_{DT_D})$, which at $m_D = 0$ is $\varepsilon_\Lambda < 1$. On the K_M-paced capability branch, $\alpha_D = \tau_M/t_D$. Substituting the own-distance capability branch into the ramp-on derivative gives $\Xi_s = \Lambda_M\varepsilon_\Lambda m_{DT_D} + (K_D/s)[\Lambda_M(\varepsilon_\Lambda - 1) + h_{M^{post}}]$, which is affine in K_D; differentiating it in K_D gives the coefficient $[\Lambda_M(\varepsilon_\Lambda - 1) + h_{M^{post}}]/s$. The coefficient is negative iff $\varepsilon_\Lambda < 1 - h_{M^{post}}/\Lambda_M$. The threshold $\alpha_D(1 - h_{M^{post}}/\Lambda_M)$ rises with K_D when $m_{DT_D} > 0$ and $\Lambda_M > h_{M^{post}}$, because $\tau_D/(\tau_D + m_{DT_D})$ increases in τ_D.

Arrival-on, with a fixed residual and $\tau_M \le t_D < L$,

$sC_{M'} = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M] + s\cdot t_{D'}(\Lambda_M - h_{M^{post}})$.

Proof. This is (0.1) with $h_{M^{post}}' = 0$. Written out, $C_M = \Lambda_M(t_D - \tau_M) + h_{M^{post}}(L - t_D)$. So $C_{M'} = \Lambda_{M'}(t_D - \tau_M) + \Lambda_M(t_{D'} - \tau_{M'}) - h_{M^{post}}\, t_{D'}$, and $s\cdot\tau_{M'} = -\tau_M$. ■ The general derivative is unsigned, and the branches sign it as follows.

- *Own distance, capability branch* ($s\cdot t_{D'} = -\tau_D$): $sC_{M'} = \Lambda_M[\varepsilon_\Lambda(\tau_D - \tau_M + m_{DT_D}) + \tau_M - \tau_D(1 - h_{M^{post}}/\Lambda_M)]$. With positive exposure $\tau_D - \tau_M + m_{DT_D} > 0$, $sC_{M'} < 0$ iff $\varepsilon_\Lambda < [\tau_D(1 - h_{M^{post}}/\Lambda_M) - \tau_M]/(\tau_D - \tau_M + m_{DT_D})$. At $h_{M^{post}} = 0$, a positive ε_Λ can meet this only if $\tau_D > \tau_M$, that is, $K_D > K_M$.
- *The K_D coefficient.* The bracket above equals $\Lambda_M\varepsilon_\Lambda(m_{DT_D} - \tau_M) + \Lambda_M\tau_M + (K_D/s)[\Lambda_M(\varepsilon_\Lambda - 1) + h_{M^{post}}]$. So $\partial(sC_{M'})/\partial K_D = [\Lambda_M(\varepsilon_\Lambda - 1) + h_{M^{post}}]/s$, the same coefficient as under ramp-on, and again affine in K_D. This holds with $\chi_D = 1$ and $\tau_M < t_D < L$ at both values of K_D, with s, m_D, T_D and the hazard fixed. *Worked instance.* Take $\Lambda_M = 0.02s^{1/2}$, $K_M = 4$, $L = 10$, $s = 1$, $m_{DT_D} = 2$ and $h_{M^{post}} = 0.005$. At $K_D = 6$, $t_D = 8$, $s\cdot t_{D'} = -6$, and $sC_{M'} = 0.02[0.5\cdot 4 + 4] - 6(0.015) = 0.12 - 0.09 = 0.03$. The branch form gives the same, $0.02[0.5\cdot 4 + 4 - 6\cdot 0.75] = 0.02\cdot 1.5 = 0.03$, so acceleration raises loss. At $K_D = 7$, $t_D = 9$ and $sC_{M'} = 0.02[0.5\cdot 5 + 4] - 7(0.015) = 0.13 - 0.105 = 0.025$. The change of −0.005 equals $[0.02(0.5 - 1) + 0.005]/1$.
- *Paced by K_M, capability branch* ($s\cdot t_{D'} = -\tau_M$): $\Lambda_M[\varepsilon_\Lambda m_{DT_D} + \tau_M] - \tau_M(\Lambda_M - h_{M^{post}}) = \varepsilon_\Lambda\Lambda_M m_{DT_D} + h_{M^{post}}\cdot\tau_M$. Under A4 with $\tau_M > 0$, this is ≥ 0, and strictly positive iff $m_{DT_D} > 0$ or $h_{M^{post}} > 0$. At $m_{DT_D} = 0$, $t_D \equiv \tau_M$ on a neighbourhood, so the plateau has zero length on both sides and the derivative is $h_{M^{post}}\cdot\tau_M$.
- *Adoption branch* ($t_{D'} = 0$): $sC_{M'} = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M] > 0$ under A4.
- *One-sided limits at $t_D = \tau_M$, own-distance clock.* On the positive-exposure side, the own-distance form above tends to $\Lambda_M\tau_M - \tau_D(\Lambda_M - h_{M^{post}})$ as $t_D - \tau_M \downarrow 0$ when Λ_M is continuously differentiable; in any case this value is s times the one-sided derivative at the boundary. On the defence-before-arrival side, the modelling choice in A0 gives $C_M = h_{M^{post}}(L - \tau_M)$, so $sC_{M'} = h_{M^{post}}\cdot\tau_M$. The two sides differ by $(\Lambda_M - h_{M^{post}})(\tau_M - \tau_D) = (\Lambda_M - h_{M^{post}})m_{DT_D}$, since $\tau_M - \tau_D = m_{DT_D}$ on the boundary. The boundary is therefore a kink unless $m_{DT_D} = 0$ or $\Lambda_M = h_{M^{post}}$, and one-sided derivatives must be used. When $K_D = K_M$ and $m_{DT_D} = 0$, the two sides agree at $h_{M^{post}}\cdot\tau_M$. *Worked instance.* Take $\Lambda_M = 0.02s^{1/2}$, $K_M = 4$, $K_D = 3$, $m_{DT_D} = 1$, $h_{M^{post}} = 0.005$, $L = 10$ and $s = 1$, so t_D

= 4 = τ_M. The positive-exposure side is *0.08 − 3(0.015) = 0.035*, and the other side is *0.005·4 = 0.02*. They differ by *0.015 = (0.02 − 0.005)·1*.

Assumptions for (iv). $\tau_M \le t_D < L$ (one-sided at equality), a fixed residual, t_D and Λ_M differentiable, and A4 only where a sign is stated. *Numerical check* (arrival-on R3, fixed residual): worst relative error 1.7×10^{-10} (54 cases).

(v) A responsive residual. Each speed-set derivative receives *(L − t_D)s·h_M^post′*. *Proof.* This is the last term of (0.1) and (0.2). The residual segment is the same set, *[t_D, L]*, under both readings once $t_D \ge \tau_M$, so the term does not depend on the reading or on the clock branch. ■ It is Leibniz's interior term. The moving boundary at t_D gives *s·t_D′(Λ_M − h_M^post)*, which is already present with a fixed residual. A4a.3 gives the level-set counterpart. Under arrival-on with defence before arrival, the modelling choice in A0 runs the residual from τ_M, and the term is *(L − τ_M)s·h_M^post′*. *Sign.* The term has the sign of *h_M^post′*. *Worked instance.* (iv)'s values at $K_D = 6$, with $h_M^{post}(s) = 0.005s^2$, give the term *(10 − 8)·0.01 = 0.02*. Under arrival-on, the derivative goes from 0.03 to 0.05. Under ramp-on, *0.02·8·0.5 − 6·0.015 = −0.01* becomes *+0.01*, so the term reverses the sign. *Numerical check:* worst relative error 1.1×10^{-10} (24 cases).

(vi) R2 with the benefit. Take the benefit's capability branch ($\chi_B = 1$, $a = t_B + \delta_T < L$), $\tau_M < L$, no closure, $\omega = 0$ and $\delta_T' = 0$, with every other marginal set to zero. The marginal form of §4a.1 is then $sW_L^{X'} = pyq - V_M \cdot sC_M'$, with $q = K_B/s$. Under level-set, A4a.1(i) gives $sW_L^{X'} > 0$ iff $pyK_B/s > V_M(K_M/s)[h_M(K_M) - \bar{h}_M]$, that is, iff $pyK_B > V_MK_M[h_M(K_M) - \bar{h}_M]$, a test free of s. Under speed-set ramp-on the exposure length is L, so $C_M = \Lambda_M L$ and $sC_M' = L \cdot s\Lambda_M' = L\Lambda_M\varepsilon_\Lambda$, giving $sW_L^{X'} > 0$ iff $pyq > V_M \cdot L \cdot \Lambda_M\varepsilon_\Lambda$ (*numerical check:* 5.5×10^{-10}, 72 draws). Under speed-set arrival-on, (i) gives $sW_L^{X'} > 0$ iff $pyq > V_M\Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$ (*numerical check:* 8.3×10^{-10}, 144 draws). Both right sides carry L and ε_Λ, and the level-set right side above carries neither; no reading makes the two tests depend on the same parameters. Under the comparison convention of §4b.1, the gated level-set test is the arrival-on speed-set test at $\varepsilon_\Lambda = 0$, and the two right sides differ by $V_M\Lambda_M\varepsilon_\Lambda(L - \tau_M) > 0$ under A4 (*numerical check:* 3.8×10^{-16}). Its right side has $\partial/\partial L = V_M\Lambda_M\varepsilon_\Lambda > 0$ under A4 and $\partial/\partial\varepsilon_\Lambda = V_M\Lambda_M(L - \tau_M) > 0$, and its left side does not depend on L while $t_B + \delta_T < L$. So with $\varepsilon_\Lambda > 0$ the sign changes at $L^* = \tau_M + [pyq/(V_M\Lambda_M) - \tau_M]/\varepsilon_\Lambda$, provided L^* exceeds $\max(t_B + \delta_T, \tau_M)$, which requires $pyq > V_M\Lambda_M\tau_M$. Growth with s is not claimed: the right side's s-derivative involves ε_Λ', which is unrestricted, and the left side moves with s. Off the capability branch, each test carries χ_B. *Worked instance.* Take (i)'s misuse values with $p = 0.5$, $y = 1$, $K_B = 3$ (so $q = 3$), $m_{BT_b} = 2$ (so $t_B = 5$) and $V_M = 10$, so that $pyq = 1.5$. At $L = 10$ the right side is 1.4, so acceleration raises the balance. At $L = 12$ it is $10 \cdot 0.02[0.5 \cdot 8 + 4] = 1.6$, so acceleration lowers it. $L^* = 4 + (7.5 - 4)/0.5 = 11$, where the right side is 1.5.

A4a.5 Mixed hazard. *Additive form*, ramp-on, hazard $h_M(c(t)) + \Lambda_M(s)$, complete closure at $\tau_M \le t_D < L$. The loss is the sum of the two components' losses, so (0.2) and (L.1) at $h_M^{post} = 0$ give

$$sC_M' = \Lambda_M t_D(\varepsilon_\Lambda - \alpha_D) + \tau_M[h_M(K_M) - \bar{h}_M] - \alpha_D t_D h_M(K_M).$$

On the *K_M*-paced capability branch, *α_Dt_D = τ_M*, so *sC_M′ = Λ_Mt_Dε_Λ − τ_M(Λ_M + h̄_M)*. Write* α_H = Λ_Mt_D/C_M* for the speed component's share of misuse loss, and *σ_M = τ_M(Λ_M + h̄_M)/C_M* for the share accrued on *[0, τ_M]*. Then *sC_M′ = C_M(α_Hε_Λ − σ_M)*, so for positive loss, acceleration lowers loss iff *α_Hε_Λ < σ_M*. At *α_H* = 1 this is *ε_Λ < τ_M/t_D*. *On the plateau* (no closure, *τ_M < L*), ramp-on gives *sC_M′ = LΛ_Mε_Λ + τ_M[h_M(K_M) − h̄_M]*, which is positive for every *ε_Λ* > 0 when the level component is nondecreasing. Arrival-on gives *Λ_M[ε_Λ(L − τ_M) + τ_M] + τ_M[h_M(K_M) − h̄_M]* if only the speed component is gated, and *Λ_Mε_Λ(L − τ_M) + [Λ_M + h_M(K_M)]τ_M* if both are. Either is positive for every *ε_Λ ≥ 0* when *K_M* > 0, *h_M* is nonnegative and nondecreasing and *Λ_M* > 0. So on the plateau no positive elasticity lowers loss, under either reading and either gating.

Separable form Λ_M(s)g_M(min(st, K_M)), **ramp-on exposure, with complete closure at τ_M ≤ t_D ≤ L**:

C_M = Λ_M[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)], *sC_M′ = Λ_M{ε_Λ[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)] + τ_M[g_M(K_M) − ḡ_M] − α_Dt_Dg_M(K_M)}*,

with *ḡ_M = K_M^{-1}∫_0^{K_M}g_M*. *Proof.* The hazard is *Λ_Mg_M(st)* on *[0, τ_M)*, *Λ_Mg_M(K_M)* on *[τ_M, t_D)*, and 0 after *t_D*. Integrating from date 0 to *t_D ≤ L*, and changing variables on the ramp as in Lemma L(i), gives *C_M = Λ_M[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)]*. By the product rule, *sC_M′ = Λ_M{ε_Λ[·] + s·d/ds[·]}*. Steps (i) and (ii) of Lemma L give *s·d/ds[·] = τ_M[g_M(K_M) − ḡ_M] + s·t_D′g_M(K_M)*, and *s·t_D′ = −α_Dt_D*. ■ For positive loss, the loss falls iff *ε_Λ < ε̄_g = [α_Dt_Dg_M(K_M) − τ_M(g_M(K_M) − ḡ_M)]/[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)]*. On the* K_M*-paced capability branch,* α_Dt_D = τ_M*, so* ε̄_g = τ_Mḡ_M/[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)]*, the ramp's share of accumulated loss. Its gap from the calendar-time share* α_D = τ_M/t_D* is *ε̄_g − α_D = τ_M(1 − α_D)(ḡ_M − g_M(K_M))/[τ_Mḡ_M + (t_D − τ_M)g_M(K_M)]. *This is zero iff* ḡ_M = g_M(K_M)* or *t_D = τ_M*. Two checks: *g_M ≡ 1* returns speed-set ramp-on R3 at complete closure, *Λ_Mt_D(ε_Λ − α_D)*, and *ε_Λ* = 0 returns level-set R3 at *h_M^post* = 0.

Conditions. Ramp-on exposure is used, because the ramp term integrates from date 0. Under arrival-on that term is absent, and both the loss and the derivative differ. *t_D ≤ L* is used: if *t_D > L*, the loss is *Λ_M[τ_Mḡ_M + (L − τ_M)g_M(K_M)]* and the *α_D* term disappears. The derivative also needs *t_D < L*, away from the horizon kink. *g_M* must be integrable, and *t_D* and *Λ_M* differentiable. *Worked instance. g_M(c) = 1 + 0.25c* on *[0, 4]*, so *g_M(K_M)* = 2 and *ḡ_M* = 1.5. *Λ_M = 0.02s^{1/2}*, *s* = 1, *K_M* = 4 and the paced capability branch with *m_DT_D* = 2 give *t_D* = 6 and *α_Dt_D* = 4. Then the bracket is *4·1.5 + 2·2 = 10*, *C_M* = 0.2, and *sC_M′ = 0.02{0.5·10 + 4(0.5) − 4·2} = −0.02*. *ε̄_g = 6/10 = 0.6 > 0.5*, so loss falls, which agrees. The gap is* 0.6 − 0.6667 = −0.0667 = 4·(1/3)·(−0.5)/10*.

A4a.6 Benefit margins. Take *B_r = p(y/r)e^{−rt_B}* and *s·t_B′ = −χ_Bq*. Then *s∂B_r/∂s = −py·e^{−rt_B}·s·t_B′ = pyqχ_Be^{−rt_B}*, whose logarithmic rate derivative is *−t_B*. Also *∂B_r/∂p = (y/r)e^{−rt_B}*, whose logarithmic rate derivative is *−(t_B + 1/r)*. So the ratio of proportional margins is *pyqχ_Be^{−rt_B}/[p(y/r)e^{−rt_B}] = rqχ_B*.

With a horizon, *L > t_B*, write *d = L − t_B*. The timing margin is unchanged, because *e^{−rL}* does not move with *t_B*. *∂B_r/∂p = (y/r)e^{−rt_B}(1 − e^{−rd})*, whose logarithmic rate

derivative is $-t_B - 1/r + d/(e^{rd} - 1)$. This is below $-t_B$, because $rd < e^{rd} - 1$. The ratio is $rq\chi_B/(1 - e^{-rd})$. Its derivative in r is $q\chi_B[1 - (1 + rd)e^{-rd}]/(1 - e^{-rd})^2$, which is positive for $\chi_B = 1$ because $e^{rd} > 1 + rd$, and zero for $\chi_B = 0$. The ratio tends to $q\chi_B/d$ as $r \to 0$.

Fixed duration ℓ_B, counted in full ($t_B + \ell_B \le L$): $*B_r = \int_{t_B}^{t_B+\ell_B} py \cdot e^{-rt} dt = p(y/r)e^{-rt_B}(1 - e^{-r\ell_B})$. *The timing margin is* $pyq\chi_B e^{-rt_B}(1 - e^{-r\ell_B})$, *which tends to zero as* $r \to 0$. *At* $r^* = 0$ the benefit is $py\ell_B$, whatever the arrival date. For $pyq\chi_B > 0$ and $r > 0$, the logarithmic rate derivative of this margin is

$$d \ln[pyq\chi_B e^{-rt_B}(1 - e^{-r\ell_B})]/dr = -t_B + \ell_B e^{-r\ell_B}/(1 - e^{-r\ell_B}) = -t_B + \ell_B/(e^{r\ell_B} - 1).$$

Sign, exactly. $\ell_B/(e^{r\ell_B} - 1)$ falls strictly in r, from $+\infty$ as $r \downarrow 0$ to 0. **For $t_B > 0$, the timing margin therefore rises with r for $r < \ln(1 + \ell_B/t_B)/\ell_B$ and falls above it; at $t_B = 0$ it rises at every r.** The margin is positive, so no absolute value is needed. *Assumptions.* $p > 0$, $y > 0$, $q > 0$ (that is, $K_B > 0$), $\chi_B = 1$ (otherwise the margin is identically zero and the logarithm is undefined), $r > 0$, ℓ_B fixed and counted in full, and t_B independent of r. *Worked instance.* Figure 16's values are $t_B = 8$ and $\ell_B = 15$, so the switch is at $\ln(2.875)/15 = 1.0560526742/15 = 0.0704035116$. At $r = 0.03$, $15/0.5683121855 = 26.3939440029$, and the derivative is $+18.3939440029$. At $r = 0.10$, $15/3.4816890703 = 4.3082537518$, and the derivative is -3.6917462482. *Numerical check* (the log rate derivative): worst relative error 7.4×10^{-10} (18 cases).

A4a.7 The facility comparison at a fixed speed. The conditions are those of §4a.5: $\tau_M < t_D < L$ with t_D independent of K_M, a constant residual, and K_D off the clock kink. Lemma 1 gives every distance derivative.

(i) The defence distance. (1.6) holds in all four cases, two specifications by two readings, under either convention: $\partial C_M/\partial K_D = (\chi_D/s)(h_M^{plat} - h_M^{post})$.

Why it holds in all four cases. The invariance is of the form, with each case's own plateau hazard substituted, and not of the value. It follows from where K_D enters: only at the deployment date $t_D = \max\{K_D/s, (1 - m_D)T_D\} + m_D T_D$, so that $\partial t_D/\partial K_D = \chi_D/s$ off the kink. Each of the three choices the paper carries governs a different object. The specification sets the *level* of the plateau hazard, which enters this derivative as h_M^{plat} and nowhere else. The exposure reading sets where counting *starts*, which K_D does not move, and t_D is interior to both windows, so the lower limit contributes nothing to the derivative under either reading. The convention sets how the hazard profile responds to K_M, which this derivative holds fixed. The agreement across the three choices is therefore a consequence of the model's structure and not three independent confirmations, and the contrast is visible in (ii) and (iii): the K_M derivative, which the conventions do govern, differs between them in form and in sign.

What the result is. Written as a translation, the content is elementary. K_D shifts the whole countermeasure schedule in calendar time, and the derivative of the loss with respect to a rigid shift of an interior block is the hazard the block enters on minus the hazard it leaves on, where each is continuous at its date. A step at t_D is the case in which those two are

h_M^plat and *h_M^post*. The same form therefore survives gradual coverage, provided the coverage schedule translates rigidly with *t_D* and completes before *L* — a case §4a.5's step condition excludes and this derivative does not need. *Numerical check:* under coverage rising linearly over *[t_D, t_D + w]*, the form matched finite differences of a direct numerical integration to 1.1×10^{-9} (1,440 case-draws, the three level-and-speed cases by two readings) [NUMCHECK].

What it needs. Completion inside the counted window is a condition and not a convenience. With coverage still rising at *L*, the derivative is *(χ_D/s)(h_M^plat − h_M^post)(L − t_D)/w*, which, on the capability branch with *h_M^plat ≠ h_M^post*, equals the invariant form only at *w = L − t_D*; with a step, completion is instantaneous and the condition is invisible. It also needs the residual fixed in *K_D*: with step coverage a responsive residual adds *(L − t_D)∂h_M^post/∂K_D*, and with gradual coverage it adds *∂h_M^post/∂K_D* times the covered time *∫_{t_D}^{L}u(t − t_D)*dt*. Numerical check:* the form matched to 9.9×10^{-10} across all six specification-by-reading cases on 144 capability-branch draws, and the derivative is exactly zero on 96 adoption-branch draws, where both sides vanish and the test is of the other half of the claim [NUMCHECK].

(ii) The misuse distance, ramp-on. For a general profile *h_M(c; K_M)* continuous at *c = K_M*, (1.2) gives *∂C_M/∂K_M = (1/s)∫₀^{K_M}∂_{K_M}h_M(c; K_M)dc + (t_D − τ_M)·(d/dK_M)h_M(K_M; K_M). *The two boundary terms cancel as in Lemma 1(b). Under A, Lemma 1(c) gives* −[ĥ_M(1) − h̄_M]/s*. Under B, Lemma 1(b) gives* (t_D − τ_M)h_M′(K_M)*.

(iii) The misuse distance, arrival-on. (1.3) holds. Under A, *dh_M^plat/dK_M* = 0, so *∂C_M/∂K_M = −ĥ_M(1)/s*. This uses no property of *ĥ_M* beyond *ĥ_M(1)* being defined, and it is negative iff *ĥ_M(1)* > 0. *Worked instance (base W).* Take *s* = 2, *K_M* = 4 (so *τ_M* = 2), *K_D* = 10, *m_D* = 0.5, *T_D* = 2, so that *χ_D* = 1 and *t_D* = 6, and *L* = 10, *h_M^post* = 0.1, *ĥ_M(x) = x*. Then *C_M(K_M) = 6.4 − K_M/2*, with *C_M(4) = 4.4* and *C_M(4.2) = 4.3*, so the quotient is exactly −0.5. *Numerical check:* this matched finite differences of a direct numerical integration of the loss to 1.2×10^{-10} (504 draws).

Under B, *∂C_M/∂K_M = (t_D − τ_M)h_M′(K_M) − h_M(K_M)/s*, for *h_M* differentiable at *K_M*. It is unsigned, even for a strictly increasing hazard. *What decides the sign.* For *h_M(K_M)* > 0, multiply by *K_M/h_M(K_M)*: the derivative is positive iff the hazard's level elasticity at the threshold, *K_Mh_M′(K_M)/h_M(K_M)*, exceeds *τ_M/(t_D − τ_M)*. *Worked instances.* With *h_M(c) = c/4*, the level elasticity is 1. - At base W (*t_D* = 6), *C_M(K_M) = 1.5K_M − K_M²/8 + 0.4*. The formula gives *(6 − 2)(1/4) − 1/2 = +0.5*. *C_M(4.1) = 4.44875* and *C_M(3.9) = 4.34875*, so the central difference is exactly 0.5. The elasticity test agrees: 1 > 2/4. - At *K_D* = 3 (*t_D* = 2.5), the formula gives *(0.5)(1/4) − 1/2 = −0.375*. The elasticity test agrees: 1 < 2/0.5.

One strictly increasing hazard gives both signs. *Numerical check:* 2.1×10^{-8} (504 draws).

Under speed-set, Lemma 1(e) gives *−Λ_M/s* under arrival-on (*numerical check:* 6.8×10^{-11}, 144 draws) and 0 under ramp-on.

(iv) The facility conditions. To first order, *ΔC_M ≈ (∂C_M/∂K_M)ΔK_M + (∂C_M/∂K_D)ΔK_D*. Multiply by *s* > 0 and substitute (i)–(iii). - *Level-set, A, ramp-on:* misuse loss falls iff *χ_Dĥ_M(1) − h_M^post* > *ĥ_M(1) − h̄_M*. - *Level-set, A, arrival-on:* misuse loss falls iff

$\chi_D\hat{h}_M(1) - h_M^{post} > \hat{h}_M(1)(-\Delta K_M)$. At $\chi_D = 1$ the boundary has slope $\hat{h}_M(1)/[\hat{h}_M(1) - h_M^{post}]$, which is Figure 11's $h/(h - P)$. The slope is positive iff $\hat{h}_M(1) > h_M^{post}$, exceeds 1 whenever $h_M^{post} > 0$, and equals 1 at $h_M^{post} = 0$. It needs convention A, because under B the weight on $(-\Delta K_M)$ is $h_M(K_M) - s(t_D - \tau_M)h_M'(K_M)$, which reduces to $h_M(K_M)$ only when $h_M'(K_M) = 0$. - *Level-set, B, ramp-on:* misuse loss falls iff $\chi_D h_M(K_M) - h_M^{post} + s(t_D - \tau_M)h_M'(K_M)(-\Delta K_M) > 0$. For nondecreasing h_M, the K_M term is ≥ 0 for a cut, so cutting K_M weakly lowers loss. The boundary slope, $-s(t_D - \tau_M)h_M'/[\chi_D(h_M - h_M^{post})]$, is ≤ 0 when $\chi_D = 1$ and $h_M(K_M) > h_M^{post}$. - *Level-set, B, arrival-on:* misuse loss falls iff $\chi_D h_M(K_M) - h_M^{post} > h_M(K_M) - s(t_D - \tau_M)h_M'(K_M)$. The weight on $(-\Delta K_M)$ is $s\cdot(-\partial C_M/\partial K_M)$, so it is negative exactly when (iii)'s level-elasticity condition holds. At Figure 11's drawn set ($s = 1$, $K_M = 1$, $K_D = 2.5$, $m_D = 0.3$, $T_D = 1$, $L = 4$, $h_M^{post} = 0$, $h_M(c) = 0.2 + c$), $t_D = 2.8$ and the weight is $1.2 - 1\cdot1.8\cdot1 = -0.6$. - *Speed-set:* under ramp-on, misuse loss falls iff $\chi_D(\Lambda_M - h_M^{post})(-\Delta K_D) > 0$. Under arrival-on, it falls iff $\chi_D(\Lambda_M - h_M^{post})(-\Delta K_D) > \Lambda_M(-\Delta K_M)$. - *The K_D weight* has the sign of $\chi_D(h_M^{plat} - h_M^{post})$. It is zero on the adoption branch and reversed when the residual exceeds the plateau.

Worked instance, B (base W, $h_M(c) = c/4$). The weight on $(-\Delta K_D)$ is $1\cdot(1 - 0.1) = 0.9$. Ramp-on, the condition is $0.9(-\Delta K_D) + 2(-\Delta K_M) > 0$. Arrival-on, the weight on $(-\Delta K_M)$ is $1 - 2\cdot4\cdot(1/4) = -1$, so any cut to K_M helps. At $K_D = 3$ ($t_D = 2.5$), the weight is $1 - 2\cdot0.5\cdot(1/4) = +0.75$, so cutting K_M alone raises loss. Directly, $C_M(4) = 1.25$ and $C_M(3.9) = 1.28625$. The change is +0.03625, against a first-order +0.0375; the gap is the second-order term $-0.1^2/8 = -0.00125$.

(v) Finite moves. Within one regime (χ_D fixed and $\tau_M < t_D < L$ at both points), t_D is affine in K_D, and τ_M is linear in K_M. Under A, (1.5) and (1.1) have coefficients that are constant in K_M, and under speed-set the hazard does not depend on either distance. So C_M is affine in (K_M, K_D) in both cases, and the first-order conditions hold exactly for real moves. The regime is an intersection of half-planes, hence convex, so the segment between the two points stays inside it. Under B the loss is not affine unless h_M is constant over the range moved: the ramp-on loss $(1/s)\int_0^{K_M} h_M + (t_D - K_M/s)h_M(K_M) + \ldots$ is nonlinear in K_M, and its cross term $t_D h_M(K_M)$ makes $\partial C_M/\partial K_D$ move with K_M. So a finite change under B must be integrated. *Worked instance (base W, $\hat{h}_M(x) = x$).* Move K_M from 4 to 2 and K_D from 10 to 5. Then $t_D = 3.5$ and $\tau_M = 1$, still in the regime. - Ramp-on, A: C_M goes from 5.4 to 3.65, a change of −1.75, which equals $(-0.25)(-2) + (0.45)(-5)$. - Arrival-on, A: C_M goes from 4.4 to 3.15, a change of −1.25, which equals $(-0.5)(-2) + 0.45(-5)$. - Ramp-on, B, with $h_M(c) = c/4$: C_M goes from 5.4 to 2.15, a change of −3.25, against an affine prediction of $(1.0)(-2) + 0.45(-5) = -4.25$.

(vi) Defence before the capability ($t_D < \tau_M$), ramp-on. - Under B the K_M derivative is **zero**. In (2.1), no limit and no integrand contains K_M, and neither does t_D. This uses nothing about h_M, and it holds whether or not $t_D < L$. - *Under A*, (2.1) becomes $C_M = (K_M/s)\int_0^{x_d} \hat{h}_M + (L - t_D)h_M^{post}$, with $x_d = st_D/K_M$ and $\partial x_d/\partial K_M = -x_d/K_M$. By the product rule and the fundamental theorem of calculus, for $\hat{h}_M$ continuous at x_d,

$$\partial C_M/\partial K_M = -(1/s)[x_d\hat{h}_M(x_d) - \int_0^{x_d}\hat{h}_M] = -(t_D/K_M)[\hat{h}_M(x_d) - x_d^{-1}\int_0^{x_d}\hat{h}_M].$$

The form needs ***t_D < L***, a condition the facility comparison's interior regime does not supply here, because this case violates $\tau_M < t_D$. If $t_D \geq L$, write $x_d = s \cdot \min(t_D, L)/K_M$. *Counterexample without it.* Take $s = 1$, $K_M = 10$, $t_D = 8$, $L = 5$ and $\hat{h}_M(x) = 0.1 + x^2$. Directly, $C_M = 0.5 + 125/(3K_M^2)$ and $\partial C_M/\partial K_M = -250/3000 = -0.083333\ldots$. The form with $x_d = 0.8$ gives −0.341333, which is wrong. With $x = sL/K_M = 0.5$ it gives −0.083333.

Sign. Under convention A, $\partial C_M/\partial K_M \leq 0$ even when the defence completes before the capability is reached, because $x_d\hat{h}_M(x_d) - \int_0^{x_d}\hat{h}_M = \int_0^{x_d}[\hat{h}_M(x_d) - \hat{h}_M(x)]dx \geq 0$ for nondecreasing $\hat{h}_M$. It is strictly negative unless $\hat{h}_M$ is constant on the interval. The compressed profile reaches higher hazard levels before t_D.

Continuity. As $x_d \to 1$, and for $\hat{h}_M$ continuous at 1 from the left, the form tends to $-(1/s)[\hat{h}_M(1) - \int_0^1 \hat{h}_M]$, the interior ramp-on A derivative of (ii). So there is no jump at $t_D = \tau_M$. Under B, (1.4) tends to 0 as $t_D \uparrow \tau_M$, so under both conventions the ramp-on K_M derivative is continuous across $t_D = \tau_M$. The second form above is R1's truncation form with t_D in place of L. Under A the ramp term depends on (s, K_M) only through K_M/s and st_D/K_M, so $K_M\partial C_M/\partial K_M = -s\partial C_M/\partial s$ at fixed t_D.

Worked instances. Take $s = 2$, $K_M = 4$, $K_D = 1$, $m_D = 1/3$, $T_D = 1.5$, so that $\chi_D = 0$ and $t_D = 1.5 < 2$, with $L = 10$ and $h_M^{post} = 0.1$. Under B, with $h_M(c) = c/4$, $C_M = 0.5625 + 0.85 = 1.4125$, with no K_M anywhere. Under A, with $\hat{h}_M(x) = x$, $x_d = 3/4$, and the formula gives $-(1/2)[0.5625 - 0.28125] = -0.140625$. Directly, $C_M = 2.25/K_M + 0.85$, whose derivative is $-2.25/16 = -0.140625$. The average form gives $-(1.5/4)(0.75 - 0.375) = -0.140625$.

Numerical check (the four level-set before-arrival rows): worst error 7.6×10^{-10} (42 draws each).

(vii) The proportional cut. Under level-set convention A, R3's derivative equals the fixed-speed derivative along a proportional cut of both distances, $s\partial C_M/\partial s = -K_M\partial C_M/\partial K_M - K_D\partial C_M/\partial K_D$, under either reading. This is (H.1). Under speed-set, (H.2) adds the intensity term: $s\partial C_M/\partial s = \Lambda_M\varepsilon_\Lambda\ell_M - K_M\partial C_M/\partial K_M - K_D\partial C_M/\partial K_D$, with $\ell_M = t_D$ under ramp-on and $\ell_M = t_D - \tau_M$ under arrival-on. The identity needs a fixed residual and m_DT_D held fixed. *Under B it fails in general*, because scaling K_M moves the hazard level reached. *Worked instances.* - Base W, A, ramp-on, from (1.5): $\partial C_M/\partial K_M = -0.25$ and $\partial C_M/\partial K_D = 0.45$, so the right side is $4 \cdot 0.25 - 10 \cdot 0.45 = -3.5$. Lemma L gives $2[1 - 0.5] - 5 \cdot 0.9 = -3.5$. - Base W, A, arrival-on: the right side is $4 \cdot 0.5 - 4.5 = -2.5$, and $s\partial C_M/\partial s = (4 - 9)/2 = -2.5$. - Base W, B, ramp-on ($h_M(c) = c/4$): the left side is −3.5, but the right side is $-4 \cdot 4 \cdot 0.25 - 4.5 = -8.5$. - Speed-set, (iv)'s values at $K_D = 6$: under arrival-on, $0.03 = 0.02 \cdot 0.5 \cdot 4 + 4 \cdot 0.02 - 6 \cdot 0.015$; under ramp-on, $-0.01 = 0.02 \cdot 0.5 \cdot 8 + 0 - 6 \cdot 0.015$.

The identity is a property of the convention, not a relation the paper uses, and it does not make speed a facility.

(viii) What the numerical check covers. [NUMCHECK] integrated the loss directly, by adaptive quadrature with breakpoints at the kinks, for seven hazard shapes, residuals above and below the plateau, both values of χ_D, both conventions and both readings. It compared the distance-derivative formulas above against central finite differences of that integral. Re-run for this version, every row of its derivative test passes, with worst

normalised error 2.1×10^{-8}, against an acceptance threshold of 10^{-6}. Finite-move comparisons use a separate error measure. Every mismatch it found lay at an excluded kink, at a hazard jump at *K_M* under B (the case Lemma 1(b) excludes), at a finite move that left its regime, or at a convention-A claim evaluated under B. That check did not test gradual coverage, discounting, more than one severity class, or the whole balance $J_I3 - J_0$; the translation check of A4a.7(i) is the one that tests rigidly translating coverage.

A4b. Tables and proofs for §4b

Table 4b.A. Where the specifications disagree, with the restrictions written into each row. "Helps" means that acceleration lowers misuse loss. Signs are those of the misuse marginal, and a divergence needs opposite strict signs. The residual is fixed, and rows 3–5 are on the capability branch ($\chi_D = 1$). The rows are conditional examples, not an exhaustive list over level profiles. Every sign entry is proved where the last column says. On the arrival-on reading, a Relation entry compares like with like only if the level-set side is gated, that is, written as the level profile that is zero below *K_M* and *h_M(K_M)* at it (§4a.2), so that $*\bar{h}_M* = 0$. Every row is now compared that way. Rows 1, 2, 4 and 6 cannot diverge on it; row 5 can; and row 7, which compares thresholds rather than signs, holds only under the comparison convention stated before the table.

The comparison convention. A *sign* comparison across the two specifications needs nothing where the rescaling is of a specification's whole hazard, residual included: every term then scales together and no sign moves. A *threshold* comparison does. The two specifications have different hazard primitives — *h_M(·)*, a hazard per unit of capability, and *Λ_M(·)*, a hazard per unit of time at a given speed — and nothing in the set-up fixes one against the other, so a statement that one threshold exceeds another has no content until something does. We therefore state the convention rather than leave it implicit: **wherever this section compares thresholds across specifications, *Λ_M* evaluated at the speed in question equals *h_M(K_M)*, the level hazard at the capability reached.** It is a normalisation, not a result, and nothing in the paper measures it. Rows 1–3 compare signs and do not use it. Rows 4 to 6 compare signs too, but the residual *h_M^post* is held common to the two specifications and is not rescaled with either primitive, so a row whose condition weighs the residual against a hazard — rows 4 and 5 carry *h_M^post/Λ_M* — can move when *Λ_M* alone is rescaled. That a divergence *exists* in those rows needs nothing; the *direction* stated for row 5 on the gated arrival-on reading uses the convention, or needs *h_M^post* = 0. *Numerical check:* over 145,885 draws inside row 5's stated case the direction reverses in 2,208 with *Λ_M* free and in none under the convention. Row 7 and the α_D threshold rule below compare thresholds and do; without it neither comparison is well posed, and that is a limitation of the comparison, not of either specification.

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
1	Plateau, no defence	sign of $*h_M(K_M) - \bar{h}_M*$	hurts under A4 (ramp-on; arrival-on)	Ramp-on: agree when *h_M(K_M	$*h_M(K_M) - \bar{h}_M*$	A4a.1(i); A4a.4(i). Arrival-on, gated:

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
				) > h̄_M* (e.g. continuous, nondecreasing, nonconstant *h_M*); diverge iff *h_M(K_M) < h̄_M. ***Arrival-on, both sides gated: cannot diverge****, since* h̄_M* = 0 makes the condition *h_M(K_M) < 0*		A4a.1(i) with *h̄_M* = 0
2	Truncation	for nondecreasing *h_M* on *[0, sL]*, weakly hurts (≥ 0), strictly iff *h_M(sL) > h̄_M(0, sL)*; otherwise sign of* h_M(sL) − h̄_M(0, sL)*	ramp-on: hurts; arrival-on: zero	Ramp-on: not opposite for nondecreasing *h_M* on *[0, sL]*; diverges iff *h_M(sL) < h̄_M(0, sL). ***Arrival-on, both sides gated: agree, and both are exactly zero —*** the gated*	*h_M(sL) − h̄_M(0, sL)*	A4a.1(ii); A4a.4(iii). Arrival-on, gated: the gated profile is identically zero on *[0, sL]* in this regime

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
				profile is zero below K_M* and the path never reaches *K_M* in this regime, so no hazard runs on either reading		
3	Risk closes at arrival, ramp-on exposure, *h_M^post* = 0, *$\bar{h}$_M* > 0	helps (*−τ_M$\bar{h}$_M*)	helps iff *ε_Λ* < 1; neutral at 1	Strict reversal iff *ε_Λ* > 1	*ε_Λ* against unity	A4a.1(iii); (0.2) with *s·t_D′ = −τ_M*, *t_D = τ_M*: *Λ_Mτ_M(ε_Λ − 1)*
4	Defence paced by *K_M*	helps iff *h_M^post < $\bar{h}$_M*	ramp-on: helps iff *ε_Λ < (τ_M/t_D)(1 − h_M^post/Λ_M)*; arrival-on: *ε_ΛΛ_Mm_DT_D + h_M^post·τ_M ≥ 0*, strict iff *m_DT_D* > 0 or *h_M^post* > 0	Ramp-on: diverge iff *(h_M^post − $\bar{h}$_M)[ε_Λ − (τ_M/t_D)(1 − h_M^post/Λ_M)] < 0, in either direction.* ***Arrival-on, both sides gated: cannot diverge****, since* $\bar{h}$_M* = 0 makes the level-set condition *h_M^post < 0*, and	level-set: *h_M^post − $\bar{h}$_M*; speed-set: the reading's threshold	A4a.3; A4a.4(iv). Arrival-on, gated: A4a.3 with *$\bar{h}$_M* = 0

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
				the gated level derivative $h_M^{post} \cdot \tau_M$ carries the same weak sign as the speed-set one		
5	Defence on its own K_D	helps iff $K_D[h_M(K_M) - h_M^{post}] > K_M[h_M(K_M) - \bar{h}_M]$	ramp-on: helps iff $\varepsilon_\Lambda < \alpha_D(1 - h_M^{post}/\Lambda_M)$; arrival-on, with $\tau_D - \tau_M + m_{DT_D} > 0$: helps iff $\varepsilon_\Lambda < [\tau_D(1 - h_M^{post}/\Lambda_M) - \tau_M]/(\tau_D - \tau_M + m_{DT_D})$	Diverge iff the two tests give opposite strict signs. Reversals can occur below unity; unity is only a sufficient bound for some speed-set cases	the two-distance inequality; the reading's elasticity threshold	A4a.3; A4a.4(iv). Arrival-on, level side gated: A4a.3 with $*\bar{h}_M* = 0$
6	Defence on adoption branch ($\chi_D = 0$)	hurts under R2's restrictions	hurts under A4	Agree on both readings. **Arrival-on, both sides gated: agree**, the gated level derivative being $h_M(K_M) \cdot \tau_M > 0$ against a	—	A4a.3; A4a.4(iv). Arrival-on, gated: A4a.3 with $*\bar{h}_M* = 0$

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
				speed-set derivative also positive under A4		
7	R2 with coupled benefit ($\chi_B = 1$, $t_B + \delta_T < L$, $\tau_M < L$)	$pyK_B > V_MK_M[h_M(K_M) - \bar{h}_M]$, free of L, ε_Λ and s on either reading	ramp-on: $pyq > V_M \cdot L \cdot \Lambda_M \varepsilon_\Lambda$; arrival-on: $pyq > V_M\Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$; both carry L and ε_Λ	**Nested, not incommensurable.** Under the comparison convention of §4b.1 the gated level-set test is the speed-set arrival-on test at $\varepsilon_\Lambda = 0$, and the two thresholds differ by exactly $V_M\Lambda_M\varepsilon_\Lambda(L - \tau_M)$, positive under A4 whenever $\tau_M < L$. So level-set passes the coupled test for every benefit in that interval and speed-set fails it, **never the reverse**.	ε_Λ, L	A4a.4(vi), ramp-on and arrival-on; level side gated, A4a.1(i) with $*\bar{h}_M* = 0$

#	Case	Level-set	Speed-set	Relation	Deciding quantity	Proof
				The gated level-set test is *pyK_B > V_MK_M·h_M(K_M)*, whose parameters are a subset of the speed-set arrival-on test's; on the ramp-on reading the two tests depend on different parameters		

In each ramp-on Relation entry, the two signs come from the cited proofs and the stated relation is their product's sign, so the entry is proved within one reading. The gated arrival-on entries are proved the same way within the arrival-on reading. Substituting *h̄_M* = 0, the level-set derivatives are *τ_Mh_M(K_M)* in row 1, *h_M^post·τ_M* in row 4, *τ_Mh_M(K_M) − τ_D[h_M(K_M) − h_M^post]* in row 5, and *τ_Mh_M(K_M)* again in rows 6 and 7, against the speed-set arrival-on derivatives of A4a.4(i), (iv) and (vi). In row 2's regime the gated hazard is zero throughout, so both sides are exactly zero. Rows 1, 2, 4 and 6 then cannot take the opposite strict sign, because a hazard is not negative; row 5 can, because its level-set test compares distances and its speed-set test compares an elasticity with a share. Row 7 compares thresholds rather than signs, and under the comparison convention above it is not a disagreement between incommensurable tests but one test at two elasticities. The gated profile is a step: the hazard is constant from arrival onward, which is what a constant *Λ_M* means. So on the arrival-on reading the gated level-set marginal is the speed-set marginal at *ε_Λ* = 0, and the two thresholds on the coupled benefit differ by exactly

V_MΛ_Mε_Λ(L − τ_M),

positive under A4 whenever *τ_M < L*. Three things follow. The speed-set threshold is the higher one, always, so a benefit between the two passes the level-set test and fails the speed-set one and never the reverse. Under A4, which imposes *ε_Λ* > 0 throughout this section, the two thresholds differ at every admissible parameter point, so there is no

frequency to report. And row 7 is an instance of the whole-balance interval rule below, read with the non-misuse remainder Υ set to *pyq* and every other term at zero: the specifications rank speed oppositely exactly when Υ lies strictly between $V_M\Xi_\ell$ and $V_M\Xi_s$. That rule holds on either reading, gated or not, so the coupled-benefit test was already inside the identification problem and this row adds no reach to it. *Numerical check* (the gated comparison, six separate numerical routes): closed forms matched finite differences of a directly integrated loss to between 3×10^{-9} and 5×10^{-10} across 740 draws for rows 1, 4 and 5, and to 3.1×10^{-9} across 288 draws for rows 2, 6 and 7; the nesting at $\varepsilon_\Lambda = 0$ and the gap $\Lambda_M\varepsilon_\Lambda(L - \tau_M)$ matched the closed forms to 3.8×10^{-16}, and the gap was positive at every admissible draw.

The thresholds. On the defence rows, with the residual fixed and assuming $h_M(K_M) > h_M\text{^}post$ and $\Lambda_M > h_M\text{^}post$:

$\bar{\alpha}_\ell = \tau_M[h_M(K_M) - \bar{h}_M]/[t_D(h_M(K_M) - h_M\text{^}post)]$, $\bar{\alpha}_s = \varepsilon_\Lambda\Lambda_M/(\Lambda_M - h_M\text{^}post)$.

Proof. Substitute $s\cdot t_D' = -\alpha_Dt_D$ into A4a.3 and into (0.2) with a fixed residual. This gives $sC_M' = \tau_M[h_M(K_M) - \bar{h}_M] - \alpha_Dt_D[h_M(K_M) - h_M\text{^}post]$ under level-set and $sC_M' = \Lambda_Mt_D\varepsilon_\Lambda - \alpha_Dt_D(\Lambda_M - h_M\text{^}post)$ under speed-set ramp-on. Dividing by the positive t_D times each hazard gap gives $sC_M' < 0 \iff \alpha_D > \bar{\alpha}_\ell$ and $\iff \alpha_D > \bar{\alpha}_s$ respectively. ■ At a given s, t_D and t_D' are separate numbers, so each is a genuine threshold on α_D.

Arrival-on. With $t_D > \tau_M$ and $\Lambda_M > h_M\text{^}post$, $\bar{\alpha}_s\text{^}arr = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M]/[t_D(\Lambda_M - h_M\text{^}post)]$. *Proof.* Substitute $s\cdot t_D' = -\alpha_Dt_D$ into A4a.4(iv). This gives $sC_M' = \Lambda_M[\varepsilon_\Lambda(t_D - \tau_M) + \tau_M] - \alpha_Dt_D(\Lambda_M - h_M\text{^}post)$. Divide by $t_D(\Lambda_M - h_M\text{^}post) > 0$. ■ The division needs $\Lambda_M > h_M\text{^}post$. If $\Lambda_M < h_M\text{^}post$, the inequality reverses: acceleration then lowers loss iff $\alpha_D < \bar{\alpha}_s\text{^}arr$. $t_D > \tau_M$ keeps the point inside the regime where the two-sided derivative exists. Under A4 the numerator is positive, so $\bar{\alpha}_s\text{^}arr > 0$. *Worked instances.* - (a) A4a.4(iv)'s values at $K_D = 6$ give $\alpha_D = 6/8 = 0.75$ and $\bar{\alpha}_s\text{^}arr = 0.12/(8\cdot0.015) = 1.0$. α_D is below the threshold, so acceleration raises loss, matching $sC_M' = +0.03$. - (b) $K_D = 10$, $m_DT_D = 1$ and $h_M\text{^}post = 0$ give $t_D = 11$, $\alpha_D = 10/11 = 0.909091$ and $\bar{\alpha}_s\text{^}arr = 0.15/0.22 = 0.681818$. α_D is above the threshold, so acceleration lowers loss, and indeed $sC_M' = 0.15 - 10\cdot0.02 = -0.05$. - (c) (a)'s clock with $h_M\text{^}post = 0.05 > \Lambda_M$ gives $\bar{\alpha}_s\text{^}arr = 0.12/(8\cdot(-0.03)) = -0.5 < \alpha_D = 0.75$, so the unqualified rule would say "helps". But $sC_M' = 0.12 - 6(0.02 - 0.05) = +0.30$. This is the reversal.

The rule pairs $\bar{\alpha}_\ell$ with $\bar{\alpha}_s$. It does not pair $\bar{\alpha}_\ell$ with $\bar{\alpha}_s\text{^}arr$. $\bar{\alpha}_\ell$ is built from the ungated level-set path, which is ramp-on, and the arrival gate is an additional exposure specification that the ungated level-set path does not imply, so that pairing would cross readings. If the denominator assumptions fail, compare Ξ_ℓ and Ξ_s directly.

A responsive residual. Each derivative receives $(L - t_D)s\cdot h_M\text{^}post'$ (A4a.3; A4a.4(v)). Each threshold then shifts by that term divided by t_D times its own hazard gap: α_D must exceed $\bar{\alpha}_\ell + (L - t_D)s\cdot h_M\text{^}post'/[t_D(h_M(K_M) - h_M\text{^}post)]$, $\bar{\alpha}_s + (L - t_D)s\cdot h_M\text{^}post'/[t_D(\Lambda_M - h_M\text{^}post)]$, or $\bar{\alpha}_s\text{^}arr + (L - t_D)s\cdot h_M\text{^}post'/[t_D(\Lambda_M - h_M\text{^}post)]$. *Proof.* Repeat each substitution and division above, with the added term on the side opposite α_D. ■ The shifted thresholds sign the derivative. The unshifted ones do not.

The threshold rule applied to rows 3 and 4. This is a substitution check and adds no result. On the *K_M*-paced capability branch, $t_D = \tau_M + m_DT_D$, so $s \cdot t_D' = -\tau_M$ and $\alpha_D = \tau_M/t_D$. Then $\alpha_D > \bar{\alpha}_s$ is $\varepsilon_\Lambda < (\tau_M/t_D)(1 - h_M\text{^}post/\Lambda_M)$, row 4's ramp-on condition, and $\alpha_D > \bar{\alpha}_\ell$ is *h̄_M > h_M^post, *row 4's level-set condition. Setting* m_DT_D* = 0 and $h_M\text{^}post = 0$ gives $\alpha_D = 1$ and $\bar{\alpha}_s = \varepsilon_\Lambda$, which is row 3. On the adoption branch $\alpha_D = 0$, and the general formulas of §4a apply.

The whole-balance interval. With $\Upsilon = s(W_L\text{^}X + V_MC_M)'$ assumed common to both specifications, and *V_M* fixed, each specification's marginal is $sW_L\text{^}X' = \Upsilon - V_M\Xi_i$, for $i \in \{\ell, s\}$. *Proof.* Two real numbers have opposite strict signs iff their product is negative. So the specifications disagree iff $(\Upsilon - V_M\Xi_\ell)(\Upsilon - V_M\Xi_s) < 0$. That product is a quadratic in Υ with roots $V_M\Xi_\ell$ and $V_M\Xi_s$ and a positive leading coefficient, so it is negative iff Υ lies strictly between the roots. ■ The specifications share a strict sign iff Υ lies strictly below both endpoints or strictly above both. At a distinct endpoint, one specification is neutral.

Table 4b.B. The parts of the §4b.2 construction, by specification. Only the hospital margin is computed, as a bound on one channel graded WEAK, and it has been tested against no witness. No other part is done under any column. The arrival-on signs this table relies on are proved in A4a.4.

Part	Level-set	Speed-set, ramp-on	Speed-set, arrival-on	Status
Target	branch join at $s = K_M/L$	no misuse kink	activation kink at $s = K_M/L$	defined; not fixed for a pair
Ω_E	same rows	same rows	same rows	rows listed; not formalised
Witnesses	rows 3–4, with facility and other added derivatives at zero: flip from hospital term or Φ only; otherwise V_M can flip through C_F. Truncation with a flat traversed profile and zero facility derivative: misuse neutral, V_M cannot flip	rows 3–4: V_M can flip; truncation: misuse positive under A4	signs proved (A4a.4(i), (iii), (iv))	none written
Restriction	on the level profile or the	on ε_Λ	on ε_Λ and the gap $K_D - K_M$	family not defined

Part	Level-set	Speed-set, ramp-on	Speed-set, arrival-on	Status
	two distances			
Hospital margin	tested against the level-set margin	tested against the speed-set margin	tested against the arrival-on margin	bound computed on one channel, WEAK; tested against no witness
Information	fixed, or learning added	fixed, or learning added	fixed, or learning added	not done

A9. Derivations for §9 and §10

None. The conditions cited in §9 and §10 are proved in A4a.3, A4a.4 and A4a.7 above.

A-open. What is not proved

"Open" here means not proved. Every result above is proved from A0, and none of them waits on anything. A second item that stood here until 27 October has also gone: the level-set half of "R3 does not determine the fixed-speed facility comparison, and the comparison does not determine R3", which was stated without a counterexample and now has one in §4a.5. One item that stood here until 27 October has gone: the arrival-on Relation entries of rows 2, 6 and 7 of Table 4b.A, which set the ungated level-set derivative beside the speed-set arrival-on one and so compared across readings. They are now gated, like the rest of the table. Rows 2 and 6 keep what they claimed — row 2 agrees with both sides exactly zero, because the gated hazard never switches on in the truncation regime, and row 6 agrees with both sides positive. Row 7 is of a different kind: it compares thresholds on a coupled health benefit, not signs, so it rests on the comparison convention stated before Table 4b.A, and under that convention the two tests are nested, the gated level-set one being the speed-set one at $\varepsilon_\Lambda = 0$. Both are now proved and stated in §4b.1 and A4a.4(vi). An earlier draft of this section said instead that the two tests were not of the same form and that the row widened the identification problem; both claims were wrong and are withdrawn.

1. **Level-set arrival-on with the defence before arrival.** The set-up does not say whether the residual runs from t_D or from τ_M in this case, and no result is claimed for it.
2. **A finite move of the whole balance across the activation kink** (§4b.2, Table 4b.B). The misuse-loss part is exact (A4a.4(iii)), but the balance part is not done.
3. **Monotonicity in *s* of the right side of the speed-set test in R2 with the benefit** (A4a.4(vi)). Its s-derivative involves ε_Λ', which is unrestricted.
4. The remaining items of §4a.8 (gradual coverage other than the K_D derivative of A4a.7(i), discounting under speed-set R3, more than one severity class, the whole facility balance, V_M varying with the level reached, and the rest) are not proved here. Which convention describes a real facility is an empirical question that no proof settles.

Supplement (not part of the argument): how the results were derived and checked

This supplement is the record of how the model's results were first obtained, how they were checked, and how they came to be proved. The argument of the paper does not rest on it. Every result the paper uses is proved in the Appendix, and a proof needs no history to be valid. The record is kept for a reader who wants to know how the work was done, and for a reader of an earlier draft who wants to map its labels onto this one.

S.1 The arrangement

The model's results were obtained in three check rounds. Each question in a round was put to two models, which worked separately and blind to each other's answers:

- the first pair under level-set [MC-S, MC-O];
- the second under speed-set [MC2-S, MC2-O];
- the third on the facility comparison at a fixed speed [MC3-S, MC3-O].

S and O denote the two models, and they were the same pair in every round. A Q-number names the question under which a result was derived. A blind read of the draft and a blind read of the assembled paper corrected several results along the way.

Two numerical checks then integrated the loss directly. [NUMCHECK] integrates the model's own definition of misuse loss, $C_M = \int_0 \hat{} L$ (hazard) dt, by adaptive quadrature with breakpoints at the kinks. It compares the printed closed forms against central finite differences of that integral, over seven hazard shapes, residuals above and below the plateau, both clock branches, both conventions and both readings. It was re-run for this version, and every row of its derivative test passes. A second implementation was written separately for this version and does not import [NUMCHECK]'s code. Under speed-set the hazard is piecewise constant in time, so this implementation builds the loss exactly from the set-up, with no quadrature, and compares a central finite difference of it against each closed form.

Finally, every result was proved. The proofs were written in two parts, one for the speed-set, responsive-residual and benefit results and one for the level-set facility derivatives. A third reader then checked them line by line, recomputed every worked instance, and asked of each step whether it follows from the one before and from the stated set-up. That check found sound every proof it covered, which was the first pass of the appendix; the proofs written later in the same session — those listed in S.5 and rows E1–E4, G and H of S.4 — have not had a line-by-line check, and S.5 says so. It found one false sentence in a justification, which is not carried; several qualifiers and one condition missing from assumption lists, all of which are now supplied; and four tagged results that no proof yet covered, which are now proved in the Appendix (S.4, rows E1–E3). The Appendix carries the corrected proofs.

S.2 Why “separate” and not “independent”

Earlier drafts called the two model checks “independent”. The word claims more than the arrangement supports. The same pair of models ran every round, and nothing establishes that their errors are uncorrelated. Two runs of related models, trained in related ways and given the same question, can make the same mistake, and when they do, their agreement is no evidence against the mistake. What the arrangement gives is two derivations obtained separately and blind. That is worth something, because a slip in one is unlikely to be repeated in exactly the same form in the other, but it is not independence in the sense that would let a reader multiply error rates. The paper now says “separate”.

The numerical checks are different in kind, and this is the distinction the apparatus exists to draw. A model check and a proof both reason from the model to a formula. A numerical check does not reason to the formula at all. It takes the model’s definition of the loss, integrates it, differences it, and compares the result with the formula. It shares no step with any derivation. It cannot reproduce an algebraic slip, because it does no algebra, and it fails at once if the formula is wrong at the points it samples. In the relevant sense, it *is* independent evidence. It has its own limits. It samples parameter points rather than covering a domain. It cannot tell whether the conditions a result states are the conditions it needs. And it can itself be wrong: a first version of the second implementation reported two false failures, both artefacts of dividing by a closed form that passes through zero. That is why the Appendix gives the proof as the warrant and the numerical check as a separate line beside it, with its worst error. The derivation history in this supplement is a third kind of record, which says who first wrote a result down. It is neither a warrant nor a test.

S.3 What the tags meant, and why they are gone

Earlier drafts marked each model result with a derivation tag.

Old tag	What it meant	What replaces it
[both]	both checks of the round derived the result	a proof in the Appendix
[both-form]	one check stated it, and it follows by substitution into a formula both derived	a proof in the Appendix, usually one line from a lemma
[O-only], **[S-only]**	one derivation only, with the result marked as awaiting independent corroboration	a proof in the Appendix. The cross-reading comparisons of Table 4b.A were the last to carry such a tag; they are now compared on one reading and proved, and A-open no longer lists them
[J]	the paper’s own judgment, not a derived result	unchanged: a judgment is not a derivation, and it is still marked as one
[NUMCHECK]	the numerical check	unchanged: it cites evidence

Old tag	What it meant	What replaces it
		of a different kind and stays as a citation

The tags recorded who had derived a result. They did not record whether it was true. A result could hold a numerical verification to ten decimal places and still be printed as awaiting corroboration, because the scheme's categories were about authorship. Once a result is proved, the proof is the warrant and the tag has nothing left to say. Where a result is not proved, the paper now says what is open, and never that only one model derived it.

S.4 The record, result by result

"Proof checked" means that the line-by-line check described in S.1 passed the proof. Rows E1–E4 were proved after that check, in the pass that wrote the Appendix, and have not yet had a line-by-line check of their own. Numerical-check figures are the worst relative or normalised error against finite differences of the directly built or integrated loss, as recorded for this version. "—" means that this version's record lists no numerical check. The last column gives the figures the proving pass's own scripts reported. Those scripts were not re-run for this version, and their figures are not carried in the Appendix.

Machine-checked in a proof assistant. Every result in the project's inventory — rows 1 to 16 — and every one of the seven rows added after it (E1 to E4, G, H and I) is formalised in Lean 4 (toolchain leanprover/lean4:v4.35.0) against Mathlib: the speed-set arrival-on family, the level-set facility derivatives, the responsive residual, the benefit margin's rate derivative, the separable mixed hazard, the proportional-cut identity, the gated arrival-on entries of Table 4b.A including the corrected row 7, and the K_D invariance of §4a.5. Three hundred and forty-six theorems check; the audit prints three hundred and forty-four lines — eighty for the inventory, the two step-integrability helpers being reached through a third, and one for every theorem in the two later modules — and every line depends on nothing beyond propext, Classical.choice and Quot.sound, the three axioms every Mathlib proof uses; no source file contains a sorry. The artefact is the project's BioPaperProofs directory: seven modules carrying theorems — StepIntegral.lean, Item1.lean, Group1.lean, LevelSet.lean and Group34.lean for the inventory, PostE.lean and PostGHI.lean for the rows added after it — with the audit in Axioms.lean, Mathlib pinned by lake-manifest.json, and a README mapping each inventory row to the theorems that carry it and pointing to the two ledger tables for the seven rows added after it. It is checked by lake build and then lake env lean BioPaperProofs/Axioms.lean, in that order: a failed build leaves the audit reading a stale object file and printing nothing, which looks like nothing to report rather than like nothing run. Both commands reach every module: the root imports Basic.lean, which imports all seven, and the audit prints the two later modules' theorems with the rest. What is **not** formalised matters as much, and the rule is the simple one: the formal warrant reaches the twenty-three rows and stops there. Everything else in this appendix carries none unless the README names a theorem for it — the level-set threshold $\bar{\alpha}_\ell$, the ramp-on Relation entries of Table 4b.A, which the second pass left outside its scope, and, among the results S.5 lists, the level-set discounted form, the speed-set ramp-on forms and the threshold $\bar{\alpha}_s$, and Result B's probability margin. Those rest on a proof, and on a numerical

check where the table below records one, as every result here did before the first pass. The items A-open lists are not proved at all, and Ω_E is an evidence set rather than a result. Six things about what the formalisation certifies are worth stating precisely.

First, misuse loss is **defined** there as the integral of the hazard, and the step to each closed form is proved rather than assumed. Under speed-set that step is the integral of a piecewise-constant hazard. Under level-set it is a change of variable turning a date integral into the capability integral $\int_0^{K_M} h_M$, which is why the level-set results carry a ramp average at all. An earlier formalisation of row 1 had taken its closed form as a definition and said so in its own notes.

Second, the derivative results are proved of the integral, which requires each closed form to hold on a neighbourhood of the point being differentiated and not only at it, and that is proved in every case. Row 11's condition $t_D < L$ is where this bites: the closed form is differentiable in K_M whatever L does, and it is the loss only while the ramp lies inside the counted window — which is what the counterexample in A4a.7(vi) shows.

Third, three steps the appendix gives in prose are now proved as steps: that the ramp integral's moving upper limit and the shrinking plateau contribute $+h_M(K_M)/s$ and $-h_M(K_M)/s$ and therefore **cancel**, leaving convention B's ramp-on derivative $(t_D - \tau_M)h_M'(K_M)$ with no level term; that a nondecreasing hazard's ramp average never exceeds its terminal value, which is what signs R1 and R2 by monotonicity alone; and that $x_d\hat{h}_M(x_d) - \int_0^{x_d}\hat{h}_M$ equals $\int_0^{x_d}[\hat{h}_M(x_d) - \hat{h}_M(x)]dx$ and is therefore nonnegative, which is what signs row 11.

Fourth, two labels are now settled by the structure of the proofs rather than by a sentence. Row 13's term arrives from differentiating the residual *inside* the interval $[t_D, L]$, with both limits fixed, so it is Leibniz's interior term and not a moving-boundary term; the project's own inventory had called it the latter, which S.7 records as a mislabel, and the formalisation puts the two kinds of term side by side in one statement. And row 15's bracket is now proved to *be* the margin's logarithmic derivative, by factoring the derivative as the margin times the bracket, rather than being an expression that resembles one.

Fifth, what a proof assistant cannot check is that the formalised statement is this paper's claim. The formalisation records the choices a reader must agree with: the hazard shapes taken for the two specifications and the two exposure readings; the writing of every speed-set statement with Λ_M' rather than dividing by Λ_M to form ε_Λ; convention A as $\hat{h}_M(c/K_M)$ and convention B as the fixed profile; and that rows 5, 6, 9, 12, 14 and 16 certify algebraic equivalences — a sign condition rearranged into a threshold, a net marginal into the printed inequality, the first-order combination of two cuts, a ratio of two partials into the facility slope, R3's inequality with the residual term added, and the exposure-response threshold $*\bar{\varepsilon}_g*$ — with the marginals themselves supplied by the rows proved of the integral. Formalising found no error in any of the sixteen results as the appendix now states them — which is the honest form of the claim, because the passes changed what two of them say: row 13's added term is relabelled, and row 11's $t_D < L$ moved from a condition the prose stated to a step its proof needs. The defects the passes

found were otherwise in the formalisations themselves, four of which compiled and passed the axiom audit while standing for something weaker than the row they were written for.

Sixth, formalising the seven post-inventory rows changed what several of their sentences say, and that is the main thing the second pass produced. Eighty-seven atomic claims were read out of those rows. Sixty-two held as the paper stated them, twenty-four were narrowed, one is an illustrative magnitude, and none was left unproved. Ten were false as printed, each with a counterexample in Lean: a deployment-date formula that holds on one clock branch and not the other; a reason given for a direction, where the direction holds and the reason does not; a residual term that takes a different form under gradual coverage; a failure claimed universally that does not occur on the adoption branch; two coupled-benefit tests that pass while the balance falls, once an attack delay pushes the benefit's actual arrival past the horizon; two positivity claims on the plateau that fail at a zero misuse capability; an exposure length that is wrong when the deployment step falls at or beyond the horizon; and an unqualified claim that an identity fails under convention B, which in fact holds for a constant profile. The other fourteen supply a side condition the prose had not stated: a positive loss per misuse event, which the set-up now fixes; a positive misuse capability, and an arrival inside the window, at the places where those are conditions rather than counterexamples; continuity where a hazard is read off at a date; continuous differentiability where a one-sided limit is taken; a capability branch with a gap between plateau and residual hazard where an "only at" claim is made; and a fixed residual and a rigidly translating coverage schedule in the abstract's invariance sentence. One further narrowing is a caption reworded rather than a condition added. Every one of them is corrected above. Read with the narrowed conditions in place, the pass changed no result's direction, no table entry's sign and no conclusion the paper draws — a statement about the corrected text, which the tables do not themselves certify: the narrowings move conditions into the statements that carry them, which is the point of formalising a paper after it is written rather than before.

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
1	R2 speed-set arrival-on, $sC_M' = \Lambda_M[\varepsilon_\Lambda(L - \tau_M) + \tau_M]$	A4a.4(i)	MC2-O [O-only]	yes	yes	4.0×10^{-9} (348)	1.1×10^{-12}
2	the same, discounted	A4a.4(ii)	MC2-O [O-only]	yes	yes	1.2×10^{-9} (96)	4.8×10^{-13}

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
3	R1 speed-set arrival-on: zero below *K_M/L*, right-hand value *Λ_ML*	A4a.4(iii)	MC2-O [O-only]	yes	yes	exactly 0 (22)	0; 3.8×10^{-9} at the kink
4	R3 speed-set arrival-on	A4a.4(iv)	MC2-O [O-only]	yes	yes	1.7×10^{-10} (54)	4.0×10^{-12}
5	threshold *ᾱ_s^arr*, needing *t_D > τ_M* and *Λ_M > h_M^post*	A4b	MC2-O [O-only]	yes	yes	—	sign rule 0 mismatches / 262; reverses 96 / 96 when *Λ_M < h_M^post*
6	arrival-on coupled benefit test	A4a.4(vi)	MC2-O [O-only]	yes	yes	—	4.1×10^{-12}
7	level-set arrival-on *∂C_M/∂K_M*; *−h_M(K_M)/s* under A	A4a.7(iii)	MC3-O [O-only]	yes	yes	1.2×10^{-10} (504)	2.2×10^{-11}

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
8	the same under B; unsigned; decided by the level elasticity	A4a.7(iii)	MC3-S [S-only]	yes	yes	2.1×10^{-8} (504)	1.7×10^{-10}
9	convention-B facility inequalities, ramp-on and arrival-on	A4a.7(iv)	MC3-S [S-only]	yes	yes	—	sign agreement 1269 / 1269
10	defence before capability, ramp-on, B: zero	A4a.7(vi)	MC3-S [S-only]	yes	yes	7.6×10^{-10} worst over the four level-set before-arrival rows (42 each)	9.1×10^{-12}
11	the same under A, with $t_D < L$ (or $\min(t_D, L)$), sign ≤ 0, no jump at $t_D = \tau_M$	A4a.7(vi)	no model check derived it; [NUMCHECK] derived and matched it (untagged)	yes	yes	as row 10	5.3×10^{-11}; counterexample without $t_D < L$ confirmed
12	arrival-on slope	A4a.7(iv), (v)	MC3-O [O-only]	yes	yes	—	affine error

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
	h/(h − P) under A; affine algebra under A						1.6×10^{-14} (ramp-on), 1.5×10^{-14} (arrival-on)
13	responsive residual adds *(L − t_D)s·h_M^post′* (the interior term)	A4a.4(v)	MC-S [S-only]	yes	yes	1.1×10^{-10} (24)	2.2×10^{-12}
14	level-set counterpart, *(L − t_D)s²h_M^post′*	A4a.3	MC-S [S-only]	yes	yes	—	2.9×10^{-12}; inequality 0 mismatches / 180
15	benefit margin's log rate derivative; rises for *r < ln(1 + ℓ_B/t_B) /ℓ_B*	A4a.6	MC-S [S-only]	yes	yes	7.4×10^{-10} (18)	3.7×10^{-12}; 1.3×10^{-8}; sign 0 mismatches / 45
16	separable mixed form, ramp-on, *τ_M ≤ t_D ≤ L*	A4a.5	MC2-S [S-only]	yes	yes	—	2.4×10^{-8}; 1.8×10^{-8}; sign rule 0 mismatches / 270

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
E1	proportional-cut identity: level-set A; speed-set with the intensity term	A4a.7(vii), Lemma H	level-set A: MC3-O [O-only]; speed-set ramp-on [both-form]; arrival-on [O-only]. The speed-set form was corrected on 27 Oct after a separate check found it wrong; the proof is of the corrected form	yes	not yet	— (an earlier [NUMCHECK] run recorded 9.7×10^{-11} and 9.0×10^{-11} under A and 8.2×10^{-11} and 5.9×10^{-11} under speed-set, with failure under B; that run is not in this version's record)	—
E2	arrival-on *K_D* coefficient *[Λ_M(ε_Λ − 1) + h_M^post]/s*	A4a.4(iv)	MC2-O [O-only]	yes	not yet	—	—
E3	one-sided limits at *t_D = τ_M*,	A4a.4(iv)	MC2-O [O-only]; the *K_D = K_M*, *m_DT_D*	yes, given the choice	not yet	—	—

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
	own-distance clock		= 0 case from the blind read of the assembled paper. The defence-before-arrival side now rests on a stated modelling choice (A0)				
E4	additive mixed form on the plateau, arrival-on: no positive elasticity lowers loss	A4a.5	MC2-O, marked in words as awaiting corroboration but not tagged	yes, under either gating of the level component	not yet	—	—
G	arrival-on Relation entries of rows 1, 4 and 5 of Table 4b.A, compared like with like with the	A4b, Table 4b.A	MC2-O [O-only], first posed across readings; gated by the author's ruling of 27 Oct, then [both-	yes, by substituting the gated profile (*h̄_M* = 0) into A4a.1(i) and A4a.3	not yet	3×10^{-9} to 5×10^{-10} (740 draws, four separate numerical routes; six routes in all once	—

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
	level-set side gated		form]			rows 2, 6 and 7 are included, row H)	
H	arrival-on Relation entries of rows 2, 6 and 7 of Table 4b.A, compared like with like with the level-set side gated	A4b, Table 4b.A	MC2-O [O-only], first posed across readings; gated by the author's ruling of 27 Oct, then [both-form]	yes, by substituting the gated profile (*h̄_M* = 0) into A4a.1 and A4a.3	not yet	rows 2 and 6 as claimed. **Row 7 was stated wrongly on the first pass and restated after review** (S.7): the two tests are nested at $\varepsilon_\Lambda = 0$, not of different form, and the row adds no reach to the identification problem	3.1×10^{-9} (288 draws); row 2 exactly zero on both sides, 0 oppositions / 72; the nesting and the gap $\Lambda_M\varepsilon_\Lambda(L - \tau_M)$ to 3.8×10^{-16}, gap positive at every admissible draw
I	the K_D invariance explained: an agreement of	A4a.7(i)	derived in this pass, after a review of 27 October	yes	not yet	9.9×10^{-10} over the six specification-by-reading cases on	—

#	Result	Appendix	First derived by (old tag)	Proved	Proof checked	Numerical check, this version	Proving pass's own script
	form at each case's own plateau hazard, holding because *K_D* enters only at the deployment date; elementary as a shift in time; extending to gradual coverage that translates rigidly with *t_D* and completes before *L*, and able to fail when it does not		asked whether the result was an identity rather than a finding			144 capability-branch draws, and exactly zero on 96 adoption-branch draws; gradual coverage 1.1×10^{-9} (1,440 case-draws); the horizon-truncated replacement matched to 2.1×10^{-9} at draws where the invariant form is wrong by construction	

S.5 Results formerly tagged [both] or [both-form]

These were derived twice, or by substitution into a twice-derived formula. Each now has a proof in the Appendix, written in the same pass as rows E1–E4 and not yet given a line-by-line check:

- the level-set plateau, truncation and closure at arrival (A4a.1, Lemma L);
- the level-set discounted form (A4a.2);
- the level-set R3 branches and the two-distance inequality (A4a.3);
- speed-set ramp-on R2, R1 and R3, with their branch thresholds and the ramp-on K_D coefficient (A4a.4);
- the additive mixed form (A4a.5);
- Result B and its corollary (A4a.6);
- the general ramp-on facility formula, the K_D derivative and the speed-set facility forms (A4a.7);
- the thresholds $\tilde{\alpha}_\ell$ and $\tilde{\alpha}_s$, and the whole-balance interval (A4b).

The cyber term in a common horizon (A4a.0) came from the blind read of the draft and is proved in the same pass. The separable form's gap from the calendar share, also from that blind read, is part of row 16's checked proof.

S.6 Arithmetic on sourced numbers

The hospital bound's anchors are arithmetic on sourced inputs, not model results, though earlier drafts gave them derivation tags too. With d = 0.568–0.905 excess deaths per attack (42/74 and 67/74, as the paper prints them):

- displaced-death ceiling: $d \times 30/365.25$ = 0.046652977… to 0.074332648…, printed as 0.047–0.074 (a ceiling, not a value: the marginal quantity $SE(a) - SE(a + \delta)$ is strictly below δ, and 65 to 87 per cent of it on a reference table calibrated to $SE(72)$ = 21.62 — PROOFCHECK/anchors.py);
- averted-death anchor, standard valuation: $d \times 21.62$ = 12.28016 to 19.5661, printed as 12.3–19.6;
- averted-death anchor, actual-expectancy valuation: $d \times 12$ = 6.816 to 10.86, printed as 6.8–10.9;
- the factor between the two standard-valuation anchors, which is the counterfactual's and not the unit's: 21.62 × 365.25/30 = 263.2235, printed as about 260;
- the upper-limit figure, more adverse specification: (0.0123 + 1.96 × 0.0047) × 5,447/74 = 0.021512 × 5,447/74 = 1.58346 deaths per attack, and 1.58346 × 21.62 = 34.234, printed as about 34κ. With the input rounded to 1.584, the product is 34.246;
- its lower limit, the other end of the same rule: (0.0123 − 1.96 × 0.0047) = 0.003088 times 5,447/74 = 0.22729 deaths per attack, and 0.22729 × 30/365.25 = 0.018666, printed as 0.019κ; the interval excludes zero, so this rule's lower region is not empty;
- the main specification, both ends: 0.0077 ± 1.96 × 0.0042 = −0.000532 to 0.015932, over an implied 5,454.5 patients from 42 ÷ 0.0077, so 0 to 1.17434 deaths per

attack; 1.17434 × 21.62 = 25.389, printed as 25.4κ, with a lower end of exactly zero because the interval includes zero;

- the displaced-death ceiling at the two standard-loss ages: $SE(a) - SE(a + \delta)$ is 65 to 87 per cent of δ = 0.082136 years on a Gompertz hazard calibrated to $SE(72)$ = 21.62, slopes 0.06 to 0.12 (PROOFCHECK/anchors.py);

All of these were recomputed and reproduce.

S.7 What proving changed

Proving the results did more than certify them. It found the following.

- **Conditions the paper had omitted.** The separable form needs ramp-on exposure and $\tau_M \le t_D \le L$. The before-arrival convention-A form needs $t_D < L$, or x_d written with $min(t_D, L)$, and a counterexample shows the printed form failing without it. The arrival-on threshold needs $\Lambda_M > h_M\text{^}post$, stated one paragraph away in earlier drafts, and the rule reverses without it.
- **Signs the paper had not stated.** Under convention A, the before-arrival derivative is ≤ 0. Under B, the arrival-on K_M derivative is signed exactly by the hazard's level elasticity against $\tau_M/(t_D - \tau_M)$. The benefit margin rises with r on an explicit interval, not only "at small enough r".
- **One understatement.** With a responsive residual, earlier drafts said the thresholds "no longer sign" the derivative. Shifted by the residual term, they do (A4b).
- **One mislabel.** The responsive-residual term is Leibniz's interior term, not a moving-boundary term.
- **One closure.** Misuse loss under speed-set arrival-on is exact across the activation kink, so that part of an open item closes. The balance part stays open.
- **The project's inventory is now machine-checked, all sixteen rows.** Five modules, eighty-two theorems, no sorry and no axiom beyond Lean's three standard ones. The inventory is not the whole appendix: S.4 lists the results that carry no formal warrant; the K_D invariance was among them until the pass that followed formalised it as row I. Three further things came out of the last pass. Row 13's added term is Leibniz's *interior* term, which the formalisation shows by putting it beside the moving-boundary term in one statement — the project's inventory had called it a moving-boundary term, and this is the second place that mislabel is corrected. Row 15's bracket is proved to be the margin's logarithmic derivative, not merely to resemble one. And row 16's closed form reuses the level-set split unchanged, because the separable mixed hazard *is* the level-set hazard scaled by $\Lambda_M(s)$: the specification sets a level and nothing else in that object. The K_D invariance of §4a.5 turns on the same kind of fact, but row I was outside that pass and nothing in it bears on the invariance.
- **What machine-checking added, the level-set pass.** Formalising the level-set facility derivatives found no error in them either, and turned three prose steps into proofs: the cancellation of the two boundary terms under convention B ramp-on, the inequality *h̄_M ≤ h_M(K_M)* that signs R1 and R2 by monotonicity alone, and the

identity that signs row 11. Row 11 is the one result in the project's inventory that no model check derived — the numerical check derived and matched it — and it is now machine-checked, integral and all. It also made one condition load-bearing in the record rather than in a sentence: row 11's $t_D < L$ is what connects its closed form to the loss.

- **What machine-checking added, the Group 1 pass.** Formalising Group 1 in a proof assistant found no error in those six results. It changed the status of one step and made two implicit ones explicit. The step: an earlier formalisation of row 1 defined misuse loss as the closed form, so the integral lay outside what was checked, and it is now inside. The two: differentiating the loss needs the closed form on a neighbourhood of the speed rather than at it, and the unreached region's zero derivative needs the loss to vanish on a neighbourhood rather than at a point. The paper's prose had both right — it says "throughout the unreached region" — but neither was written as a step.
- **What machine-checking added, the post-inventory pass.** Formalising the seven rows that postdate the inventory — E1 to E4, G, H and I — narrowed twenty-four of their eighty-seven atomic claims and, once those narrowings are in place, left every direction and sign standing. Ten were false as printed. The one that mattered most was a formula for the deployment date, $\chi_DK_D/s + m_DT_D$, written in a passage explaining *why* the K_D invariance holds: it agrees with the model's own clock on the capability branch and not on the adoption branch, where the clock is T_D and the formula gives m_DT_D. The invariance itself is proved; the explanation had quoted the wrong clock. The others are of a kind: a reason offered for a direction that holds without it, a residual term that changes form when coverage is gradual rather than stepped, two benefit tests that pass while the balance falls once an attack delay pushes arrival past the horizon, two positivity claims that fail at a zero misuse capability, and a failure asserted for every case that occurs in some. The fourteen narrower ones add a condition rather than correct a claim — a positive loss per misuse event, continuity, a fixed residual. S.4's sixth point lists both sets.
- **Two claims of novelty, narrowed.** The paper called the K_D result new and said it held under every choice. The invariance is real but structural: K_D enters the loss at one point, each of the three choices governs another object, and written as a shift in time the derivative is elementary, so what is the paper's own is the capability-set and adoption-set division of the wait and the floor that follows. Deriving it also found a condition the claim had not carried, that the countermeasure schedule complete inside the counted window, and an extension the step had hidden, to gradual coverage (A4a.7(i)). Truncation was called the paper's most novel result on the ground that it exists under one specification only; it rests equally on the objective stopping at a finite horizon, without which its regime $sL < K_M$ is empty (§4a.4).
- **One false justification, not carried.** A draft proof said that two boundary terms cancel "whether or not h_M is continuous". They cancel because h_M is continuous at K_M, and with a jump they differ by $[h_M(K_M^-) - h_M(K_M)]/s$. This is consistent with [NUMCHECK]'s mismatches at a hazard jump under B.

- **One question proofs could not settle, now settled by a ruling and a substitution.** The arrival-on Relation entries were first posed across readings, which is not a comparison a proof can make. The author ruled on 27 October that they be compared like with like, with the level-set side gated. On that reading one of the three divergence claims survives, row 5; gating removed the arrival-on divergences of rows 1 and 4, and row 3 has no arrival-on version. Of the three remaining entries, rows 2 and 6 hold as claimed. Row 7 took two passes to state correctly. The first said its two tests were not of the same form and that the row widened the identification problem. A review of 27 October showed both to be wrong: the same criterion would have condemned the row's other half, the gated level-set test is the speed-set test at $\varepsilon_\Lambda = 0$ rather than an incommensurable object, and the row is an instance of the whole-balance interval rule already in §4b.1. The second pass states the nesting, the signed gap $V_M\Lambda_M\varepsilon_\Lambda(L - \tau_M)$, and the comparison convention that any cross-specification threshold comparison needs. Two reported numbers went with the first pass and are gone: a frequency of disagreement across parameter draws, which counted draws at $\varepsilon_\Lambda = 0$ that A4 excludes, and a widest-gap figure that was mostly the harness's hazard scaling rather than the term at issue.